Honours Analysis 1 (Math 254) Tutorial Notes

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1 First Tutorial

As we are just beginning the course, this first tutorial will be somewhat introductory. We begin with some basic statements that can be established via direct proof, then briefly review the principle of mathematical induction at an informal/intuitive level. Following this, we provide several detailed examples of proof by induction with solutions. As the semester progresses, we shall encounter more techniques that may be used to prove mathematical claims. Nonetheless, it is typically a good idea to approach a new problem with an open mind, and to avoid getting stuck in a single method of proof. For instance, as we shall see shortly, there are statements that are quickly proven via a direct argument despite seeming prime candidates for mathematical induction.

Our first two examples are very basic and simply meant to illustrate the logical flow from assumption to deduction. In other words, these first few examples are included so we may familiarize ourselves with direct proofs, from hypothesis to conclusion.

Example 1.1. Prove the following statements, where $x, y \in \mathbb{Z}$.

1. If x and y are even, then $x + y$ is even.
2. If x and y are odd, then $x + y$ is even.
3. If one of x and y is even, and the other odd, then $x + y$ is odd.

Proof.

1. We are given that x, y are both even integers. Thus, by definition, there exist $m, n \in \mathbb{Z}$ such that $x = 2m$ and $y = 2n$. Then, we find that $x+y = 2m+2n = 2(m+n)$, where $m + n$ is itself an integer since $\mathbb{Z}$ is closed[1] under addition. In other words, defining $k := n + m \in \mathbb{Z}$, we have written $x + y = 2k$, whence we see that $x + y$ is an even integer.
2. We approach this in the same way as the previous point. First, since x and y are *odd*, there exist integers m, n such that $x = 2m + 1$ and $y = 2n + 1$. Then, we calculate

$$x + y = (2m + 1) + (2n + 1) = 2m + 2n + 2 = 2\underbrace{(m + n + 1)}_{\in\mathbb{Z}}.$$

   Thus, this form shows that $x + y$ is an even integer.

[1] The statement that $\mathbb{Z}$ is *closed* under addition merely asserts that $x + y$ is always an integer, for all $x, y \in \mathbb{Z}$.

3. Finally, we show that $x + y$ is odd if only one of x and y is even. Without loss of generality, by interchanging x and y if necessary, we may assume that x is even and y is odd.[2] That is, there exist integers m, n such that

$$x = 2m, \quad y = 2n + 1.$$

Then, $x + y = 2m + 2n + 1 = 2(m + n) + 1$. But, since $m + n \in \mathbb{Z}$, this implies that $x + y$ is an odd integer.

□

A similar statement holds for multiplication. As before, this can be solved via a direct argument.

Example 1.2. Prove the following statements, where $x, y \in \mathbb{Z}$.

1. If one of x and y is even, then xy is even.
2. If x and y are both odd, then xy is odd.

Proof.

1. As before, by swapping the roles of x and y if necessary, we may assume without loss of generality that x is even (why?). Then, $x = 2m$ for some $m \in \mathbb{Z}$. It follows that $xy = (2m)y = 2(my)$, where $my \in \mathbb{Z}$ since the product of any two integers is again an integer. This implies that xy is an even integer, as was asserted.
2. Using that x and y are both assumed to be odd, we may write $x = 2m + 1$ and $y = 2n + 1$, for suitable integers m, n. Following this, we write

$$xy = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Define $k := 2mn + m + n$; since $\mathbb{Z}$ is closed under addition and multiplication, we have that $k \in \mathbb{Z}$. Thus, $xy = 2k + 1$, where $k \in \mathbb{Z}$ whence we conclude that xy is an odd integer.

□

Example 1.3. Let $x, y \in \mathbb{Z}$ be such that $x^2 + y^2$ is even. Prove that $x + y$ is even.

Proof. Notice that x^2, y^2 are integers. Moreover, $x^2 + y^2$ is even by assumption. Now, by recalling Example 1.1, we see that the only way $x^2 + y^2$ can be even, is if either

[2] Indeed, if we found ourselves in the "other case" (i.e. x is odd and y is even), then we would simply apply the argument above to conclude that $y + x$ is odd, which is the same as saying that $x + y$ is odd.

(1) x^2 and y^2 are both even;

(2) x^2 and y^2 are both odd.

Indeed, if x^2 is even but y^2 is odd, then Example 1.1-(iii) would imply that $x^2 + y^2$ is odd, contradicting our assumption (the same applies if x^2 is odd but y^2 is even).

In the first case, if $x^2 = xx$ and $y^2 = yy$ are even, we infer from Example 1.2 that x and y must be even (why?). But then, citing once again Example 1.1, we see that $x + y$ is even as the sum of two even integers. In the second case where x^2 and y^2 are both odd, Example 1.2 implies instead that x and y are odd. In this case as well, Example 1.1 implies that $x + y$ is even. Since we may conclude that $x + y$ is even in both cases, the claim is proven. □

1.1 Proof by Induction

When it comes to mathematical induction, I think that the best way to understand the concept is to work out examples. Nonetheless, it is important to understand the logic behind this method. Say we have a statement $P(n)$, depending on $n \in \mathbb{N}$, that we wish to prove. More precisely, let's assume we want to show that $P(n)$ is true for all $n \geq 1$. For example, $P(n)$ could be the proposition

$$\text{“}5^n - 1 \text{ is divisible by } 4\text{”}.$$

In any case, our (hoped for) proof by induction should include the following:

(i) The *base case*. That is, we should check that $P(n)$ is true for the “base value” $n = 1$.

(ii) The *inductive step*. Here, we assume that $P(n)$ is true for some $n \geq 1$. Using this assumption, we then try to prove that $P(n + 1)$ is *also* true.

If we can establish both these parts, we will have proven that $P(n)$ holds true for all $n \geq 1$. Why? Since we know that $P(1)$ is true by our base case, the inductive step with $n = 1$ ensures that $P(1 + 1) = P(2)$ is also true. But then, the inductive step would again imply that $P(2 + 1) = P(3)$ is true, and so on.

Having given this explanation, we are ready to give some detailed examples in which the technique of mathematical induction proves to be very useful.

Proposition 1.1. $5^n - 1$ *is divisible by* 4 *for all* $n \in \mathbb{N}$.

Proof. Since we wish to prove a claim for all $n \geq 1$, we should take $n = 1$ as the base case.

Base case. Clearly, $5^1 - 1 = 4$ which is certainly divisible by 4. Hence, the base case is true.

Inductive step. Assume that $5^n - 1$ is divisible by 4 for some[3] $n \geq 1$ (this is the inductive hypothesis). We must show that $5^{n+1} - 1$ is also divisible by 4. But, this is easily seen by writing

$$\begin{aligned} 5^{n+1} - 1 = 5 \cdot 5^n - 1 &= (4+1) \cdot 5^n - 1 \\ &= 4 \cdot 5^n + (5^n - 1). \end{aligned}$$

Clearly, $4 \cdot 5^n$ is divisible by 4. By the induction hypothesis, so is $5^n - 1$. Hence, 4 must be a divisor of their sum, which is equal to $5^{n+1} - 1$ by the calculations above. □

Proposition 1.2. *For all integers $n \geq 4$, one has $2^n < n!$.*

Proof. We proceed by induction on $n \geq 4$. Since we want to prove a statement for all $n \geq 4$, our base case should be verifying the case $n = 4$.

Base case. For $n = 4$ we manually verify the claim:

$$2^4 = 16 < 24 = 4!.$$

That is, the statement $2^n < n!$ is valid when $n = 4$. This completes our base case.

Inductive step. Now we make the induction hypothesis. For some $n \geq 4$, let us assume that

$$2^n < n!. \tag{1.1}$$

We want to show that the statement also holds for $n + 1$. More precisely, we want to prove the following inequality:

$$2^{n+1} < (n+1)!$$

based off of the information in (1.1). By the inductive hypothesis (1.1), we have

$$(n+1)! = (n+1) \cdot n! \overset{(1.1)}{>} (n+1) \cdot 2^n.$$

Because $n \geq 4$, we also get $(n+1) \geq 5 \geq 2$. Using this together with the equation above yields

$$(n+1)! > (n+1) \cdot 2^n \geq 2 \cdot 2^n = 2^{n+1}.$$

This completes the inductive step. □

[3] We are *not* assuming that $5^n - 1$ is divisible by 4 for *all* $n \in \mathbb{N}$. Rather, we are just supposing that $5^n - 1$ is divisible by 4 for a single natural number n. For this natural number n, we demonstrate that $5^{n+1} - 1$ is also divisible by 4 *using the inductive hypothesis*, i.e. $5^n - 1$ is divisible by 4. Then, taking $n = 1$, since $5^1 - 1$ is divisible by 4, the inductive step shows that $5^2 - 1$ is divisible by 4. Next, taking $n = 2$, this step shows that $5^3 - 1$ is divisible by 4 and so on.

Proposition 1.3. *There holds*

$$\sum_{k=1}^{n} (8k-5) = 3 + 11 + \cdots + (8n-5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Proof. For $n \in \mathbb{N}$, the proposition/statement depending on n we wish to prove is the equality

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n.$$

Since we wish to prove the above for all $n \geq 1$, our base case should be $n = 1$.

Base case. We verify directly that the statement is true for $n = 1$:

$$3 = 8 \cdot 1 - 5 = 4 \cdot 1^2 - 1.$$

Inductive step. Our induction hypothesis will be that the statement is true for some $n \geq 1$. Our job is then to show that the claim holds for $n+1$. The induction hypothesis means that for this n:

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n. \tag{1.2}$$

We will deduce that:

$$3 + 11 + \cdots + (8(n+1)-5) = 4(n+1)^2 - (n+1).$$

First, our inductive hypothesis in (1.2) gives

$$\begin{aligned} 3 + 11 + \cdots + (8(n+1)-5) &= \underbrace{3 + 11 + \cdots + (8n-5)}_{=4n^2-n \text{ by } (1.2)} + (8(n+1)-5) \\ &= 4n^2 - n + 8(n+1) - 5 \\ &= 4n^2 + 7n + 3. \end{aligned}$$

On the other hand,

$$\begin{aligned} 4(n+1)^2 - (n+1) &= 4\left(n^2 + 2n + 1\right) - n - 1 \\ &= 4n^2 + 8n + 4 - n - 1 \\ &= 4n^2 + 7n + 3. \end{aligned}$$

Thus,

$$3 + 11 + \cdots + (8(n+1)-5) = 4n^2 + 7n + 3 = 4(n+1)^2 - (n+1).$$

This completes the inductive step. □

Direct Proof. If one recalls from class that $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$, for all $n \in \mathbb{N}$, it is possible to give a short direct proof of this last statement. Indeed, using this, we find that

$$\begin{aligned}\sum_{k=1}^{n} (8k-5) = 8\sum_{k=1}^{n} k - 5\sum_{k=1}^{n} 1 &= 8\left(\frac{n(n+1)}{2}\right) - 5n \\ &= 4n(n+1) - 5n \\ &= 4n^2 + 4n - 5n \\ &= 4n^2 - n.\end{aligned}$$

□

Proposition 1.4. *For every $n \in \mathbb{N}$, one has* $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

Proof. We establish the claim by way of induction. For the base case, we verify that the statement holds for $n = 1$. Indeed, for $n = 1$, the left hand side becomes

$$\sum_{k=1}^{1} k^2 = 1^2 = 1$$

whilst the right hand side reduces to

$$\frac{1(1+1)(2+1)}{6} = \frac{6}{6} = 1.$$

Thus, the statement holds at the "base case" $n = 1$.

We now perform the inductive step. To this end, let us assume the statement holds for some $n \in \mathbb{N}$, i.e.

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}. \tag{1.3}$$

We shall prove this implies the validity of the claim for $n + 1$. More precisely, using the truth of (1.3), we shall establish

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1)(n+1+1)(2(n+1)+1)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}.$$

To see this, we first write

$$\sum_{k=1}^{n+1} k^2 = \sum_{k=1}^{n} k^2 + (n+1)^2.$$

Using the inductive hypothesis (1.3), this becomes

$$\begin{aligned}
\sum_{k=1}^{n+1} k^2 = \sum_{k=1}^{n} k^2 + \left(n^2 + 2n + 1\right) &\overset{(1.3)}{=} \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \\
&= (n+1)\left[\frac{n(2n+1)}{6} + (n+1)\right] \\
&= \frac{n+1}{6}\left[n(2n+1) + 6(n+1)\right] \\
&= \frac{n+1}{6}\left[2n^2 + n + 6n + 6\right] \\
&= \frac{n+1}{6}\left[2n^2 + 7n + 6\right] \\
&= \frac{n+1}{6}\left[(n+2)(2n+3)\right].
\end{aligned}$$

This completes the inductive step and our proof is complete. □

Example 1.4. Prove that

$$\sum_{k=1}^{n} \frac{1}{\sqrt{k}} = \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n > 1$.

Proof. Although an inductive proof is certainly possible, we find it instructive (and more elegant) to give a direct proof based off a result seen in the lectures. Indeed, recall that $\sqrt{x}$ is strictly increasing on $[0, \infty)$. This means that, whenever $0 \leq x < y$, we have $\sqrt{x} < \sqrt{y}$. Now, notice that $k < n$ for all $k \in \{1, 2, \ldots, n-1\}$. Therefore, since $\sqrt{x}$ is strictly increasing, we have that

$$0 < \sqrt{k} < \sqrt{n}, \quad \forall k \in \{1, 2, \ldots, n-1\}.$$

This implies that

$$0 < \frac{1}{\sqrt{n}} < \frac{1}{\sqrt{k}}, \quad \forall k \in \{1, 2, \ldots, n-1\}.$$

Consequently,

$$\sum_{k=1}^{n} \frac{1}{\sqrt{k}} = \sum_{k=1}^{n-1} \overbrace{\frac{1}{\sqrt{k}}}^{>\frac{1}{\sqrt{n}}} + \frac{1}{\sqrt{n}} > \sum_{k=1}^{n-1} \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} 1 + \frac{1}{\sqrt{n}}$$
$$= \frac{(n-1)}{\sqrt{n}} + \frac{1}{\sqrt{n}}$$
$$= \frac{n}{\sqrt{n}}$$
$$= \sqrt{n}.$$

□

2 Second Tutorial

To get started, let us treat two important inequalities related to the absolute value function. Recall that for any $x \in \mathbb{R}$, its absolute value, denoted $|x|$, is given by the rule

$$|x| = \max\{x, -x\} = \begin{cases} x & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

By definition, $\pm x \leq |x|$ for all $x \in \mathbb{R}$. Some elementary properties are listed below, the proofs of which are left to the reader as exercises.

- $|xy| = |x|\,|y|$ for all $x, y \in \mathbb{R}$.
- $-|x| \leq x \leq |x|$, for all $x \in \mathbb{R}$.
- $|-x| = |x|$, for all $x \in \mathbb{R}$.
- $\sqrt{x^2} = |x|$, for all $x \in \mathbb{R}$.

Let us now establish the so called triangle inequality, which is of significant importance in this course. In fact, the triangle inequality (in other contexts) will play a central role in the study of metric spaces in Honours Analysis 2 (Math 254) next semester.

Theorem 2.1. *For all $x, y \in \mathbb{R}$, there holds that $|x + y| \leq |x| + |y|$.*

First Proof. By definition, given $x, y \in \mathbb{R}$, it holds that $\pm x \leq |x|$ and $\pm y \leq |y|$. Therefore,

$$x + y \leq |x| + y \leq |x| + |y|\,, \tag{2.1}$$
$$-x - y \leq |x| - y \leq |x| + |y|\,. \tag{2.2}$$

From (2.1), we infer that $x + y \leq |x| + |y|$ whilst (2.2) implies that $-(x + y) \leq |x| + |y|$. Combining these, it follows that $\max\{\pm(x + y)\} \leq |x| + |y|$ whence we conclude that $|x + y| \leq |x| + |y|$, as was asserted. □

We also provide a second proof, which instead uses a nice trick.

Second Proof. Given any $x, y \in \mathbb{R}$, we observe that

$$\begin{aligned} x^2 + 2xy + y^2 = |x|^2 + 2xy + |y|^2 &\leq |x|^2 + 2|xy| + |y|^2 \\ &= |x|^2 + 2|x||y| + |y|^2. \end{aligned}$$

More concisely,

$$x^2 + 2xy + y^2 \leq |x|^2 + 2|x||y| + |y|^2$$

which is equivalent to the statement $(x + y)^2 \leq (|x| + |y|)^2$, where both these quantities are non-negative. By taking square-roots of both sides and using that the square-root function is monotone increasing, it follows that

$$|x + y| = \sqrt{(x + y)^2} \leq \sqrt{(|x| + |y|)^2} = ||x| + |y|| = |x| + |y|.$$

□

2.1 Sets and Functions Between Them

Informally, a set should be understood as an unambiguous collection of objects. For example, $\mathbb{N} = \{1, 2, 3, \dots\}$ is the set of all natural numbers. Given two sets X and Y, a *function* $f : X \to Y$ is a rule which assigns to each $x \in X$ a *unique* element $f(x) \in Y$. Note that we do not require that f be described by an explicit formula here. Let us also take a brief moment to recall some basic definitions regarding functions. If X and Y are sets, a *bijection* from X to Y is a function $f : X \to Y$ satisfying each of the following:

(i) f is *injective*, i.e. $f(x) = f(x')$ implies $x = x'$;

(ii) f is *surjective*, i.e. for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.

Put otherwise, a function $f : X \to Y$ is a bijection *if and only if* for every $y \in Y$ there exists a *unique* point $x \in X$ such that $f(x) = y$.

Example 2.1. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ is not a bijection because it is neither injective nor surjective. Certainly, injectivity fails because $f(-1) = 1 = f(1)$. Surjectivity does not hold because there does not exist $x \in \mathbb{R}$ such that $f(x) = -1$.

Even so, we must be careful as a function that is not surjective can become surjective if we restrict its range. Consider the function

$$f : \mathbb{R} \to [0, \infty), \quad f(x) = x^2.$$

This is the same function as in the previous example, but with a "smaller" range. Although the function is still not injective ($f(-1) = f(1)$), it is surjective! Indeed, given $y \in [0, \infty)$, we have that $f(\sqrt{y}) = y$. Similarly, a function can become injective when we restrict its domain. Certainly, the function

$$f : [0, \infty) \to [0, \infty), \quad f(x) = x^2$$

is actually a bijection.

Remark 2.1 (Inverse Functions). The concept of an inverse function arises naturally when discussing bijective functions. If X and Y are sets and $f : X \to Y$ is a bijection, there exists for each $y \in Y$ a *unique* point x with the property that $f(x) = y$. Therefore, we can construct a function $f^{-1} : Y \to X$ by simply defining $f^{-1}(y) := x$. This map f^{-1} is called the *inverse function* of f.

Definition 2.1 (Image & Pre-image)**.** Let X and Y be sets and $f : X \to Y$ a function. For any set $A \subseteq X$, we define

$$f(A) := \{f(x) : x \in A\}$$

which is a subset of Y. This set $f(A)$ is called the *image of A under f*. Similarly, given $B \subseteq Y$, the *pre-image of B under f* is the set

$$f^{-1}(B) := \{x \in X : f(x) \in B\}.$$

That is, $f^{-1}(B)$ consists of those $x \in X$ that f takes to B.

Remark 2.2. The notation f^{-1} should **not** be confused with the inverse function of f, which in general does not exist. Even when f does not have an inverse function, we can still make sense of $f^{-1}(B)$! However, if the inverse function f^{-1} does exist, then the pre-image of a set $B \subseteq Y$ is precisely the *image* of B under the inverse function f^{-1}.

In practice one often has to deal with a large collection of sets. For such purposes, the concept of an index set is *very* useful. Informally, an index set I is a set that operates as the labeling scheme for a family of sets. By way of an example, consider for each $x \in \mathbb{R}$ the singleton set $\{x\}$.[4] Then, we may want to consider the set of all these singletons, i.e.

$$\Sigma := \big\{\{x\} : x \in \mathbb{R}\big\}.$$

[4]Here, $\mathbb{R}$ denotes the real number line $(-\infty, \infty)$.

Here, Σ is a set whose elements are those sets $\{x\}$ with $x \in \mathbb{R}$. However, when dealing with more complicated sets, this notation rapidly becomes cumbersome and awkward. Instead, we could use the more general concept of an index set. Defining $A_x := \{x\}$ for $x \in \mathbb{R}$, the set Σ can instead be written as $\{A_x\}_{x\in\mathbb{R}}$. Here, the real line $\mathbb{R}$ serves as our index set. The familiar concepts of unions and intersections carry over nicely when dealing with index sets. Let I be an index set and $\{X_\alpha\}_{\alpha\in I}$ be an indexed family of sets[5], we define:

$$\bigcup_{\alpha\in I} X_\alpha := \{x : \text{there exists } \alpha \in I \text{ such that } x \in X_\alpha\},$$
$$\bigcap_{\alpha\in I} X_\alpha := \{x : x \in X_\alpha \text{ for all } \alpha \in I\}.$$

Finally, let us recall a piece of notation: given a set X and a subset $A \subseteq X$, we denote by A^c the set $X \setminus A$. This set is often referred to as the *complement* of A in X, or simply the complement of A when the set X is implicitly understood. That is,

$$A^c := X \setminus A = \{x \in X : x \notin A\}.$$

Especially, we have $X = A \cup A^c$ with $A \cap A^c = \varnothing$. It should also be noted that A^c depends on the "parent set" X. Equipped with these concepts, we are ready to prove de Morgan's laws:

Theorem 2.2 (de Morgan's Laws)**.** *Let X be a set and let $\{X_\alpha\}_{\alpha\in I}$ be an indexed family of subsets of X, i.e. $X_\alpha \subseteq X$ for each $\alpha \in I$. Then,*

$$\left(\bigcup_{\alpha\in I} X_\alpha\right)^c = \bigcap_{\alpha\in I} X_\alpha^c, \tag{2.3}$$
$$\left(\bigcap_{\alpha\in I} X_\alpha\right)^c = \bigcup_{\alpha\in I} X_\alpha^c. \tag{2.4}$$

Proof. We begin by establishing (2.3). Here, we have the following two inclusions to demonstrate:

$$\left(\bigcup_{\alpha\in I} X_\alpha\right)^c \subseteq \bigcap_{\alpha\in I} X_\alpha^c \quad \text{and} \quad \bigcap_{\alpha\in I} X_\alpha^c \subseteq \left(\bigcup_{\alpha\in I} X_\alpha\right)^c.$$

(1) Let $x \in (\bigcup_{\alpha\in I} X_\alpha)^c$ be given. By definition, this means that $x \notin \bigcup_{\alpha\in I} X_\alpha$. Consequently, $x \notin X_\alpha$ for each $\alpha \in I$.[6] Put otherwise, we have $x \in X_\alpha^c$ for every $\alpha \in I$. Therefore, $x \in \bigcap_{\alpha\in I} X_\alpha^c$. Hence, we have proven that $(\bigcup_{\alpha\in I} X_\alpha)^c \subseteq \bigcap_{\alpha\in I} X_\alpha^c$.

[5]Remember, this only means that we have associated to each $\alpha \in I$ a set X_α. That is, we are using the elements of I to label the X_α's. This is particularly general because, as we shall see later on, some sets are so large that they cannot be exhaustively enumerated!

[6]Indeed, if x were an element of some X_α, it would also have to be an element of $\bigcup_{\alpha\in I} X_\alpha$.

(2) For the reverse inclusion, let $x \in \bigcap_{\alpha\in I} X_\alpha^{\mathsf{c}}$. By definition, this implies that $x \in X_\alpha^{\mathsf{c}}$ for every $\alpha \in I$. Equivalently,

$$x \notin X_\alpha \text{ for each index } \alpha \in I.$$

Now, $x \in \bigcup_{\alpha\in I} X_\alpha$ implies $x \in X_\alpha$ for some α. Since this contradicts the above, we must have $x \notin \bigcup_{\alpha\in I} X_\alpha$. By definition, this is the statement $x \in (\bigcup_{\alpha\in I} X_\alpha)^{\mathsf{c}}$. It follows that $\bigcap_{\alpha\in I} X_\alpha^{\mathsf{c}} \subseteq (\bigcup_{\alpha\in I} X_\alpha)^{\mathsf{c}}$.

This completes the proof of (2.3). Next, we turn our attention to (2.4). As above, we have two inclusions to prove:

(1) Fix a point $x \in (\bigcap_{\alpha\in I} X_\alpha)^{\mathsf{c}}$ and note that, by definition, $x \notin \bigcap_{\alpha\in I} X_\alpha$. Hence, there must exist an index $\alpha \in I$ such that $x \notin X_\alpha$. Otherwise, we would have $x \in X_\alpha$ for all $\alpha \in I$ whence $x \in \bigcap_{\alpha\in I} X_\alpha$ – which is a clear contradiction. Now, since $x \notin X_\alpha$ for some $\alpha \in I$, we also have $x \in X_\alpha^{\mathsf{c}}$ for this same α. In particular, $x \in \bigcup_{\alpha\in I} X_\alpha^{\mathsf{c}}$. This proves the inclusion $(\bigcap_{\alpha\in I} X_\alpha)^{\mathsf{c}} \subseteq \bigcup_{\alpha\in I} X_\alpha^{\mathsf{c}}$.

(2) Conversely, let $x \in \bigcup_{\alpha\in I} X_\alpha^{\mathsf{c}}$ be given. By definition of the union, this means that $x \in X_\alpha^{\mathsf{c}}$ for some $\alpha \in I$. That is, $x \notin X_\alpha$ for some index α. Consequently, we cannot have $x \in \bigcap_{\alpha\in I} X_\alpha$. Hence, $x \in (\bigcap_{\alpha\in I} X_\alpha)^{\mathsf{c}}$. We have therefore shown that $\bigcup_{\alpha\in I} X_\alpha^{\mathsf{c}} \subseteq (\bigcup_{\alpha\in I} X_\alpha)^{\mathsf{c}}$.

With this, (2.4) has been established. □

For the purposes of better understanding how functions apply to sets and the operations that act upon them, let us establish this next foundational result as an exercise:

Proposition 2.3. *Let X, Y be sets and $\{X_\alpha\}_{\alpha\in I}$ be an indexed family of subsets of X. If $f : X \to Y$ is a function, then*

$$f\left(\bigcup_{\alpha\in I} X_\alpha\right) = \bigcup_{\alpha\in I} f(X_\alpha). \tag{2.5}$$

Proof. We begin with the "$\subseteq$" inclusion. To this end, let $y \in f\left(\bigcup_{\alpha\in I} X_\alpha\right)$. This implies that there exists $x \in \bigcup_{\alpha\in I} X_\alpha$ with the property that $f(x) = y$. In other words, there is some index $\alpha \in I$ such that there exists $x \in X_\alpha$ satisfying $f(x) = y$. But, for this index α, it follows that $y \in f(X_\alpha)$. Consequently, $y \in \bigcup_{\alpha\in I} f(X_\alpha)$. Since $y \in f\left(\bigcup_{\alpha\in I} X_\alpha\right)$ was arbitrary, it follows that

$$f\left(\bigcup_{\alpha\in I} X_\alpha\right) \subseteq \bigcup_{\alpha\in I} f(X_\alpha).$$

Conversely, let $y \in \bigcup_{\alpha\in I} f(X_\alpha)$ be given. Then, there exists $\alpha \in I$ such that $y \in f(X_\alpha)$. For this index α, there must exist $x \in X_\alpha$ with $f(x) = y$. But then, since $x \in \bigcup_{\alpha\in I} X_\alpha$ as well, we conclude from $y = f(x)$ that $y \in f\left(\bigcup_{\alpha\in I} X_\alpha\right)$. This shows that

$$\bigcup_{\alpha\in I} f(X_\alpha) \subseteq f\left(\bigcup_{\alpha\in I} X_\alpha\right)$$

and the proof is complete. □

Having digested this, it is natural to ask whether the same can be said for intersections. Unfortunately, there is a caveat here. To understand why, consider first the following example: let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$ and consider the sets $A := [0, \infty)$ and $B := (-\infty, 0]$. Clearly, $A \cap B = \{0\}$ and so

$$f(A \cap B) = \{f(0)\} = \{0^2\} = \{0\}.$$

However, in contrast, $f(A) = [0, \infty)$ and $f(B) = [0, \infty)$ so that

$$f(A) \cap f(B) = [0, \infty) \supsetneq \{0\} = f(A \cap B).$$

Therefore, in general, we **cannot** assert that $f\left(\bigcap_{\alpha\in I} X_\alpha\right) = \bigcap_{\alpha\in I} f(X_\alpha)$. Nonetheless, the next statement can be made.

Proposition 2.4. *Let X, Y be sets and $f : X \to Y$ a function. Given any indexed family $\{X_\alpha\}_{\alpha\in I}$ of subsets of X, there holds*

$$f\left(\bigcap_{\alpha\in I} X_\alpha\right) \subseteq \bigcap_{\alpha\in I} f(X_\alpha). \tag{2.6}$$

Proof. Let $y \in f\left(\bigcap_{\alpha\in I} X_\alpha\right)$. Then, $y = f(x)$ for some $x \in \bigcap_{\alpha\in I} X_\alpha$. In other words, there exists x such that $x \in X_\alpha$, for all $\alpha \in I$, and $y = f(x)$. In particular, for each index $\alpha \in I$, we have that $y \in f(X_\alpha)$, since $y = f(x)$ for $x \in X_\alpha$. Hence, $y \in f(X_\alpha)$, for all $\alpha \in I$, i.e. $y \in \bigcap_{\alpha\in I} f(X_\alpha)$. This concludes the proof. □

Remark 2.3. What happens if we try to reverse the proof above to show the inverse inclusion, i.e. try to show that $\bigcap_{\alpha\in I} f(X_\alpha) \subseteq f\left(\bigcap_{\alpha\in I} X_\alpha\right)$? Clearly, by virtue of our counterexample, we *know* such an attempt at a proof must fail, for this claim is not in general true. But, which step is it precisely that fails? To see where, let us try to prove this erroneous inclusion of sets.

If $y \in \bigcap_{\alpha\in I} f(X_\alpha)$, then for all $\alpha \in I$ we have $y \in f(X_\alpha)$. This guarantees that, for each $\alpha \in I$, there exists $x_\alpha \in X_\alpha$ with $f(x_\alpha) = y$. Notice, however, that the selection of x_α *depends* on the index α, meaning we may pick up a distinct x_α for each new index. This means we cannot feasibly guarantee that there is some x in common to all the X_α's such that $f(x) = y$ and we can go no further.

Despite the fact that the opposite inclusion does not hold for images of intersections, understanding *where* the proof attempt went wrong provides insight into which property one should impose on f to enforce this equality. In our case, the issue came from the fact that we may be finding distinct points x_α that f sends to y. This would be rectified by requiring that f be injective. Indeed, this is established below.

Proposition 2.5. *Let X, Y be sets and $f : X \to Y$ an injective function. Given any indexed family $\{X_\alpha\}_{\alpha \in I}$ of subsets of X, there holds*

$$f\left(\bigcap_{\alpha \in I} X_\alpha\right) = \bigcap_{\alpha \in I} f(X_\alpha). \tag{2.7}$$

Proof. By Proposition 2.4, it suffices to show that $\bigcap_{\alpha \in I} f(X_\alpha) \subseteq f\left(\bigcap_{\alpha \in I} X_\alpha\right)$. To this end, let $y \in \bigcap_{\alpha \in I} f(X_\alpha)$ be given. By definition, this means that $y \in f(X_\alpha)$ for all indices $\alpha \in I$. Hence, for each $\alpha \in I$, there exists a point $x_\alpha \in X_\alpha$ satisfying $f(x_\alpha) = y$.

Choose any of these x_α's and label it x. Fix $\alpha \in I$ and observe that $f(x) = y = f(x_\alpha)$. Using that f is injective, it follows that $x = x_\alpha$. Since $\alpha \in I$ was arbitrary, it follows that $x = x_\alpha$ for all α whence $x \in X_\alpha$ for every $\alpha \in I$. To summarize, we have shown that there exists $x \in \bigcap_{\alpha \in I} X_\alpha$ such that $f(x) = y$. In other words, $y \in f\left(\bigcap_{\alpha \in I} X_\alpha\right)$. □

Example 2.2. Let X and Y be sets and $f : X \to Y$ a function. Let $A \subseteq X$ and $B \subseteq Y$. Prove the following statements.

(1) $f(f^{-1}(B)) \subseteq B$ with equality if f is surjective.

(2) $A \subseteq f^{-1}(f(A))$ with equality if f is injective.

Proof.

(1) First, we prove $f(f^{-1}(B)) \subseteq B$ in generality. Let $y \in f(f^{-1}(B))$ so that $y = f(x)$ for some $x \in f^{-1}(B)$. By definition of the pre-image, $x \in f^{-1}(B)$ is equivalent to the statement that $f(x) \in B$. Since $y = f(x)$, it follows that $y \in B$ so that $f(f^{-1}(B)) \subseteq B$.

    For the converse, let us assume in addition that f is surjective and let $y \in B$ be arbitrary. Using that f is surjective, and $y \in B \subseteq Y$, there exists at least one point $x \in X$ such that $f(x) = y$. But then, $x \in f^{-1}(B)$ by definition of pre-images. Thus, $y = f(x) \in f(f^{-1}(B))$ and we conclude that $B \subseteq (f^{-1}(B))$.

(2) Let $x \in A$ be given. Obviously, $f(x) \in f(A)$ which implies that $x \in f^{-1}(f(A))$. Conversely, assuming that f is injective, we observe that every $x \in f^{-1}(f(A))$ must satisfy $f(x) \in f(A)$, i.e. there exists $a \in A$ such that $f(x) = f(a)$. Appealing to the injectivity of f, we find that $x = a$ whence $x \in A$. This proves that $f^{-1}(f(A)) \subseteq A$.

□

2.2 Additional Examples

Here we collect some additional problems (selected from Bartle & Sherbert §1.1) relating to the material discussed in this tutorial (namely basic problems involving sets and functions). Although we will likely not have the time to discuss every problem here, we include them as a resource for additional practice.

Proposition 2.6. *For any sets A and B, we have*

$$(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B).$$

Remark 2.4. Given sets A and B, the **symmetric difference** of A and B, denoted $A \triangle B$, is defined by either side of the equation above. Descriptively, $A \triangle B$ consists of all elements that belong to A or B, but not both.

Proof. We have two inclusions to prove:

$$(A \setminus B) \cup (B \setminus A) \subseteq (A \cup B) \setminus (A \cap B)$$
$$(A \cup B) \setminus (A \cap B) \subseteq (A \setminus B) \cup (B \setminus A).$$

(1) We begin with the first inclusion. Let $x \in (A \setminus B) \cup (B \setminus A)$ so that it belongs to $(A \setminus B)$ or $(B \setminus A)$. Without loss of generality, we can assume that $x \in A \setminus B$.[7] Then, $x \in A$ but $x \notin B$. This means that

$$x \in A \cup B \quad \text{and} \quad x \notin A \cap B.$$

Hence, $x \in (A \cup B) \setminus (A \cap B)$.

(2) Conversely, let $x \in (A \cup B) \setminus (A \cap B)$. This simply means that

$$x \in A \cup B \quad \text{and} \quad x \notin A \cap B.$$

Without loss of generality, we can assume that $x \in A$. Since $x \notin A \cap B$, we cannot have $x \in B$. That is, $x \notin B$. This implies that $x \in A \setminus B$ whence

$$x \in (A \setminus B) \subseteq (A \setminus B) \cup (B \setminus A).$$

This completes the proof. □

Proposition 2.7. *Let X, Y and Z be sets. Suppose that*

$$f : X \to Y \quad \textit{and} \quad g : Y \to Z$$

are functions. Then, for every $A \subseteq Z$, we have

$$(g \circ f)^{-1}(A) = f^{-1}\left(g^{-1}(A)\right).$$

Here, $g \circ f$ denotes the composite function $x \mapsto g(f(x))$.

[7] Otherwise, $x \in B \setminus A$. In this case, we just relabel A as B and B as A!

Proof. First, observe that the claim makes sense because $g \circ f$ is a function $X \to Z$. Let $x \in (g \circ f)^{-1}(A)$ be given. This is to say that

$$(g \circ f)(x) = g(f(x)) \in A.$$

Hence, $f(x) \in g^{-1}(A)$. But this implies that $x \in f^{-1}(g^{-1}(A))$. This verifies the inclusion

$$(g \circ f)^{-1}(A) \subseteq f^{-1}(g^{-1}(A)).$$

Conversely, fix $x \in f^{-1}(g^{-1}(A))$. It follows that

$$f(x) \in g^{-1}(A) \implies g(f(x)) \in A.$$

Put otherwise, $(g \circ f)(x) = g(f(x)) \in A$. Thus, $x \in (g \circ f)^{-1}(A)$. Since x was arbitrary, it follows that

$$f^{-1}(g^{-1}(A)) \subseteq (g \circ f)^{-1}(A).$$

□

Proposition 2.8. *Let A, B and C be sets, and let $f : A \to B$ and $g : B \to C$ be functions. There hold the following:*

(1) if f and g are injective, then so is $g \circ f$;

(2) if $g \circ f$ is injective, then so is f;

(3) if f and g are surjective, then so is $g \circ f$;

(4) if $g \circ f$ is surjective, then so is g.

Proof.

(1) Let $a, a' \in A$ and suppose that $(g \circ f)(a) = (g \circ f)(a')$; our goal is to prove that $a = a'$. Now, this equation is equivalent to $g(f(a)) = g(f(a'))$. Because g is injective, this implies $f(a) = f(a')$. But, f is also injective! Consequently, it follows that $a = a'$.

(2) Suppose that $g \circ f$ is injective and let $a, a' \in A$ be such that $f(a) = f(a')$. We want to deduce that $a = a'$. Let us now apply the function g to both sides of the equality $f(a) = f(a')$. Doing so gives

$$(g \circ f)(a) = g(f(a)) = g(f(a')) = (g \circ f)(a').$$

Because $g \circ f$ is injective, it follows that $a = a'$. We conclude that f is injective.

(3) Let f and g both be surjective. For any point $c \in C$, we can choose $b \in B$ such that $g(b) = c$. Since f is surjective, there exists $a \in A$ with the property that $f(a) = b$. Then,

$$(g \circ f)(a) = g(f(a)) = g(b) = c.$$

(4) If $g \circ f$ is surjective, given any point $c \in C$ there exists $a \in A$ such that $(g \circ f)(a) = c$. This is simply the statement that $g(f(a)) = c$. Taking $b := f(a) \in B$, we see that $g(b) = c$. Hence, g is surjective.

□

Remark 2.5. Despite the above, there do exist functions f and g such that

(i) f is not surjective;
(ii) g is not injective;
(iii) and $g \circ f$ is a bijection.

Indeed, let $\mathbb{E}$ denote the set of all even natural numbers $\{2, 4, 6, \ldots, \}$. Consider the functions:

$$f : \mathbb{E} \to \mathbb{N}, \quad f(n) = n,$$
$$g : \mathbb{N} \to \mathbb{E}, \quad g(n) = \begin{cases} n & \text{if } n \text{ is even} \\ 2n & \text{if } n \text{ is odd.} \end{cases}$$

Clearly, f is injective but not surjective. Because $g(1) = g(2) = 2$, we see that g is not injective. But, $g \circ f$ is simply the identity map

$$g \circ f : \mathbb{E} \to \mathbb{E}, \quad (g \circ f)(n) = n$$

which is a bijection.

3 Third Tutorial

We begin with a rapid review of the definitions. Let X be a subset of $\mathbb{R}$ (a priori, possibly empty). A real number u is said to be an *upper bound* for X if $x \leq u$ for all $x \in X$. If X has an upper bound, we say that X is bounded from above. We call a point $s \in \mathbb{R}$ a *least upper bound* for X if both of the following hold:

(I) s is an upper bound for X;

(II) if u is any upper bound for X, then $s \leq u$.

A given subset of $\mathbb{R}$ may or may not have a least upper bound. If one exists, however, it must be unique. To see this, let s and s' be least upper bounds for a set X. In particular, they are both upper bounds. Thus, we must have $s \leq s'$ and $s' \leq s$ by (II) above. Of course, this forces $s = s'$.

Completeness Property of $\mathbb{R}$. Let X be a non-empty subset of $\mathbb{R}$ that is bounded from above. Then X has a least upper bound, which we denote by $\sup X$.

In your assignment, you have proven that a non-empty subset of $\mathbb{R}$ that is bounded from below has an *infimum*, i.e. a greatest lower bound. If X is such a set, we denote this quantity by $\inf X$. As before, $\inf X$ can be shown to be unique. With this in mind, we can state a converse to the completeness property of $\mathbb{R}$.

Proposition 3.1. *Let X be a subset of $\mathbb{R}$.*

1. *If X has a least upper bound (i.e. a supremum), then X is non-empty and bounded from above.*
2. *If X has a greatest lower bound (i.e. an infimum), then X is non-empty and bounded from below.*

Proof. We only prove the first claim, and leave the second as an exercise. Suppose that X has a supremum s in $\mathbb{R}$. Then, s is an upper bound for X. By definition, X must be bounded from above. All that remains is to check that X is non-empty. For this, it is enough to show that the empty set has no least upper bound.

Consider the empty set $\emptyset$ and suppose (for a contradiction) that $\emptyset$ has a least upper bound u in $\mathbb{R}$. Then, $x \leq u$ for all $x \in \emptyset$. But, we also have $x \leq u - 1$ for all $x \in \emptyset$. This means that $u - 1 < u$ is an upper bound for $\emptyset$, contradicting the choice of u as the least upper bound. □

Let X be a non-empty set that is bounded from above and let s denote its supremum in $\mathbb{R}$. Let $\varepsilon > 0$ be given and note that $s - \varepsilon < s$ cannot be an upper bound for X (since s is the *least* upper bound for X). Thus, there must exist $x_\varepsilon \in X$ such that $s - \varepsilon < x_\varepsilon$. Since s is an upper bound for X, this gives

$$s - \varepsilon < x_\varepsilon \leq s. \tag{3.1}$$

Conversely, let s be an upper bound for X and suppose that for each $\varepsilon > 0$ one can find $x_\varepsilon \in X$ such that (12.1) holds true. We claim that $s = \sup X$. As s is an upper bound for X, it is enough to show that $s \leq u$ for all upper bounds u of X. If u is an upper bound for X with $u < s$, then $\varepsilon := (s - u) > 0$ whence there exists $x_\varepsilon \in X$ with

$$s - \varepsilon < x_\varepsilon \leq s.$$

But, $s-\varepsilon = s-(s-u) = u$. This gives $u < x_\varepsilon$ which contradicts the fact that u is an upper bound for X. Thus, we must have $s \leq u$. Since u was an arbitrary upper bound, we have $s = \sup X$. This gives the following result from Bartle.

Theorem 3.2. *Let X be a non-empty set and s an upper bound for X in $\mathbb{R}$. The following statements are equivalent.*

1. *$s = \sup X$;*
2. *for every $\varepsilon > 0$ there exists $x_\varepsilon \in X$ such that $s - \varepsilon < x_\varepsilon \leq s$.*

For the infimum, a similar argument to what we used above yields an analogous result.

Theorem 3.3. *Let X be a non-empty set and v an lower bound for X in $\mathbb{R}$. The following statements are equivalent.*

1. *$v = \inf X$;*
2. *for every $\varepsilon > 0$ there exists $x_\varepsilon \in X$ such that $v \leq x_\varepsilon < v + \varepsilon$.*

3.1 The Existence of $\sqrt{2}$

We will elaborate on the argument used in Theorem 2.4.7 of Bartle's *Introduction to Real Analysis*. We claim that there exists a real number x such that $x^2 = 2$. To show this, we consider the set

$$X = \left\{ s \in [0, \infty) : s^2 < 2 \right\}.$$

Clearly, X is non-empty for $1^2 < 2$ implies $1 \in X$. Also, if $s > 2$ then $s^2 > 4$ implies $s \notin X$. Hence, $s \leq 2$ for all $s \in X$. This means that X is a non-empty bounded subset of $\mathbb{R}$. It therefore has a supremum x in $\mathbb{R}$. Since $1 \in X$ and x is an upper bound for X, we immediately have

$$0 < 1 \leq x.$$

We will show that both cases $x^2 < 2$ and $x^2 > 2$ are impossible. Clearly, this will force $x^2 = 2$ whence x is a square-root of 2.

1. First, suppose that $x^2 < 2$. For every $n \in \mathbb{N}$ we have $\frac{1}{n^2} \leq \frac{1}{n}$ so that

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}. \tag{3.2}$$

   By the assumption that $x^2 < 2$, we see that

$$\frac{2 - x^2}{2x + 1} > 0.$$

Thus, we can choose $n \in \mathbb{N}$ such that

$$0 < \frac{1}{n} < \frac{2 - x^2}{2x + 1}.$$

This implies that

$$\frac{2x + 1}{n} < 2 - x^2.$$

Returning to (3.2) yields

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x + 1}{n} < x^2 + 2 - x^2 = 2. \tag{3.3}$$

Because $x > 0$, we get that $x + \frac{1}{n}$ is an element of X. Since x is an upper bound for X we must have $x + \frac{1}{n} \leq x$, which is a contradiction. Therefore, the case $x^2 < 2$ is impossible.

2. Next, we show that the case $x^2 > 2$ is impossible. Since we know that $x \geq 1$ and $x^2 > 2$, we actually have $x > 1$. Given $m \in \mathbb{N}$ we calculate

$$\left(x - \frac{1}{m}\right)^2 = x^2 - \frac{2x}{m} + \frac{1}{m^2} > x^2 - \frac{2x}{m}. \tag{3.4}$$

Since $x^2 > 2$, $\frac{x^2 - 2}{2x}$ is positive. We can therefore choose $m \in \mathbb{N}$ such that

$$0 < \frac{1}{m} < \frac{x^2 - 2}{2x} \iff 0 < \frac{2x}{m} < x^2 - 2.$$

Of course, this implies

$$2 - x^2 < -\frac{2x}{m}.$$

Using this choice of m in (3.4), we obtain

$$\left(x - \frac{1}{m}\right)^2 > x^2 - \frac{2x}{m} > x^2 + 2 - x^2 = 2.$$

Since $x > 1$, we have $x - \frac{1}{m} > 0$. If $s > x - \frac{1}{m}$ then

$$s^2 > \left(x - \frac{1}{m}\right)^2 > 2.$$

Hence, $s \leq x - \frac{1}{m}$ for all $s \in X$. This means that $x - \frac{1}{m}$ is an upper bound for X, contradicting the least upper bound property of x.

3.2 The Construction of $\mathbb{R}$ via Dedekind Cuts

We have seen the Completeness property of the real numbers as an axiom, i.e. a statement that is taken to be true rather than proven. However, this property is generally not an axiom in standard modern formalisations of mathematics. Instead, the real numbers are constructed and the completeness property is proven.

For completeness of your mathematical education, we find it beneficial to provide a brief overview of one method that can be used to construct the real line from the rationals $\mathbb{Q}$. Furthermore, we will outline how to deduce the completeness property.

In what follows, we illustrate the construction of $\mathbb{R}$ via Dedekind cuts. We will *not* produce the construction in detail, i.e. our main goal is to describe how the real numbers may be built from the rationals, at a top level, without getting into the technical details. The main takeaway here should be the necessity of the construction *and* the guarantee of the least upper bound property in $\mathbb{R}$.

We would nonetheless like to point out that we shall provide slightly more detail here than in the tutorial. This is largely because the theory of Dedekind cuts is considered optional in this course; although you are likely to encounter them again in future algebra or set theory courses. For the curious student, we reference the famous textbook of Walter Rudin, titled *Principles of Mathematical Analysis*; a detailed construction of $\mathbb{R}$ via Dedekind cuts may be found within the appendix of Chapter 1.

Definition 3.1 (Dedekind Cut)**.** A subset $\alpha \subset \mathbb{Q}$ shall be called a *Dedekind cut* provided each of the following hold true:

1) $\varnothing \neq \alpha \neq \mathbb{Q}$.

2) for all $p \in \alpha$, if $q \in \mathbb{Q}$ is satisfies $q < p$, then $q \in \alpha$.

3) for each $p \in \alpha$, there exists $q \in \mathbb{Q}$ with $q > p$ and $q \in \alpha$.

There a few observations to be made here:

(d1) A Dedekind must be a proper subset of $\mathbb{Q}$ with no largest element.

(d2) Let $\alpha \subset \mathbb{Q}$ be a Dedekind cut and let $p \in \alpha$. If $q \notin \alpha$, then $q > p$.

(d3) Let $\alpha \subset \mathbb{Q}$ be a Dedekind cut and suppose $p, q \in \mathbb{Q}$ are such that $p < q$. If $p \notin \alpha$, then $q \notin \alpha$.

At the end of our construction, every rational number will be represented by a Dedekind cut as follows: for $q \in \mathbb{Q}$ set

$$\alpha_q := \{p \in \mathbb{Q} : p < q\}.$$

It is in this way that we shall produce a new number field containing a copy of $\mathbb{Q}$. Now, it is easy to see that q^* is indeed a Dedekind cut:

1) $\varnothing \neq q^* \neq \mathbb{Q}$.

2) Let $p \in q^*$ and suppose $r \in \mathbb{Q}$ satisfies $r < p$. Since $p \in q^*$ we have $p < q$ so, since $r < p$, there holds $r < q$. Therefore, $r \in q^*$.

3) For each $p \in q^*$, the rational number $r := (p + q)/2$ satisfies $p < r < q$. In particular, $r > p$ and $r \in q^*$.

Example 3.1. Consider the set $0^* := \{p \in \mathbb{Q} : p < 0\}$. Then, 0^* is a Dedekind cut. In fact, this will be the Dedekind cut that plays the role of *zero* in our new number system.

Using the set of rational numbers $\mathbb{Q}$, we must somehow conjure up a new set (denoted $\mathbb{R}$) upon which we may prescribe our familiar operations of arithmetic. Even more, we must define an *ordering* such that this new set $\mathbb{R}$ has the least upper bound property.

3.2.1 The Construction

Step 1. We define a new set, denoted $\mathbb{R}$, consisting of all possible Dedekind cuts. That is, in symbolic terms,

$$\mathbb{R} := \{\alpha \subset \mathbb{Q} : \alpha \text{ is a Dedekind cut}\}.$$

From a technical standpoint, note that $\mathbb{R}$ is not a set containing the rationals. Instead, $\mathbb{R}$ contains the representations q^* ($q \in \mathbb{Q}$) of rational numbers by Dedekind cuts. It is in this way that we view $\mathbb{R}$ as a set containing (a copy of) the rational numbers.

At this point in the construction, although the notation '$\mathbb{R}$' is very familiar, we *do not* yet know how to add, subtract, multiply or even compare any two elements of this set. In fact, it is precisely our goal to make sense of these notions in this new setting. Even so, the elements of this new set $\mathbb{R}$ (i.e. the Dedekind cuts) *will* play the role of real numbers.

As our next step, we define what it means for a Dedekind cut to be less than another given cut. This will induce a notion of "order" upon this set $\mathbb{R}$ thereby producing something called an ordered set. We define our order and emphasize the important properties satisfied below:

Step 2. Given two elements $\alpha, \beta \in \mathbb{R}$, we shall write $\alpha < \beta$ provided $\alpha \subset \beta$ is a proper subset (sometimes written $\alpha \subsetneq \beta$). In other words, we shall say that a cut α is *strictly less* than another cut β if $\alpha \subseteq \beta$ but there exists $p \in \beta \setminus \alpha$. Of course, we write $\alpha \leq \beta$ if either $\alpha < \beta$ or $\alpha = \beta$.

Lemma 3.4. *The binary relation defined above satisfies the following properties.*

1. *We have $\alpha \leq \alpha$ for any $\alpha \in \mathbb{R}$.*

2. *If $\alpha, \beta, \gamma \in \mathbb{R}$ satisfy $\alpha \leq \beta$ and $\beta \leq \gamma$, then $\alpha \leq \gamma$.*

3. *If $\alpha, \beta \in \mathbb{R}$ satisfy $\alpha \leq \beta$ and $\beta \leq \alpha$, then $\alpha = \beta$.*

4. *Given any $\alpha, \beta \in \mathbb{R}$, one of the following hold true:*

$$\alpha < \beta, \quad \alpha = \beta, \quad \beta < \alpha.$$

Proof of Lemma. The first three points are automatic from the definitions. To verify the last claim, let $\alpha, \beta \in \mathbb{R}$ and assume that $\alpha \not< \beta$ and $\alpha \neq \beta$. Then, because $\alpha \not< \beta$ and $\alpha \neq \beta$, we have that $\alpha \not\subseteq \beta$. So, there exists $p \in \alpha$ such that $p \notin \beta$. Now, given any $q \in \beta$, we must have $q < p$ by property (d2). However, this implies (since α is a Dedekind cut) that $q \in \alpha$. As $q \in \beta$ was arbitrary, it follows that $\beta \subset \alpha$, i.e. $\beta < \alpha$. □

Having defined an order on the set $\mathbb{R}$, we may now compare elements within this set. This naturally leads us to the notion of a set which is bounded from above. What is special regarding this construction, is the existence of a least upper bound. Although several constructions of $\mathbb{R}$ exist (Cantor's construction via Cauchy sequences was published alongside Dedekind's proof in 1872), it is Dedekind's construction that leads to the supremum property the fastest. This is explicitly proven below.

Step 3. Let $A \subseteq \mathbb{R}$ be a non-empty set with an upper bound. Then, A has a least upper bound in $\mathbb{R}$.

Proof. Let us consider the subset of $\mathbb{Q}$ given by $\gamma := \bigcup_{\alpha \in A} \alpha$. We first claim that α is indeed a real number (i.e. a Dedekind cut).

1) First, notice that A is itself a non-empty set consisting of Dedekind cuts and therefore each $\alpha \in A$ is non-empty, and such an α exists. Since this α is a subset of γ, we see that γ is non-empty. To see that $\gamma \neq \mathbb{Q}$, let $\beta \in \mathbb{R}$ be an upper bound of A, i.e. $\alpha \leq \beta$ for all $\alpha \in A$. Then, $\alpha \subseteq \beta$ for all $\alpha \in A$ whence $\gamma = \bigcup_{\alpha \in A} \alpha \subseteq \beta$ where $\beta \neq \mathbb{Q}$ since β is a Dedekind cut. It follows that $\varnothing \neq \gamma \neq \mathbb{Q}$.

2) Next, let $p \in \gamma$ and suppose $q \in \mathbb{Q}$ is such that $q < p$. Since $p \in \gamma$, there exists $\alpha \in A$ with $p \in \alpha$. Using that $\alpha \ni p$ is a Dedekind cut, we infer that $q \in \alpha \subseteq \gamma$ whence $q \in \gamma$. This shows that γ is closed from below.

3) To verify that γ indeed does not possess a largest element, let $p \in \gamma$ be given. There exists $\alpha \in A$ with $p \in \alpha$. Since α has no largest element, there is $q \in \alpha$ with $q > p$. But then, $q \in \gamma$ which concludes the proof that $\gamma \in \mathbb{R}$.

We conclude that γ is indeed a Dedekind cut. It remains only to verify that γ is the least upper bound of A in $\mathbb{R}$. To see that it is an upper bound of A, simply observe that $\alpha \subseteq \gamma$ for each $\alpha \in A$ by definition, i.e. $\alpha \leq \gamma$ for all $\alpha \in A$. Now, if $\delta \in \mathbb{R}$ is such that $\delta < \gamma$, then δ *cannot* be an upper bound of A. Indeed, since $\delta < \gamma$, we may find some $p \in \gamma \setminus \delta$. But then, $p \in \alpha$ for some $\alpha \in A$. By definition, it follows that $\alpha \not\leq \delta$ and so δ is *not* an upper bound of A. □

Step 4. Next, we define addition and the additive inverse of a real number. These notions combined will also produce the concept of subtraction. Given any two real numbers (i.e. Dedekind cuts) $\alpha, \beta \in \mathbb{R}$, we associate a new element of $\mathbb{R}$ given by

$$\alpha + \beta := \{p + q : p \in \alpha,\ q \in \beta\}\,.$$

It can be shown that $\alpha + \beta \in \mathbb{R}$, and this new function

$$+ : \mathbb{R} \times \mathbb{R} \to \mathbb{R},$$

satisfies the following:

- $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{R}$;
- $\alpha + \beta = \beta + \alpha$, for all $\alpha, \beta \in \mathbb{R}$.
- Defining $0^* := \{p \in \mathbb{Q} : p < 0\}$, we obtain a Dedekind cut such that

$$\alpha + 0^* = \alpha, \quad \forall \alpha \in \mathbb{R}.$$

- Given any $\alpha \in \mathbb{R}$, the cut

$$(-\alpha) := \{p \in \mathbb{Q} : \exists q \in \mathbb{Q}_{>0} \text{ such that } -p - q \notin \alpha\}$$

  satisfies $\alpha + (-\alpha) = 0^*$.

Moreover, is $\alpha, \beta, \gamma \in \mathbb{R}$ and $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$.

Step 5. Here, we define multiplication. First, if $\alpha, \beta \in \mathbb{R}$ are such that $\alpha, \beta \geq 0^*$, we put

$$\alpha\beta := \{p \in \mathbb{Q} : \exists r, s \in \mathbb{Q}_{>0} \text{ such that } r \in \alpha, s \in \beta \text{ and } p \leq rs\}\,.$$

As was done for 0^*, we define $1^* := \{p \in \mathbb{Q} : p < 1\}$. Finally, multiplication for negative real numbers is defined in terms of the above:

$$\alpha\beta := \begin{cases} -[(-\alpha)\beta] & \text{if } \alpha < 0^*,\ \beta \geq 0^*, \\ -[\alpha(-\beta)] & \text{if } \alpha \geq 0^*,\ \beta < 0^*, \\ (-\alpha)(-\beta) & \text{if } \alpha, \beta < 0^*. \end{cases}$$

Multiplication can be seen to be commutative and associative. Furthermore, the distributive law hols. That is, for $\alpha, \beta, \gamma \in \mathbb{R}$

$$\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma \,.$$

Finally, omitting details, we note that every $\alpha \in \mathbb{R}$, $\alpha \neq 0^*$ has a multiplicative inverse; i.e. an element $\alpha^{-1} \in \mathbb{R}$ such that $\alpha\alpha^{-1} = \alpha^{-1}\alpha = 1^*$.

Remark 3.1. Earlier in the tutorial, we discussed the existence of $\sqrt{2}$, i.e. the existence of a (non-negative) real number α such that $\alpha^2 = 2$. Having discussed Dedekind cuts, we note in our construction of $\mathbb{R}$ the square root of 2 is

$$\alpha := \mathbb{Q}_{\leq 0} \cup \left\{ q \in \mathbb{Q}_{>0} : q^2 < 2 \right\} \,.$$

Indeed, α is a Dedekind cut and it is possible to show that $\alpha^2 = 2^*$ (where $2^* = \{q \in \mathbb{Q} : q < 2\}$ is the Dedekind cut representing 2).

4 Fourth Tutorial

Let X and Y be non-empty sets. We say that X and Y have the same *cardinality* if there exists a bijection $f : X \to Y$. In this case, we write $|X| = |Y|$. If there does *not* exist a surjective function $X \to Y$, we write $|X| < |Y|$. Remember that the set X is called *countably infinite* if $|X| = |\mathbb{N}|$. More generally, the set X is said to be **countable** if it is either finite or countably infinite.

A useful (and natural) characterization of countability is the following, which should be well understood as it will be invoked liberally throughout the next results/examples. Of course, the main work was done within the lectures.

Theorem 4.1. *Let X be a non-empty set. Then, X is countable if and only if there exists a surjection $f : \mathbb{N} \to X$.*

Proof. The difficult imlpication " $\Longleftarrow$ " was proven in the lectures. The other (" $\Longrightarrow$ ") is very easy. Indeed, if X is countable, then it is either countably infinite or finite. In the former case, there exists a *bijection* $f : \mathbb{N} \to X$ which is, especially, a surjection. In the latter case, where X is finite, we may write $X = \{x_1, \ldots, x_N\}$ at which point the function

$$f : \mathbb{N} \to X, \quad n \mapsto \begin{cases} x_n & \text{if } 1 \leq n < N, \\ x_N & \text{if } n \geq N \end{cases}$$

is easily seen to be a surjection. □

Remark 4.1. It should be noted that a characterization of countability in terms of injective functions also exists. Namely, a non-empty set X is countable if and only if there exists an injection $X \to \mathbb{N}$. For more details, see Theorem 1.3.10 in Bartle & Sherbert.

Proposition 4.2. *The set $\mathbb{N} \times \mathbb{N}$ is countable.*

First Proof. Clearly, $\mathbb{N} \times \mathbb{N}$ is not finite. Thus, our only option is to show that $\mathbb{N} \times \mathbb{N}$ is countably infinite. Next, we observe that the elements of $\mathbb{N} \times \mathbb{N}$ can be listed according to the following table:

$$\begin{array}{cccccc}
(1,1) & (1,2) & (1,3) & \cdots & (1,n) & \cdots \\
(2,1) & (2,2) & (2,3) & \cdots & (2,n) & \cdots \\
(3,1) & (3,2) & (3,3) & \cdots & (3,n) & \cdots \\
(4,1) & (4,2) & (4,3) & \cdots & (4,n) & \cdots \\
\vdots & \vdots & \vdots & \ddots & \vdots & \cdots \\
(m,1) & (m,2) & (m,3) & \cdots & (m,n) & \cdots \\
\vdots & \vdots & \vdots & & \vdots & \ddots
\end{array}$$

We then enumerate the elements of $\mathbb{N} \times \mathbb{N}$ (i.e. the elements of the list above) according to the so-called "diagonal rule". Namely, we list the elements of $\mathbb{N} \times \mathbb{N}$ in the following order:

$$(1,1),\ (2,1),\ (1,2),\ (3,1),\ (2,2),\ (1,3), \ldots$$

Pictorially, we are drawing arrows through the diagonals of the list and "enumerating" the elements of $\mathbb{N} \times \mathbb{N}$ by following the arrows. In doing so, we are actually constructing a function $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$. This function would satisfy

$$1 \mapsto (1,1), \quad 2 \mapsto (2,1), \quad 3 \mapsto (1,2) \ldots.$$

Now, this function is clearly surjective[8]. Thus, we have shown that $\mathbb{N} \times \mathbb{N}$ is countable. □

Second Proof. We construct an explicit surjection $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$. By the fundamental theorem of arithmetic, any $n \in \mathbb{N}$ may be factored uniquely into products of primes, i.e.

$$n = 2^{k_1} 3^{k_2} 5^{k_3} \cdots$$

with each $k_j \in \mathbb{N}_0$ being uniquely determined by n. In other words, $k_1 \in \mathbb{N}_0$ is the maximal number such that $2^{k_1} \mid n$ and, similarly, $k_2 \in \mathbb{N}_0$ is the maximal number such that $3^{k_2} \mid n$. Then, we define a mapping

$$f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}, \quad n \mapsto (k_1 + 1, k_2 + 1).$$

[8] Since every element of $\mathbb{N} \times \mathbb{N}$ can be found on one of these finite diagonal lines, our function will "eventually reach" any given element of $\mathbb{N} \times \mathbb{N}$.

Clearly, f is surjective because, given a pair $(m, \ell) \in \mathbb{N} \times \mathbb{N}$, we see that

$$f(2^{m-1}3^{\ell-1}) = (m, \ell).$$

It follows that there exists a surjection $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$ and our proof is complete. □

Third Proof. We define a function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ according to the formula

$$f(m, n) = 2^m 3^n.$$

Suppose that $2^m 3^n = 2^r 3^s$ for natural numbers m, n, r, s. By the uniqueness of factorization into primes, we must have $m = r$ and $n = s$. Thus, f is injective whence $\mathbb{N} \times \mathbb{N}$ is countable. □

Using the countability of $\mathbb{N} \times \mathbb{N}$, we are able to deduce the following easy fact.

Proposition 4.3. *If X and Y are countable sets, then so is $X \times Y$.*

Proof. Because X and Y are both countable, we can find surjective functions

$$f : \mathbb{N} \to X \quad \text{and} \quad g : \mathbb{N} \to Y.$$

Now consider the function $h : \mathbb{N} \times \mathbb{N} \to X \times Y$ given by the formula

$$h(m, n) = (f(m), g(n)).$$

Let $(x, y) \in X \times Y$ be a point. Because f and g are surjective, there exist $m, n \in \mathbb{N}$ such that $f(m) = x$ and $g(n) = y$. Thus, $h(m, n) = (x, y)$. Hence, h is surjective. By the previous proposition, we know there exists a surjection $\psi : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$. Then, the composite $h \circ \psi$ will be a surjective function $\mathbb{N} \to X \times Y$. □

Using a simple inductive argument, we can actually conclude that finite products of countable sets are again countable. However, we should first discuss some of the unfortunate intricacies of notation.

4.1 Cartesian Products

Let $X_1, \ldots, X_n$ be sets. We define $X_1 \times \cdots \times X_n$ to be the set of all n-tuples

$$\{(x_1, \ldots, x_n) : x_j \in X_j,\ 1 \leq j \leq n\}.$$

To simplify notation, we often denote the product $X_1 \times \cdots \times X_n$ by

$$\prod_{k=1}^{n} X_k.$$

However, if we have $(n+1)$-many sets $X_1, \ldots, X_n, X_{n+1}$,

$$X_1 \times \cdots \times X_n \times X_{n+1} \neq (X_1 \times \cdots \times X_n) \times X_{n+1}. \tag{4.1}$$

Indeed, the set on the left has elements the form $(x_1, \ldots, x_{n+1})$ but the product on the right consists of tuples of the form $((x_1, \ldots, x_n), x_{n+1})$. Nonetheless, there always exists a bijection between these two sets. Certainly, consider the function

$$f : (X_1 \times \cdots \times X_n) \times X_{n+1} \to X_1 \times \cdots \times X_{n+1}$$

given by $f((x_1, \ldots, x_n), x_{n+1}) = (x_1, \ldots, x_n, x_{n+1})$. Clearly, f is surjective. To see that it is injective, suppose that

$$(x_1, \ldots, x_{n+1}) = \left(x_1', \ldots, x_{n+1}'\right).$$

Then, $x_j = x_j'$ for all $1 \leq j \leq n+1$. In particular, $x_{n+1} = x_{n+1}'$ and

$$(x_1, \ldots, x_n) = (x_1', \ldots, x_n').$$

Therefore, $((x_1, \ldots, x_n), x_{n+1}) = \left((x_1', \ldots, x_n'), x_{n+1}'\right)$. This shows that f is injective, and hence a bijection.

Since both the sets in (4.1) are always in bijection, there is no reason to distinguish between the two when discussing cardinality.

Proposition 4.4. *If $X_1, \ldots, X_n$ are countable sets, then so is $X_1 \times \cdots \times X_n$.*

Proof. We argue by induction on n. The base case $n = 1$ is obvious. Thus, assume that the claim holds true for $n \geq 1$ and let $X_1, \ldots, X_n, X_{n+1}$ be countable sets. By the inductive hypothesis, the set

$$X_1 \times \cdots \times X_n$$

is countable. By Proposition 4.3, so must be the product

$$(X_1 \times \cdots \times X_n) \times X_{n+1} \text{ “=” } X_1 \times \cdots \times X_n \times X_{n+1}.$$

This completes the proof. Here we are writing “=” instead of = because, although the sets are not equal per se, there exists a bijection between the two. Thus, these sets can be treated as the “same set” when discussing cardinality. □

Proposition 4.5. *Let X and Y be non-empty sets with $Y \subseteq X$. There exists a surjective function $\phi : X \to Y$ such that $\phi(x) = x$ for all $x \in X$.*

Proof. This is not difficult to prove. Fix any point $y_0 \in Y$ and define a function

$$\phi : X \to Y, \quad \phi(x) := \begin{cases} x & \text{if } x \in Y, \\ y_0 & \text{if } x \notin Y. \end{cases}$$

Obviously, ϕ is the desired surjective map. □

Proposition 4.6. *Let X and Y be sets with $Y \subseteq X$. If X is countable, then so is Y.*

Proof. Because X is countable (i.e. finite or countably infinite) there exists a surjective function $f : \mathbb{N} \to X$. By Proposition 4.5, we may choose a surjective map $g : X \to Y$. Then, the composite $g \circ f$ is a surjective map $\mathbb{N} \to Y$. Thus, Y is countable. □

Proposition 4.7. *Let $A_1, A_2, A_3, \ldots$ each be countable sets. Then, $\bigcup_{n\in\mathbb{N}} A_n$ is countable.*

Proof. Since each A_n is assumed to be countable, there exists a surjection $f_n : \mathbb{N} \to A_n$. Then, the function

$$\varphi : \mathbb{N} \times \mathbb{N} \to \bigcup_{n\in\mathbb{N}} A_n, \quad (n, m) \mapsto f_n(m)$$

defines a surjection. Indeed, given $x \in \bigcup_{n\in\mathbb{N}} A_n$, we must have $x \in A_n$ for some $n \in \mathbb{N}$. Using that $f : \mathbb{N} \to A_n$ is surjective, there exists $m \in \mathbb{N}$ such that $f_n(m) = x$. Then, $\varphi(n, m) = x$ and we see that φ is indeed a surjection. Finally, recall that $\mathbb{N}\times\mathbb{N}$ is countable and there exists a surjective mapping

$$\psi : \mathbb{N} \to \mathbb{N} \times \mathbb{N}.$$

By composition, $\varphi \circ \psi$ defines a surjection $\mathbb{N} \to \mathbb{N} \times \mathbb{N} \to \bigcup_{n\in\mathbb{N}} A_n$ and the proof is complete. □

Remark 4.2. In particular, this result states that the *finite* union of countable sets is countable. Indeed, given finitely many $A_1, \ldots, A_N$ that are each countable, we may define new sets

$$B_n := \begin{cases} A_n & \text{if } 1 \leq n < N, \\ A_N & \text{if } n \geq N. \end{cases}$$

Then, $\bigcup_{n=1}^{N} A_n = \bigcup_{n\in\mathbb{N}} B_n$ where $\bigcup_{n\in\mathbb{N}} B_n$ is countable by the last result. Consequently, so is $\bigcup_{n=1}^{N} A_n = A_1 \cup A_2 \cup \cdots \cup A_N$.

Corollary 4.8. *The set of all irrationals in $\mathbb{R}$ is uncountable.*

Proof. We know that every interval $[a, b] \subset \mathbb{R}$ is uncountable, and thus so is $\mathbb{R}$ (this follows from Proposition 4.6). Arguing by contradiction, let us assume that the set consisting of all irrational numbers, $\mathbb{Q}^c = \mathbb{R} \setminus \mathbb{Q}$, is countably infinite. Then, $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$ would be countable (see the latest remark), thereby contradicting the uncountability of $\mathbb{R}$. It follows that the set of irrationals must be uncountable. □

4.2 The Power Set

Let X be a set (possibly empty). The *power set of* X, denoted $\mathcal{P}(X)$, is defined to be the **set** of all subsets of X. Symbolically,

$$\mathcal{P}(X) := \{A : A \subseteq X\}.$$

Note that $\varnothing$ is always a subset of X, regardless of the set X. Even if X is empty, $\varnothing$ will be a subset of X. This means that $\varnothing \in \mathcal{P}(X)$ for all sets X. In particular, $\mathcal{P}(X)$ is never empty.

Example 4.1. We compute the power set for very small sets. First, we consider $\mathcal{P}(\varnothing)$. We have already seen that $\varnothing \in \mathcal{P}(\varnothing)$. Because $\varnothing$ is empty, it cannot have any non-empty subsets. Therefore, $\mathcal{P}(\varnothing) = \{\varnothing\}$.[9]

Example 4.2. We compute the power set of a singleton $\{x\}$. As always, $\varnothing$ is a subset of $\{x\}$ and hence $\varnothing \in \mathcal{P}(X)$. But, $\{x\}$ is also a subset of X. Thus,

$$\mathcal{P}(\{x\}) = \{\varnothing, \{x\}\}.$$

This argument shows that for a general set X there holds $\varnothing, X \in \mathcal{P}(X)$.

These examples tell us that $|\mathcal{P}(\varnothing)| = 1$ and $|\mathcal{P}(\{x\})| = 2$. The next problem generalizes this property.

Proposition 4.9. *If X is a finite set, then $|\mathcal{P}(X)| = 2^{|X|}$. Here, $|X|$ denotes the cardinality of the set X.*

Proof. As this is a statement depending on $n \in \mathbb{N}_0$, it is a good idea to prove the claim by induction.

Base case(s). In the previous examples we verified directly the cases $n = 0$ and $n = 1$. Therefore, our base case is already complete.

Inductive Step. We now make the induction hypothesis: assume that $|\mathcal{P}(X)| = 2^n$ for all sets X having n-elements. We want to show that $|\mathcal{P}(Y)| = 2^{n+1}$ for all sets Y having $(n+1)$-elements. Let Y be a set with $(n+1)$-elements. To count the elements of $\mathcal{P}(Y)$ is to count the subsets of Y. Fix a point $y \in Y$ and define $X := Y \setminus \{y\}$; this set X has n-elements. Now, there are **exactly** two types of subsets of Y:

(i) subsets of Y that contain y;

[9] Be warned that $\{\varnothing\}$ is a set containing the empty set. It is **not** the empty set, but rather a set whose single element is the empty set!

(ii) subsets of Y that do not contain y.

Clearly,

$$\begin{aligned}|\mathcal{P}(Y)| &= \#\text{subsets of } Y \\ &= \#\text{subsets of } Y \text{ that contain } y + \#\text{subsets of } Y \text{ not containing } y.\end{aligned}$$

Let $A \subseteq Y$ be a subset *not* containing y. Then, $A \subseteq X = Y \setminus \{y\}$. By our induction hypothesis, $|\mathcal{P}(X)| = 2^n$. Thus, there are exactly 2^n possibilities for A or, equivalently, 2^n-subsets of Y that don't contain y.

On the other hand, suppose that B is a subset of Y that does contain y. Then, $B \setminus \{y\}$ does not contain y. By the previous part, there are exactly 2^n-distinct possibilities for $B \setminus \{y\}$. But this means that there are precisely 2^n-distinct possibilities for B. Hence, there are 2^n-subsets of Y that contain y. This implies that

$$\begin{aligned}|\mathcal{P}(Y)| &= \#\text{subsets of } Y \text{ that contain } y + \#\text{subsets of } Y \text{ not containing } y \\ &= 2^n + 2^n = 2^{n+1}.\end{aligned}$$

This completes the inductive step. □

Corollary 4.10. $\displaystyle\sum_{k=0}^{n} \binom{n}{k} = 2^n$ *for all* $n \geq 0$.

Theorem 4.11 (Cantor)**.** *Let X be a set. There does not exist a surjection $X \to \mathcal{P}(X)$. In particular, no bijection $X \to \mathcal{P}(X)$ exists. Hence, $|X| < |\mathcal{P}(X)|$.*[10]

Proof. We argue by contradiction. Let $f : X \to \mathcal{P}(X)$ be a surjection. Consider the following subset of X:

$$A := \{x \in X : x \notin f(x)\}.$$

Because f is surjective, there exists $a \in X$ such that $f(a) = A$. There are now two cases to be considered.

(1) Suppose $a \in A$. By definition of A, this forces $a \notin f(a)$. But, $f(a) = A$. Therefore, $a \notin f(a)$ implies $a \notin A$. This is a contradiction.

(2) Assume $a \notin A$. That is, $a \notin f(a)$ whence $a \in A$. This is also a contradiction.

In either case we have a contradiction. Hence, no surjective function f can exist. □

[10] The map $f : X \to \mathcal{P}(X)$ given by $f(x) = \{x\}$ is clearly an injection from $X \to \mathcal{P}(X)$. Since no bijection $X \to \mathcal{P}(X)$ exists, we may indeed write $|X| < |\mathcal{P}(X)|$.

This subtle theorem has grandiose implications. Essentially, it states that one can build arbitrarily large sets by simply taking more and more power sets. Consider for instance the infinite set $\mathbb{N}$. You have probably seen that there does not exist a bijection $\mathbb{N} \to \mathbb{R}$. In this sense, we say that $\mathbb{R}$ is of a "larger infinity" than $\mathbb{N}$.[11] Cantor's theorem states that $\mathcal{P}(\mathbb{N})$ is of a "larger infinity" than $\mathbb{N}$. But then, $\mathcal{P}(\mathcal{P}(\mathbb{N}))$ is of an "even larger infinity" than $\mathcal{P}(\mathbb{N})$. Continuing in this way, we see that there are "infinitely many distinct infinities".

Corollary 4.12 (Russel's Paradox)**.** *The set of all sets does not exist.*

Proof. We argue by contradiction. Suppose that the set of all sets exists, and call it X. Then, $\mathcal{P}(X)$ is a set consisting of subsets of X. Hence, every element of $\mathcal{P}(X)$ is a set. It follows that $\mathcal{P}(X) \subseteq X$. By Proposition 4.5, there exists a surjective function $X \to \mathcal{P}(X)$. But this contradicts Cantor's theorem. □

Proposition 4.13. *Let $\mathcal{F}(\mathbb{N})$ denote the set of all finite subsets of $\mathbb{N}$. Then, $\mathcal{F}(\mathbb{N})$ is countably infinite.*

Proof. First, observe that $\mathcal{F}(\mathbb{N})$ is infinite. Indeed, it is clear that $\{n\} \subset \mathcal{F}(\mathbb{N})$, for each $n \in \mathbb{N}$, whence $\mathcal{F}(\mathbb{N})$ contains infinitely many finite subsets of $\mathbb{N}$. In addition, note that

$$\mathcal{F}(\mathbb{N}) = \bigcup_{n\in\mathbb{N}} \mathcal{P}(\{1,\dots,n\}). \tag{4.2}$$

To see this, first notice that any set $X \in \mathcal{P}(\{1,\dots,n\})$ is a subset of $\{1,\dots,n\}$, and hence finite. Thus, $X \in \mathcal{F}(\mathbb{N})$ whence $\mathcal{P}(\{1,\dots,n\}) \subseteq \mathcal{F}(\mathbb{N})$ so that

$$\bigcup_{n\in\mathbb{N}} \mathcal{P}(\{1,\dots,n\}) \subseteq \mathcal{F}(\mathbb{N}).$$

To justify the reverse inclusion, let $X \in \mathcal{F}(\mathbb{N})$, i.e. let X be a finite subset of $\mathbb{N}$. Denoting by $m \in \mathbb{N}$ the maximal element from X (this exists because X is finite), we must have $X \subseteq \{1,\dots,m\}$, from which we infer that $X \in \mathcal{P}(\{1,\dots,m\})$ and therefore $X \in \bigcup_{n\in\mathbb{N}} \mathcal{P}(\{1,\dots,n\})$. This concludes the proof of identity (4.2).

Having established this equality of sets, the proof is nearly complete. By virtue of Proposition 4.9, it is known that each $\mathcal{P}(\{1,\dots,n\})$ has finite cardinality. In particular, each $\mathcal{P}(\{1,\dots,n\})$ is countable. Then, $\mathcal{F}(\mathbb{N})$ must also be countable as the countable union of countable sets. □

Example 4.3. A binary sequence is a "list of points"

$$(a_1, a_2, a_3, \dots, a_n, \dots)$$

[11] This can be formalized in the sense of cardinal numbers. However, this is a topic for discussion in a dedicated set theory or logic course.

such that each a_j belongs to the set $\{0, 1\}$. Usually, one denotes such a sequence by (a_n). Let $\mathcal{B}$ be the set of all binary sequences. Then, $\mathcal{B}$ is uncountable.

Proof. We argue by contradiction. Suppose that $\mathcal{B}$ is countable, then we can enumerate the elements of $\mathcal{B}$:

$$\begin{aligned}(a_n^{(1)}) &= (a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \dots)\\ (a_n^{(2)}) &= (a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \dots)\\ &\vdots\\ (a_n^{(m)}) &= (a_1^{(m)}, a_2^{(m)}, a_3^{(m)}, \dots)\\ &\vdots\end{aligned}$$

Here, each $(a_n^{(m)})$ represents a sequence in $\mathcal{B}$. To derive a contradiction, we will construct an element in $\mathcal{B}$ (i.e. a binary sequence) that does not belong to the list above. For a given $n \in \mathbb{N}$, let b_n be given by the following formula:

$$b_n := \begin{cases} 1 & \text{if } a_n^{(n)} = 0, \\ 0 & \text{if } a_n^{(n)} = 1. \end{cases}$$

Grouping together all the b_n into a list gives us a sequence (b_n) in $\mathcal{B}$. We claim that (b_n) is not equal to any $(a_n^{(m)})$ appearing in the list above. If $(b_n) = (a_n^{(m)})$ for some $m \geq 1$, then $b_m = a_m^{(m)}$ for this m. But this would directly contradict the construction of (b_n). Hence, $(b_n) \notin \mathcal{B}$ which contradicts our assumption that $\mathcal{B}$ was countable. □

Remark 4.3. The set $\mathcal{B}$ treated in this last problem, consisting of all possible binary seqeunces, happens to be equal to the *infinite* cartesian product $\prod_{n=1}^{\infty} \{0, 1\}$. Hence, we see that the countable product of *finite sets* is not in general countable.

5 Fifth Tutorial

We begin by reiterating the definition of a limit for sequences. Recall that a sequence of real numbers is a function $x : \mathbb{N} \to \mathbb{R}$, where we denote by x_n the value $x(n)$. It has become convention to denote a sequence by either (x_n) or $\{x_n\}_{n\in\mathbb{N}}$; I prefer the former. Intuitively, we like to picture a sequence as an "infinite list" of numbers, that may have repetitions. For this reason, it is important not to confuse a sequence with a set of real numbers, as sets do not allow for repetitions.

Definition 5.1. Let (x_n) be a sequence (of real numbers) and fix $x \in \mathbb{R}$. We say that x_n *converges* to x (as $n \to \infty$) if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$. In symbolic terms:

$$(\forall \varepsilon > 0)\,(\exists N \in \mathbb{N})\,(n \geq N \implies |x_n - x| < \varepsilon)\,.$$

If this is the case, then we write $x_n \to x$, $x_n \xrightarrow{n\to\infty} x$, or

$$\lim x_n = x.$$

To help solidify the definition of convergence, we give a few explicit examples of convergent sequences. Although in practice one does not argue directly from the definition, it is important to fully understand the logic behind this definition.

Example 5.1. Show that

$$\lim \frac{2n}{n+1} = 2.$$

Proof. Let $\varepsilon > 0$ be given, we must find $N \in \mathbb{N}$ such that

$$\left|\frac{2n}{n+1} - 2\right| < \varepsilon, \quad \forall n \geq N.$$

First, we calculate for all $n \in \mathbb{N}$:

$$\left|\frac{2n}{n+1} - 2\right| = \left|\frac{2n - 2(n+1)}{n+1}\right| = \frac{2}{n+1} \leq \frac{2}{n}.$$

By the Archimedean property, we can find $N \in \mathbb{N}$ such that $N > \frac{2}{\varepsilon}$. Then, for all $n \geq N$ there holds

$$\frac{2}{n} \leq \frac{2}{N} < 2 \cdot \frac{\varepsilon}{2} = \varepsilon.$$

Hence, if $n \geq N$, we see that

$$\left|\frac{2n}{n+1} - 2\right| \leq \frac{2}{n} < \varepsilon,$$

which is what had to be shown. □

Example 5.2. Prove that

$$\lim \frac{3n+1}{2n+5} = \frac{3}{2}.$$

Proof. Let $\varepsilon > 0$ be given, we seek $N \in \mathbb{N}$ such that

$$\left|\frac{3n+1}{2n+5} - \frac{3}{2}\right| < \varepsilon$$

for all $n \geq N$. Naturally, for each $n \in \mathbb{N}$, we have the following estimate:

$$\begin{aligned}\left|\frac{3n+1}{2n+5} - \frac{3}{2}\right| = \left|\frac{2(3n+1) - 3(2n+5)}{2(2n+5)}\right| &= \left|\frac{2-15}{2(2n+5)}\right| \\ &= \frac{13}{2(2n+5)} \\ &\leq \frac{13}{2n} \\ &\leq \frac{13}{n}.\end{aligned}$$

Choose $N \in \mathbb{N}$ so large that $\frac{1}{N} < \frac{\varepsilon}{13}$ (why can we do this?), if $n \geq N$ the above implies that

$$\left|\frac{3n+1}{2n+5} - \frac{3}{2}\right| \leq \frac{13}{n} \leq \frac{13}{N} < 13 \cdot \frac{\varepsilon}{13} = \varepsilon.$$

By definition, this means that

$$\frac{3n+1}{2n+5} \xrightarrow{n\to\infty} \frac{3}{2}.$$

□

Remark 5.1. We would like to point out our general strategy for computing limits via the definition. If we wish to show that a sequence (x_n) converges to $x \in \mathbb{R}$, we start by fixing $\varepsilon > 0$. Our goal is then to show that there exists $N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ whenever $n \geq N$. To *find* such an N, we estimate $|x_n - x|$ for **all** n. This is done in the hopes of bounding the (often complex) expression $|x_n - x|$ by some term, depending on n, that is easier to estimate.

With regards to the last remark, we emphasize that by estimating the expression $|x_n - x|$, one often obtains a new sequence (a_n) bounding $|x_n - x|$, i.e. $|x_n - x| \leq a_n$ for all $n \in \mathbb{N}$. Whether or not this (a_n) converges to 0 is what determines if the estimate we have found is suitable. This is made precise below.

Theorem 5.1 (Null Sequence Criterion)**.** *Let (x_n) be a sequence and $x \in \mathbb{R}$. Then, x_n converges to x if and only if there exists a sequence (a_n) converging to 0 such that*

$$|x_n - x| \leq a_n, \quad \forall n \in \mathbb{N}.$$

5.1 More Limit Laws

We begin with the following:

Lemma 5.2. *Let (x_n) be a sequence of real numbers such that $\lim x_n = x \neq 0$. Then, there exists $N \in \mathbb{N}$ such that*

$$\frac{1}{2}|x| < |x_n| < \frac{3}{2}|x|,$$

for all $n \in \mathbb{N}$. In particular, the sequence is eventually non-zero.

Proof. Using that $x \neq 0$, we notice that $|x|/2 > 0$. Then, for $\varepsilon := |x|/2 > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon = \frac{|x|}{2}, \quad \forall n \geq N.$$

By the reverse triangle inequality, it follows that

$$||x_n| - |x|| < \frac{|x|}{2}, \quad \forall n \geq N.$$

Equivalently,

$$-\frac{|x|}{2} < |x_n| - |x| < \frac{|x|}{2}, \quad \forall n \geq N$$

whence a rearrangement yields

$$\frac{|x|}{2} < |x_n| < \frac{3|x|}{2}, \quad \forall n \geq N.$$

□

Proposition 5.3. *Let (x_n) be a sequence of real numbers such that $x_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim x_n = x \neq 0$. Then,*

$$\lim \frac{1}{x_n} = \frac{1}{x}.$$

Proof. Let $\varepsilon > 0$ be given; for each $n \in \mathbb{N}$ a straightforward calculation yields

$$\left|\frac{1}{x_n} - \frac{1}{x}\right| = \left|\frac{x - x_n}{x_n x}\right| = \frac{|x_n - x|}{|x_n|\,|x|}.$$

By the previous result, there exists $N_1 \in \mathbb{N}$ such that

$$\frac{1}{2}|x| < |x_n| < \frac{3}{2}|x|, \quad \forall n \geq N_1.$$

Meaning, for all $n \geq N_1$,

$$\left|\frac{1}{x_n} - \frac{1}{x}\right| = \frac{|x_n - x|}{|x_n|\,|x|} < \frac{2|x_n - x|}{|x|^2}.$$

Since $x_n \to x$, as $n \to \infty$, there exists by definition $N_2 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon |x|^2}{2}, \quad \forall n \geq N_2.$$

Then, if $n \geq N := \max(N_1, N_2)$, it follows that

$$\left| \frac{1}{x_n} - \frac{1}{x} \right| < \frac{2|x_n - x|}{|x|^2} < \varepsilon.$$

□

Corollary 5.4. *Let (x_n) and (y_n) be convergent sequences, such that $y_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim y_n \neq 0$. Assume further that (x_n) is convergent. Then, (x_n/y_n) is convergent and, moreover,*

$$\lim \frac{x_n}{y_n} = \frac{\lim x_n}{\lim y_n}$$

Proof. For each $n \in \mathbb{N}$, we may write

$$\frac{x_n}{y_n} = x_n \left(\frac{1}{y_n} \right),$$

where x_n converges to x by hypothesis and $\frac{1}{y_n}$ converges to $\frac{1}{y}$ by the last proposition. Thus, x_n/y_n is the *product* of convergent sequences and the product limit law implies that the product

$$\frac{x_n}{y_n} = x_n \left(\frac{1}{y_n} \right),$$

is convergent with limit equal to

$$\lim \frac{x_n}{y_n} = \lim \left[x_n \left(\frac{1}{y_n} \right) \right] = \lim x_n \, \lim \left(\frac{1}{y_n} \right) = x \left(\frac{1}{y} \right) = \frac{x}{y}.$$

□

5.2 Other Properties and Limits

We note that the convergence of a sequence (x_n) does imply that convergence of the sequence $(|x_n|)$. Indeed, we prove this below.

Proposition 5.5. *Let (x_n) be a sequence of real numbers converging to $x \in \mathbb{R}$. Then, $|x_n|$ converges to $|x|$, as $n \to \infty$.*

Proof. Let $\varepsilon > 0$ be given, since $\lim x_n = x$, we can find $N \in \mathbb{N}$ so large that $|x_n - x| < \varepsilon$ for all $n \geq N$. By the reverse triangle inequality, for all $n \geq N$ there then holds

$$||x_n| - |x|| \leq |x_n - x| < \varepsilon.$$

Hence, $\lim |x_n| = |x|$. □

The converse need not hold in general. Indeed, consider the sequence (x_n) given by $x_n := (-1)^n$. This sequence does **not** converge to any real number x, but the sequence given by $|x_n|$ is always equal to 1. Since constant sequences converge, this means that $(|x_n|)$ is convergent. Despite this, a partial converse to the above does hold.

Proposition 5.6. *Let (x_n) be a sequence of real numbers. If $(|x_n|)$ converges to 0 as $n \to \infty$, then so does (x_n).*

Proof. Given $\varepsilon > 0$, we can find $N \in \mathbb{N}$ so large that $||x_n| - 0| < \varepsilon$, whenever $n \geq N$. But this means that $|x_n| < \varepsilon$ for all $n \geq N$. This proves the claim. □

Proposition 5.7. *If $a > 1$ is a real number, then*

$$\lim \left(\frac{1}{a^n} \right) = 0.$$

Moreover, if $a \in (-1, 1)$, then $\lim a^n = 0$.

Proof. First suppose that $a > 1$. Then, $a = 1 + t$ for some $t > 0$. By Bernoulli's inequality, we know that

$$a^n = (1 + t)^n \geq 1 + nt, \quad \forall n \geq 1.$$

Therefore, for $n \in \mathbb{N}$, we have the estimate:

$$\left| \frac{1}{a^n} \right| = \frac{1}{a^n} \leq \frac{1}{1 + nt} \leq \frac{1}{nt}.$$

Choose $N \in \mathbb{N}$ such that $N > \frac{1}{t\varepsilon} > 0$. For all $n \geq N$, we have

$$\left| \frac{1}{a^n} - 0 \right| \leq \frac{1}{nt} \leq \frac{1}{Nt} < \frac{t\varepsilon}{t} = \varepsilon.$$

This shows that $\frac{1}{a^n} \to 0$, as $n \to \infty$. Now, we handle the case where $a \in (0, 1)$. Consider $b := \frac{1}{a} > 1$; for every $n \in \mathbb{N}$ we have

$$a^n = \frac{1}{b^n}.$$

By the first part, it follows that

$$\lim a^n = \lim \left(\frac{1}{b^n} \right) = 0.$$

If $a = 0$, then obviously $\lim a^n = 0$. Thus, it only remains to handle the case $a \in (-1, 0)$. In this case, note that $|a| \in (0, 1)$ and

$$|a^n| = |a|^n, \quad \forall n \in \mathbb{N}.$$

By the previous case (the case of $a \in (0, 1)$), we see that

$$\lim |a^n| = \lim \left(|a|^n \right) = 0.$$

Invoking Proposition 5.6, we get that $\lim a^n = 0$ in this case as well. □

6 Sixth Tutorial

Let (x_n) be a sequence and suppose that x_n converges to $x \in \mathbb{R}$. It was seen in the lectures that any subsequence (x_{n_k}) of (x_n) must *also* converge to this point x. Let us also recall the following limit, which was discussed in class:

$$e := \lim_{n \to \infty} \left(1 + \frac{1}{n} \right)^n. \tag{6.1}$$

We should also point out that this is our *definition* of e. Let us warm up by practicing with subsequences and limit-laws.

Example 6.1. We show that

$$\lim \left(1 - \frac{1}{n} \right)^n = \frac{1}{e}.$$

To see this, note that for $n \geq 2$:

$$\left(1 - \frac{1}{n} \right)^{-n} = \left(\frac{n-1}{n} \right)^{-n} = \left(\frac{n}{n-1} \right)^n.$$

Or, equivalently,

$$\left(1 - \frac{1}{n} \right)^{-n} = \left(1 + \frac{1}{n-1} \right)^n = \left(1 + \frac{1}{n-1} \right)^{n-1} \left(1 + \frac{1}{n-1} \right).$$

Therefore,

$$\begin{aligned}\lim\left(1-\frac{1}{n}\right)^{-n} &= \lim\left(1+\frac{1}{n-1}\right)^{n-1}\lim\left(1+\frac{1}{n-1}\right)\\ &= \lim\left(1+\frac{1}{n-1}\right)^{n-1}\\ &= e,\end{aligned}$$

where we have used (12.1) in this last step. Now, we write

$$\lim\left(1-\frac{1}{n}\right)^{n} = \lim\frac{1}{\left(1-\frac{1}{n}\right)^{-n}} = \frac{1}{\lim\left(1-\frac{1}{n}\right)^{-n}} = \frac{1}{e}.$$

Example 6.2. Similarly, we will show that

$$\lim\left(1+\frac{1}{2n}\right)^{n} = \sqrt{e}.$$

Here, we will instead make use of subsequences. For each $n \in \mathbb{N}$, we can write

$$\left(1+\frac{1}{2n}\right)^{n} = \left[\left(1+\frac{1}{2n}\right)^{2n}\right]^{\frac{1}{2}}. \tag{6.2}$$

Now,

$$\left(1+\frac{1}{2n}\right)^{2n}$$

is a subsequence of $\left(1+\frac{1}{n}\right)^{n}$, which we know to be convergent with limit e. This tells us that the sequence given above converges:

$$\lim\left(1+\frac{1}{2n}\right)^{2n} = \lim\left(1+\frac{1}{n}\right)^{n} = e.$$

Using this in (8.6), the limit laws tell us that

$$\lim\left(1+\frac{1}{2n}\right)^{n} = \lim\left[\left(1+\frac{1}{2n}\right)^{2n}\right]^{\frac{1}{2}} = \sqrt{e}.$$

Example 6.3. Find the flaw in the following "proof" that

$$\lim\left(1+\frac{2}{n}\right)^{n} = e^{2}.$$

"Proof". For every $n \in \mathbb{N}$ we can write

$$\left(1+\frac{2}{n}\right)^n = \left[\left(1+\frac{1}{n/2}\right)^{n/2}\right]^2.$$

Since we know that

$$\lim\left(1+\frac{1}{n}\right)^n = e,$$

we get that

$$\lim\left(1+\frac{1}{n/2}\right)^{n/2} = e.$$

Thus,

$$\lim\left(1+\frac{2}{n}\right)^n = \lim\left[\left(1+\frac{1}{n/2}\right)^{n/2}\right]^2 = e^2.$$

□

What exactly goes wrong with the proof above? Well, everything is correct up until the statement

$$\lim\left(1+\frac{1}{n/2}\right)^{n/2} = e.$$

This does **not** follow from the fact that

$$\lim\left(1+\frac{1}{n}\right)^n = e,$$

Indeed, although it seems to be at a first glance,

$$\left(1+\frac{1}{n/2}\right)^{n/2}$$

is not a subsequence of

$$\left(1+\frac{1}{n}\right)^n.$$

This is because $n/2$ need not be a natural number for every $n \in \mathbb{N}$.

Example 6.4. We claim that

$$\lim \sqrt[n]{n} = 1.$$

As a first step, notice that Bernoulli's inequality implies that

$$\left(1+\frac{1}{\sqrt{n}}\right)^n \geq 1+\frac{n}{\sqrt{n}} = 1+\sqrt{n} > \sqrt{n}, \quad \forall n \in \mathbb{N}. \tag{6.3}$$

Squaring both sides gives that

$$\left(1+\frac{1}{\sqrt{n}}\right)^{2n} > n, \quad \forall n \in \mathbb{N}.$$

Then, taking n^{th}-roots on both sides, we find that

$$\left(1+\frac{1}{\sqrt{n}}\right)^{2} > \sqrt[n]{n}, \quad \forall n \in \mathbb{N}.$$

Since we obviously have $1 \leq \sqrt[n]{n}$ for all $n \in \mathbb{N}$, it follows that

$$1 \leq \sqrt[n]{n} < \left(1+\frac{1}{\sqrt{n}}\right)^{2}, \quad \forall n \in \mathbb{N}.$$

Now, by the limit laws, the sequence on the right hand side is convergent with limit

$$\lim\left(1+\frac{1}{\sqrt{n}}\right)^{2} = \lim\left(1+\frac{1}{\sqrt{n}}\right)\lim\left(1+\frac{1}{\sqrt{n}}\right) = 1.$$

Consequently, it follows from the squeeze theorem that $\sqrt[n]{n} \to 1$, as $n \to \infty$.

Example 6.5. Consider the sequence $(x_n) \coloneqq \left(n^{1/n^2}\right)$. For each $n \in \mathbb{N}$, there holds

$$1 \leq (n)^{1/n^2} \leq (n^n)^{1/n^2} = n^{1/n} = \sqrt[n]{n}.$$

Thus, by the squeeze theorem, $x_n \to 1$.

Example 6.6. Let us try an adapt this reasoning to determine the limit of $(n!)^{1/n^2}$, as $n \to \infty$. As before, we have

$$1 \leq n \leq n! \leq n^n, \quad \forall n \in \mathbb{N}.$$

Therefore,

$$1 \leq (n!)^{1/n^2} \leq (n^n)^{1/n^2} = \sqrt[n]{n}, \quad \forall n \in \mathbb{N}.$$

Once again, by the squeeze theorem, we conclude that our given sequence is convergent with limit equal to 1.

6.1 Approximating square roots

Let $a > 0$ be given; we will construct a sequence in $\mathbb{R}$ converging to $\sqrt{a}$. In fact, as we shall soon see, this sequence will not require that we know the numerical value of $\sqrt{a}$. As a first step, pick **any** $x_1 > 0$ from $\mathbb{R}$. For $n \geq 1$, we define

$$x_{n+1} := \frac{1}{2}\left(x_n + \frac{a}{x_n}\right) > 0 \tag{6.4}$$

This gives rise to a recursive sequence (x_n), with $x_n > 0$ for every $n \in \mathbb{N}$. As a first step, we will show that $x_n^2 \geq a$ for all $n \geq 2$. For any n, note that x_n is a root of the following equation:

$$x^2 - 2x_{n+1}x + a = 0.$$

In particular, the quadratic equation above has a real root. It follows from the quadratic formula that the discriminant

$$4x_{n+1}^2 - 4a$$

must be non-negative, i.e. $4x_{n+1}^2 - 4a \geq 0$. Thus, $x_{n+1}^2 \geq a$ whence $x_{n+1} \geq \sqrt{a}$.

As a second step, we show that the sequence (x_n) is decreasing for all $n \geq 2$. Namely, we show that $x_{n+1} \leq x_n$, whenever $n \geq 2$. To see this, we simply calculate

$$\begin{aligned} x_n - x_{n+1} = x_n - \frac{1}{2}\left(x_n + \frac{a}{x_n}\right) &= \frac{x_n}{2} - \frac{a}{2x_n} \\ &= \frac{1}{2}\left(\frac{x_n^2 - a}{x_n}\right) \geq 0. \end{aligned}$$

Here, we have used that $x_n^2 \geq a$ whenever $n \geq 2$. Thus, $x_{n+1} \geq x_n$ whenever $n \geq 2$. Combined, we have shown that (x_n) is an eventually decreasing sequence with $x_n \geq \sqrt{a}$ for $n \geq 2$. By the monotone convergence theorem, we know that (x_n) will converge to some $x \geq \sqrt{a}$. In fact, using the limit laws with (6.4) shows that x will satisfy

$$x = \frac{1}{2}\left(x + \frac{a}{x}\right) \geq 0.$$

Rather,

$$x = \frac{a}{x} \geq 0.$$

Thus, we have $x^2 = a$ as was required. This recursive procedure is in general quite fruitful and simple. For this reason, we give one final example before moving on to more general results.

6.2 The Bolzano-Weierstrass Theorem

In this section, we prove one of the most fundamental and useful theorems in real analysis. Although the result heavily hinges upon our cumulative efforts thus far, the theorem will easily follow from one simple observation.

Definition 6.1. Let (x_n) be a sequence of real numbers. The Nth term x_N is called a *peak* of the sequence (x_n) if, for every $n \geq N$, there holds $x_n \leq x_N$.

In light of this definition, we have a relatively simple proof of our key lemma.

Lemma 6.1. *Any sequence of real numbers has a monotone subsequence.*

Proof. Let (x_n) be a sequence of real numbers. We distinguish the only two possible cases.

1. Suppose (x_n) has *infinitely many* peaks. Then, we can find a subsequence (x_{n_k}) of (x_n) consisting entirely of peaks. For every index k, x_{n_k} is a peak of the sequence (x_n). In particular, x_{n_1} is a peak of the sequence (x_n). Since $n_2 > n_1$, we must then have $x_{n_1} \geq x_{n_2}$. Similarly, since x_{n_2} is a peak and $n_3 > n_2$, there holds

   $$x_{n_2} \geq x_{n_3}$$

   and so forth. In general, we have

   $$x_{n_1} \geq x_{n_2} \geq \cdots \geq x_{n_k} \geq \cdots$$

   That is, (x_{n_k}) is a monotone subsequence of (x_n).

2. Otherwise, (x_n) has only finitely many peaks. Thus, we can find $N \in \mathbb{N}$ so large that x_n is not a peak of the sequence (x_n) for all $n \geq N$. Define $n_1 := N$. Since x_{n_1} is not a peak of the sequence (x_n), there is some x_j in our sequence, with $j \geq n_1$, such that $x_j > x_{n_1}$. In particular, $j \neq n_1$. Put then

   $$n_2 := j$$

   so that $x_{n_1} < x_{n_2}$ and $N \leq n_1 < n_2$. Again, x_{n_2} is not a peak of the sequence (x_n). By a similar argument, we can find $j > n_2$ such that $x_j > x_{n_2}$. Define then $n_3 := j$ so that

   $$x_{n_1} < x_{n_2} < x_{n_3}.$$

   Continuing in this way, we construct a monotone increasing subsequence (x_{n_k}) of (x_n).

In either case, we have found a monotone subsequence. □

Armed with this lemma, we can easily establish the following powerhouse of a theorem.

Theorem 6.2 (Bolzano-Weierstrass)**.** *Any bounded sequence in $\mathbb{R}$ has a monotone convergent subsequence.*

Remark 6.1. If we forgo the requirement that the subsequence be monotone, the theorem above continues to hold in $\mathbb{R}^m$ and $\mathbb{C}^n$ for all $m, n \in \mathbb{N}$. Such theorems are often known as *compactness* theorems and are extremely powerful.

Proof of Theorem. By our lemma, any sequence has a monotone subsequence. If (x_n) is bounded, then any monotone subsequence of (x_n) is also bounded. By the monotone convergence theorem, such a subsequence converges. □

6.3 A Quick Look at Series

Let (x_k) be a sequence of real numbers. For every $n \in \mathbb{N}$, let us define

$$S_n := \sum_{k=1}^{n} x_k.$$

This gives us a *new* sequence (S_n) in $\mathbb{R}$. We then say the infinite series $\sum_{k=1}^{\infty} x_k$ converges if there exists a real number S such that

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{k=1}^{n} x_n = S.$$

In this case, we will denote this limit by the quantity $\sum_{k=1}^{\infty} x_k$. The series $\sum_{k=1}^{\infty} x_k$ is said to be *absolutely convergent* if the series

$$\sum_{k=1}^{\infty} |x_k|$$

is convergent.

Example 6.7. You have most likely seen before that the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ does not converge because its partial sums are unbounded. Here, we show that the series

$$\sum_{k=1}^{\infty} \frac{1}{k^2}$$

is convergent (although we will not determine its value). Consider the sequence (S_n) of partial sums given by

$$S_n = \sum_{k=1}^{n} \frac{1}{k^2}.$$

Our goal is to show that (S_n) is convergent, as $n \to \infty$. First, observe that the partial sums (S_n) are strictly monotone increasing. Indeed,

$$S_{n+1} = \sum_{k=1}^{n+1} \frac{1}{k^2} = \sum_{k=1}^{n} \frac{1}{k^2} + \frac{1}{(n+1)^2} > \sum_{k=1}^{n} \frac{1}{k^2} = S_n.$$

By the monotone convergence theorem, it will follow that (S_n) (and, by definition, the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$) is convergent *provided* we can show that (S_n) is bounded. To this end, observe that

$$k(k-1) \leq k^2 \leq k(k+1), \quad \forall k \in \mathbb{N}.$$

Consequently,

$$\frac{1}{k(k+1)} \leq \frac{1}{k^2} \leq \frac{1}{k(k-1)}, \quad \forall k \geq 2. \tag{6.5}$$

Now, for each $n > 1$,

$$\begin{aligned} S_n = \sum_{k=1}^{n} \frac{1}{k^2} = 1 + \sum_{k=2}^{n} \frac{1}{k^2} &\leq 1 + \sum_{k=2}^{n} \frac{1}{k(k-1)} \\ &= 1 + \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right), \end{aligned}$$

where we have used (6.5) for the inequality and the fact that

$$\frac{1}{k-1} - \frac{1}{k} = \frac{1}{k(k-1)}.$$

It follows that

$$\begin{aligned} S_n \leq 1 + \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right) &= 1 + \sum_{k=2}^{n} \frac{1}{k-1} - \sum_{k=2}^{n} \frac{1}{k} \\ &= 1 + \sum_{k=1}^{n-1} \frac{1}{k} - \sum_{k=2}^{n} \frac{1}{k} \\ &= 1 + 1 + \sum_{k=2}^{n-1} \frac{1}{k} - \sum_{k=2}^{n} \frac{1}{k} \\ &= 2 - \frac{1}{n} \\ &\leq 2. \end{aligned}$$

Therefore, we conclude that (S_n) is monotone increasing and bounded from above. It follows that (S_n) converges.

7 Review Tutorial

In preparation for the upcoming midterm exam, we shall make this tutorial a review of the concepts we have seen thus far, together with some new worked out examples. Please note that there are already problems (contained in these notes) that we have not had the time to cover within the tutorials themselves. In addition, it is important to work out all assignment problems (especially the ungraded exercises).

7.1 Countability

Let X and Y be sets. Recall that a function $f : X \to Y$ is a rule assigning to each $x \in X$ a unique point $f(x) \in Y$. The set X in question would be called the domain of f, while the target Y is called the range. Given a subset $A \subseteq X$, its image under f is the subset of Y defined by

$$\{f(x) : x \in A\}$$

The function f is called injective if $f(x) = f(x')$ implies that $x = x'$, i.e. if each point in the image $f(X)$ is only mapped to from a unique point in X. Furthermore, we say this function f is surjective if for each $y \in Y$, there exists $x \in X$ such that $f(x) = y$, i.e. if $f(X) = Y$. Finally, we call f a bijection if f is both injective and surjective. In this case, f describes a *one-to-one* correspondence between the elements of X and those in Y. In reference to this, we say that two sets X and Y have the same cardinality provided there exists a bijection between X and Y, which we write $|X| = |Y|$. More generally, we shall say that $|X| \leq |Y|$ if there exists an injection $f : X \to Y$ (this would mean that X is in bijection with a subset of Y).

A given set X is said to be *countably infinite* provided $|X| = |\mathbb{N}|$. More generally, we say that X is *countable* if X is finite or countably infinite. Perhaps our most useful result in justifying countability is the following:

Theorem 7.1. *Let X be a non-empty set. Then, the following statements are equivalent.*

(1) X is countable;

(2) there exists a surjection $\mathbb{N} \to X$.

(3) there exists an injection $X \to \mathbb{N}$.

We remark that the surjection in point (2) simply describes an enumeration of the elements of the set X, possibly with repetition. Point (3), instead, states that there exists an injection $f : X \to \mathbb{N}$ whence X is bijection with its image $f(X) \subseteq \mathbb{N}$.

7.2 Completeness of $\mathbb{R}$

We saw in the Dedekind construction of the real numbers that the real line $\mathbb{R}$ possesses the *least upperbound property*. Recall that a number $u \in \mathbb{R}$ is called an upper bound for a set $A \subseteq \mathbb{R}$ if $a \leq u$ for all $a \in A$. We say that u is the *least* upper bound of A if $u \leq v$ for *any* upper bound v of A. When this least upper bound exists, it is called the *supremum* of A and denoted $\sup A$. Formally, the completeness of $\mathbb{R}$ can be stated as follows:

Completeness Property of $\mathbb{R}$. *Let $A \subseteq \mathbb{R}$ be non-empty and bounded from above. Then, A has a supremum in $\mathbb{R}$.*

A useful characterization of supremum is given by the following theorem:

Theorem 7.2. *Let $A \subseteq \mathbb{R}$ be non-empty and let u be an upper bound of A. The following statements are equivalent:*

(1) $u = \sup A$*;*

(2) for all $\varepsilon > 0$, there exists $a_\varepsilon \in A$ such that

$$u - \varepsilon < a_\varepsilon.$$

Remark 7.1. This is problem 1 from assignment 2, and is very important/useful. It should noted that the proof may also be found in the discussion preceding Theorem 3.2. A similar characterization also holds for the infimum (see Theorem 3.2 and assignment 2).

Problem 1. *Let A, B be non-empty subsets of $\mathbb{R}$ and suppose $B \subseteq A$. If A is bounded from above, prove that B is bounded from above and* $\sup B \leq \sup A$.

Solution. Observe that $\sup A$ is an upper bound for B. Indeed, if $b \in B$ then, since $B \subseteq A$, $b \in A$ so $b \leq \sup A$. In particular, the least upper bound $\sup B$ exists and satisfies $\sup B \leq \sup A$. □

Problem 2. *Let $X \subseteq \mathbb{R}$ be a non-empty set and define*

$$-X := \{-x : x \in X\}.$$

(a) Show that u is an upper bound for X if and only if $-u$ is a lower bound for $-X$.

(b) Show that X is bounded from above if and only if $-X$ is bounded from below.

(c) If X is bounded from above, show that $\inf(-X) = -\sup X$.

(d) Let X be a non-empty set bounded from below. show that X has an infimum in $\mathbb{R}$.

Solution.

(a) Suppose first u is an upper bound for X. Then, $x \leq u$ for all $x \in X$. Hence, $-u \leq -x$ for all $x \in X$. It follows that $-u \leq z$ for all $z \in -X$. Conversely, if $-u$ is a lower bound for $-X$ then $-u \leq z$ for all $z \in -X$. Thus, $-u \leq -x$ for all $x \in X$ so $x \leq u$ for all $x \in X$.

(b) If X is bounded from above, let u be an upper bound for X. Then, by (a), $-u$ is a lower bound for $-X$. Conversely, suppose $-X$ is bounded from below. Then let v be a lower bound for $-X$. With $u = -v$, we see that $-u$ is a lower bound for $-X$ so u is an upper bound for X. Ergo, X is bounded above.

(c) Write $u = \sup X$. Since u is an upper bound for X, by (a), $-u$ is a lower bound for $-X$. We claim that $-u$ is the greatest lower bound of $-X$. Indeed, suppose v is another lower bound for $-X$. Then, $-v$ is an upper bound for X. Since u is the least upper bound, $u \leq -v$ or, equivalently, $v \leq -u$.

(d) Suppose X is a non-empty set bounded from below. Then, $X = -(-X)$ is bounded from below so, by (b), $-X$ is bounded from above. Furthermore, by (c), we have $\inf(-(-X)) = -\sup(-X)$. Using once again that $-(-X) = X$, we conclude that X has an infimum in $\mathbb{R}$ (given by $-\sup(-X)$).

□

Problem 3. *Consider the set*

$$X = \left\{ \frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N} \right\}.$$

Compute $\sup X$ *and* $\inf X$.

Solution. We first check that X is bounded (bounded from above and below). By the triangle inequality, for all $n, m \in \mathbb{N}$:

$$\left| \frac{1}{n} - \frac{1}{m} \right| \leq \frac{1}{n} + \frac{1}{m} \leq 1 + 1 = 2.$$

Hence, X is indeed bounded. By the completeness of $\mathbb{R}$, both $\sup X$ and $\inf X$ exist in $\mathbb{R}$. Do note that

$$\frac{1}{n} - \frac{1}{m} \leq \frac{1}{n} \leq 1,$$

for all $n, m \in \mathbb{N}$. Hence, 1 is an upper bound for X. Informally, we should try to look for the largest elements in X and see what they approach. Here, we should take $n = 1$ and try to make $\frac{1}{m}$ as small as possible. In doing so, $1 - \frac{1}{m}$ seems to approach 1. With this in

mind, a good idea is to try and show that $\sup X = 1$. We give two proofs of this fact, one using the definition and the other via Theorem 7.2.

First Proof. Since we have already shown that 1 is an upper bound, it remains to check that 1 is the least upper bound. Suppose that $v < 1$ is an upperbound of X; we claim that v is *not* an upper bound of X. Indeed, by the Archimedean Property, there exists $m \in \mathbb{N}$ such that $\frac{1}{m} < 1 - v$. Then, notice that

$$1 - \frac{1}{m} > 1 - (1 - v) = v.$$

Since $1 - \frac{1}{m} \in X$ by definition, and v was assumed to be an upperbound for X, this is a contradiction. To summarize, we have shown that 1 is an upper bound of X and no $v < 1$ can be an upper bound of X. It follows that 1 is the *least* upper bound of X.

Second Proof. Let $\varepsilon > 0$ be given, we want to show that there exists $x_\varepsilon \in X$ such that

$$1 - \varepsilon < x_\varepsilon \leq 1.$$

Choose $m \in \mathbb{N}$ such that $\frac{1}{m} < \varepsilon$ (how?) and note that

$$x_\varepsilon := 1 - \frac{1}{m} \in X.$$

Then, $-\varepsilon < -\frac{1}{m}$ whence

$$1 - \varepsilon < 1 - \frac{1}{m} = x_\varepsilon \leq 1.$$

By Theorem 3.2, we see that $\sup X = 1$.

For $\inf X$, we will use a trick from Problem 2 Observe that

$$X = \left\{ \frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N} \right\} = \left\{ \frac{1}{m} - \frac{1}{n} : n, m \in \mathbb{N} \right\} = -X.$$

Hence, by Problem 2 we get $\inf X = \inf(-X) = -\sup X = -1$. □

7.3 Density

We have seen in class that the rationals $\mathbb{Q}$ are dense in $\mathbb{R}$. Following this, you demonstrated in your assignment that the irrationals $\mathbb{R} \setminus \mathbb{Q}$ were also dense in $\mathbb{R}$. For the sake of review, let us recall that a set $S \subseteq \mathbb{R}$ is said to be dense in $\mathbb{R}$ provided, for each $a < b$, there exists $s \in S$ with $a < s < b$. In other words, S is called dense if $S \cap (a, b) \neq \varnothing$ for all non-empty open intervals $(a, b) \subseteq \mathbb{R}$. Let us offer a different proof, based on the uncountability of $(0, 1)$, that the irrationals are dense in $\mathbb{R}$.

Proposition 7.3. *The irrationals $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$. In fact, every open interval contains uncountably many irrationals.*

Proof. Let $a < b$ be real numbers; we must show that (a, b) contains an irrational number. To this end, recall from class that $[a, b]$ is uncountable. If (a, b) were countable, then writing $[a, b] = (a, b) \cup \{a\} \cup \{b\}$, it would follow that $[a, b]$ is the finite union of countable sets and hence countable itself. Thus, (a, b) is uncountable. However,

$$(a, b) = [\mathbb{Q} \cap (a, b)] \cup [(\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)]$$

where $\mathbb{Q} \cap (a, b) \subset \mathbb{Q}$ is countable as a subset of $\mathbb{Q}$. If $(\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)$ were countable (e.g. empty) then (a, b) would be the union of two countable sets, and hence countable itself. This impossibility implies that $(\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)$ is uncountable and, in particular, non-empty. □

7.4 Limit of Sequences

Example 7.1. Prove, using only the definition, that

$$\lim \frac{n^2 - 1}{2n^2 + 3} = \frac{1}{2}.$$

Proof. Consider the expression:

$$\begin{aligned} \left| \frac{n^2 - 1}{2n^2 + 3} - \frac{1}{2} \right| &= \left| \frac{2(n^2 - 1) - (2n^2 + 3)}{2(2n^2 + 3)} \right| \\ &= \left| \frac{5}{2(2n^2 + 3)} \right| \\ &\leq \frac{5}{4n^2} \\ &\leq \frac{4}{n}. \end{aligned}$$

Now, it will be easier to find our candidate for N. Let $N \in \mathbb{N}$ be such that

$$N > \frac{2}{\sqrt{\varepsilon}},$$

which we know to exist by the Archimedean property. If $n \geq N$, then our estimate above tells us that

$$\left| \frac{n^2 - 1}{2n^2 + 3} - \frac{1}{2} \right| \leq \frac{4}{n^2} \leq \frac{4}{N^2} < 4 \cdot \frac{\varepsilon}{4} = \varepsilon.$$

This is what had to be shown. □

We give one final example.

Example 7.2. Prove that

$$\lim \frac{\sqrt{n}}{n+1} = 0.$$

Proof. For every $n \in \mathbb{N}$, we have

$$\left| \frac{\sqrt{n}}{n+1} - 0 \right| = \frac{\sqrt{n}}{n+1} \leq \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}.$$

Since $1/\sqrt{n} \to 0$, the claim now follows by the null sequence criterion Theorem 5.1. □

Example 7.3. Prove that

$$\lim \left(\frac{(-1)^n + \sqrt{n}}{\sqrt{n}+1} \right) = 1.$$

Proof. For each $n \in \mathbb{N}$, we have

$$\begin{aligned} \left| \frac{(-1)^n + \sqrt{n}}{\sqrt{n}+1} - 1 \right| = \left| \frac{(-1)^n + \sqrt{n} - (\sqrt{n}+1)}{\sqrt{n}+1} \right| &= \left| \frac{(-1)^n - 1}{\sqrt{n}+1} \right| \\ &= \frac{|(-1)^n - 1|}{\sqrt{n}+1} \\ &\leq \frac{|(-1)^n| + |-1|}{\sqrt{n}+1} \\ &= \frac{2}{\sqrt{n}+1} \\ &< \frac{2}{\sqrt{n}}. \end{aligned}$$

Since $2/\sqrt{n} \to 0$, we conclude by the null sequence criterion (Theorem 5.1) that the given sequence converges to 1. □

Example 7.4. We claim that

$$\lim \sqrt[n]{n} = 1.$$

As a first step, notice that Bernoulli's inequality implies that

$$\left(1 + \frac{1}{\sqrt{n}}\right)^n \geq 1 + \frac{n}{\sqrt{n}} = 1 + \sqrt{n} > \sqrt{n}, \quad \forall n \in \mathbb{N}. \tag{7.1}$$

Squaring both sides gives that

$$\left(1 + \frac{1}{\sqrt{n}}\right)^{2n} > n, \quad \forall n \in \mathbb{N}.$$

Then, taking n^{th}-roots on both sides, we find that

$$\left(1+\frac{1}{\sqrt{n}}\right)^2 > \sqrt[n]{n}, \quad \forall n \in \mathbb{N}.$$

Since we obviously have $1 \leq \sqrt[n]{n}$ for all $n \in \mathbb{N}$, it follows that

$$1 \leq \sqrt[n]{n} < \left(1+\frac{1}{\sqrt{n}}\right)^2, \quad \forall n \in \mathbb{N}.$$

Now, by the limit laws, the sequence on the right hand side is convergent with limit

$$\lim\left(1+\frac{1}{\sqrt{n}}\right)^2 = \lim\left(1+\frac{1}{\sqrt{n}}\right)\lim\left(1+\frac{1}{\sqrt{n}}\right) = 1.$$

Consequently, it follows from the squeeze theorem that $\sqrt[n]{n} \to 1$, as $n \to \infty$.

7.5 Monotone Convergence

As we've previously discussed, it is difficult to establish the convergence of a sequence via the definition when a candidate for the limit is not clear. In this case, we have other (less direct) methods to help establish the convergence. Following this, we can try to identify the limit explicitly using the limit laws. One such useful result is the monotone convergence theorem, which we reiterate below.

Theorem 7.4 (Monotone Convergence Theorem)**.** *Let (x_n) be a bounded sequence of real numbers.*

(i) If (x_n) is increasing[12]*, then (x_n) converges to $\sup\{x_n : n \in \mathbb{N}\}$.*

(ii) If (x_n) is decreasing[13]*, then (x_n) converges to $\inf\{x_n : n \in \mathbb{N}\}$.*

Let us now give an example.

Example 7.5. Let $x_1 \geq 2$ be given and define for $n \geq 1$:

$$x_{n+1} := 1 + \sqrt{x_n - 1}. \tag{7.2}$$

We will show that the recursive sequence (x_n) converges and calculate its limit. First, let us show (by induction) that $x_n \geq 2$ for all $n \in \mathbb{N}$. The base case $n = 1$ is trivial. Assuming that $x_n \geq 2$, we then have

$$x_{n+1} = 1 + \sqrt{x_n - 1} \geq 1 + \sqrt{2-1} = 2.$$

[12] This means that $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$.
[13] This means that $x_n \geq x_{n+1}$ for all $n \in \mathbb{N}$.

Here, we have used that the function $f(x) := \sqrt{x}$ is monotone increasing on $[0, \infty)$.[14] We conclude that $x_n \geq 2$ for all $n \in \mathbb{N}$. In particular, $x_n - 1 \geq 1$ for all $n \in \mathbb{N}$. Thus, given $n \in \mathbb{N}$, we have

$$x_{n+1} = 1 + \sqrt{x_n - 1} \leq 1 + (x_n - 1) = x_n.$$

In this last step, we have used that $\sqrt{t} \leq t$ for all $t \geq 1$.[15] Hence, (x_n) is a monotone decreasing sequence that is bounded below by 2. It therefore converges to some point $x \geq 2$. Using the limit laws and (7.2), this point x must satisfy

$$x = 1 + \sqrt{x - 1}.$$

It follows that $(x - 1) = \sqrt{x - 1}$, where $x - 1 \geq 1$. Hence, $\sqrt{x - 1} = 1$ whence $x = 2$. We conclude that $\lim x_n = 2$.

Example 7.6. Let (a_n) be an increasing sequence and (b_n) a decreasing sequence such that $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then, (a_n) and (b_n) both converge and $\lim a_n \leq \lim b_n$.

Proof. Note that $a_n \leq b_n \leq b_1$, for all $n \in \mathbb{N}$. Therefore, (a_n) is bounded from above and convergent. Similarly, $a_1 \leq a_n \leq b_n$, whence (b_n) is bounded from below and convergent. Now, the sequence (x_n) defined by $x_n := b_n - a_n \geq 0$ and is convergent (by the limit laws) with limit

$$\lim x_n = \lim b_n - \lim a_n.$$

Since $x_n \geq 0$ for all $n \in \mathbb{N}$, we know from class that $0 \leq \lim x_n = \lim b_n - \lim a_n$, i.e. $\lim a_n \leq \lim b_n$ as was asserted. □

Remark 7.2. It is possible to deduce the nested interval property from Theorem 7.6 (and hence from the monotone convergence theorem). Indeed, let

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots \supseteq [a_n, b_n] \supseteq \ldots$$

[14] To see why this is true, suppose that $0 \leq x < y$. Then, we calculate

$$\sqrt{y} - \sqrt{x} = \frac{(\sqrt{y} - \sqrt{x})(\sqrt{y} + \sqrt{x})}{\sqrt{y} + \sqrt{x}} = \frac{y - x}{\sqrt{x} + \sqrt{y}} \geq 0.$$

Thus, $\sqrt{x} \leq \sqrt{y}$ whenever $0 \leq x < y$. Since this also holds for $0 \leq x = y$, we see that $f(x) = \sqrt{x}$ is monotone increasing on $[0, \infty)$.

[15] To justify this inequality, let $t \geq 1$ and note that

$$t - \sqrt{t} = \sqrt{t}(\sqrt{t} - 1) \geq \sqrt{t} - 1 \geq 0.$$

be nested sequence of closed and bounded intervals. Then, the sequences (a_n) and (b_n) defined by the endpoints of these intervals satisfy the requirements of the previous example. Consequently, $\lim a_n, \lim b_n$ exist and satisfy $\lim a_n \leq \lim b_n$. Let $\xi \in \mathbb{R}$ be any real number such that $\lim a_n \leq \xi \leq \lim b_n$. Then, for each $n \in \mathbb{N}$,

$$a_n \leq \sup\{a_k : k \in \mathbb{N}\} = \lim a_n \leq \xi \leq \lim b_n = \inf\{b_k : k \in \mathbb{N}\} \leq b_n.$$

Hence, $\xi \in [a_n, b_n]$ for all $n \in \mathbb{N}$ so that $\xi \in \bigcap_{n=1}^{\infty}[a_n, b_n]$.

7.6 Subsequences

Let (x_n) be a sequence and let $(n_k)_{k\in\mathbb{N}}$ be a *strictly* increasing sequence of natural numbers.[16]. Then, we say that the sequence (x_{n_k}) is a subsequence of (x_n). Recall also that if (x_n) is convergent, then every subsequence of (x_n) is convergent with the same limit as (x_n). We should also recall that, in class, we defined

$$e := \lim\left(1 + \frac{1}{n}\right)^n.$$

Example 7.7. Prove that

$$\lim\left(1 + \frac{1}{n}\right)^n = e.$$

Proof. For each $n \in \mathbb{N}$, let us write

$$\left(1 + \frac{1}{n}\right)^{n-1} = \left(1 + \frac{1}{n}\right)^n \left(1 + \frac{1}{n}\right)^{-1}.$$

Then, since

$$\left(1 + \frac{1}{n}\right)^n \to e \quad \text{and} \quad 1 + \frac{1}{n} \to 1,$$

the result follows from the limit laws. □

Problem 4. *Let (x_n) be a sequence with $x_n \geq 0$, for all $n \in \mathbb{N}$. Suppose that $\lim (-1)^n x_n$ exists. Prove that (x_n) converges.*

Proof. Let x denote the limit of the sequence $(-1)^n x_n$. Then, all subsequences of $(-1)^n(x_n)$ converge to this point x. In particular,

$$x = \lim (-1)^{2n} x_{2n} = \lim x_{2n} \geq 0$$

[16] In other words, $n_k = n(k)$ is a strictly increasing function $\mathbb{N} \to \mathbb{N}$.

Since $x_n \geq 0$ for all $n \in \mathbb{N}$. Similarly,

$$x = \lim (-1)^{2n+1} x_{2n+1} = \lim (-x_{2n+1}) \leq 0.$$

Therefore, the only possibility is to have $x = 0$. That is, $\lim (-1)^n x_n = 0$. Thus, for any $\varepsilon > 0$ we can find $N \in \mathbb{N}$ such that

$$|x_n| = |(-1)^n x_n - 0| < \varepsilon$$

for all $n \geq N$. This means that $\lim x_n = 0$. □

Proposition 7.5. *Let (x_n) be a sequence of real numbers such that both the subsequences (x_{2k}) and (x_{2k-1}) converge with*

$$\lim x_{2k} = \lim x_{2k-1}.$$

Then, (x_n) is convergent with limit $\lim x_{2k} = \lim x_{2k-1}$.

Proof. Define $x := \lim x_{2k} = \lim x_{2k-1}$ and let $\varepsilon > 0$ be given. Since (x_{2k}) converges to x, there exists $N_1 \in \mathbb{N}$ with the property that

$$|x_{2k} - x| < \varepsilon, \quad \forall k \geq N_1.$$

Similarly, using that (x_{2k-1}) converges to x, we may find $N_2 \in \mathbb{N}$ such that

$$|x_{2k-1} - x| < \varepsilon, \quad \forall k \geq N_2.$$

Put $N := 2 \max(N_1, N_2)$ and let $n \geq N$.

If n is even, we may write it as $n = 2k \geq 2N_1$ for some $k \in \mathbb{N}$ with $k \geq N_1$. Therefore,

$$|x_n - x| = |x_{2k} - x| < \varepsilon\,.$$

Similarly, if n is odd, we may write $n = 2k - 1$ with $n \geq 2N_2$. Thus, $2k - 1 \geq 2N_2$ so that $k > N_2$ and

$$|x_n - x| = |x_{2k-1} - x| < \varepsilon\,.$$

To summarize, we have shown that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

As $\varepsilon > 0$ was arbitrary, we conclude that (x_n) is convergent with limit

$$x = \lim x_{2k} = \lim x_{2k-1}.$$

□

Proposition 7.6. *Let (x_n) be a sequence of real numbers such that the subsequences (x_{2k}), (x_{2k-1}) and (x_{3k}) each converge. Then, (x_n) is also convergent.*

Proof. By virtue of the Proposition 7.5, it suffices to show that (x_{2k}) and (x_{2k-1}) have the same limit. To this end, consider the subsequence (x_{6k}) of (x_n). Since $6k = 2(3k) = 3(2k)$, this is a subsequence of both (x_{2k}) and (x_{3k}). Now, because (x_{2k}) is convergent, any of its subsequences must converge to the same limit, i.e.

$$\lim x_{6k} = \lim x_{2k}.$$

However, (x_{6k}) is *also* a subsequence of (x_{3k}), and so $\lim x_{6k} = \lim x_{3k}$ as well. Combining these, we infer that

$$\lim x_{2k} = \lim x_{6k} = \lim x_{3k}.$$

Finally, let us consider the subsequence $(x_{3(2k-1)})$ of (x_{3k}). For each $k \in \mathbb{N}$, we have that

$$3(2k-1) = 6k - 3 = 6k - 2 - 1 = 2(3k-1) - 1$$

is an odd number. Hence, $(x_{3(2k-1)})$ is a subsequence of (x_{2k-1}) and must therefore share the same limit, i.e. $\lim x_{3(2k-1)} = \lim x_{2k-1}$. On the other hand $(x_{3(2k-1)})$ is also a subsequence of (x_{3k}), so we have

$$\lim x_{3(2k-1)} = \lim x_{3k} = \lim x_{2k}.$$

Putting all this together yields that

$$\lim x_{2k-1} = \lim x_{3(2k-1)} = \lim x_{2k}.$$

□

Example 7.8. Let (x_n) be an unbounded sequence. We claim that there exists a subsequence (x_{n_k}) such that

$$\lim \frac{1}{x_{n_k}} = 0.$$

Since (x_n) is unbounded, we know that for every $M > 0$, one can find some $n \in \mathbb{N}$ such that $|x_n| > M$. In fact, we can find *infinitely many* n such that $|x_n| > M$. Otherwise, we would have

$$|x_n| \leq M$$

for all but finitely many $n \in \mathbb{N}$. If this were the case, then the sequence (x_n) would be bounded.

Now, let us choose $n_1 \in \mathbb{N}$ such that $x_{n_1} > 1$. Next, given $k \geq 1$, we can guarantee the existence of $n_{k+1} \in \mathbb{N}$ such that

$$\left|x_{n_{k+1}}\right| > k + 1 \quad \text{and} \quad n_{k+1} > n_k.$$

In this way, we construct a subsequence (x_{n_k}) such that $\left|x_{n_k}\right| > k$, for every $k \in \mathbb{N}$. It follows that,

$$\left|\frac{1}{x_{n_k}} - 0\right| = \frac{1}{\left|x_{n_k}\right|} < \frac{1}{k}, \quad \forall k \geq 1.$$

Hence, $\lim \frac{1}{x_{n_k}} = 0$.

7.7 Cauchy and Contractive Sequences

We now come to the topics of Cauchy and contractive sequences. Both of these are useful, as they provide means of establishing the convergence of sequences (x_n) without needing to know a priori what the limit will be. In practice, it's often used in a similar way to the monotone convergence theorem.

Definition 7.1. Let (x_n) be a sequence. We say that (x_n) is Cauchy if, for each $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x_m| < \varepsilon$$

for all $n, m \geq N$.

For the purposes of review, let us reiterate that every Cauchy sequence is bounded. In fact, we showed in class that a sequence of real numbers if convergent if and only if it is Cauchy. This is a very important point (and you should know how it is proven).

Example 7.9. Let (x_n) be a sequence of real numbers and let $r \in (0, 1)$. In addition, suppose that $|x_{n+1} - x_n| < r^n$, for all $n \in \mathbb{N}$. We show that (x_n) is convergent. Based on the given information, it is not clear what $\lim x_n$ should be. Thus, our only real option is to check that (x_n) is Cauchy.[17]

To this end, let $\varepsilon > 0$ be given, and choose $N \in \mathbb{N}$ so large that

$$\frac{r^N}{1 - r} < \varepsilon.$$

This can be done because $r \in (0, 1)$ implies $\lim r^n = 0$. Now, given any natural numbers $m > n \geq N$,

$$\begin{aligned} |x_m - x_n| &= |x_m - x_{m-1} + x_{m-1} + \cdots - x_{n+1} + x_{n+1} - x_n| \\ &\leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \end{aligned}$$

[17] Since $\mathbb{R}$ is complete, a sequence in $\mathbb{R}$ converges if and only if it is Cauchy. The importance of this fact should not be underestimated.

$$
\begin{aligned}
&\leq \sum_{k=n}^{m-1} r^k \\
&= \sum_{k=0}^{m-1} r^k - \sum_{k=0}^{n-1} r^k \\
&= \frac{1-r^m}{1-r} - \frac{1-r^n}{1-r} \\
&= \frac{r^n - r^m}{1-r} \\
&< \frac{r^n}{1-r} \\
&\leq \frac{r^N}{1-r}.
\end{aligned}
$$

Hence, $|x_m - x_n| < \varepsilon$ whenever $m \geq n \geq N$. This shows that (x_n) is Cauchy.

The astute reader may have noticed that the argument utilized above is reminiscent of that used to show that a contractive sequence was convergent. Indeed, to prove that a contractive sequence is necessarily Cauchy (thence convergent), we first show that contractive sequences satisfy $|x_{n+1} - x_n| < Cr^n$ for some constants $C > 0$ and $r \in (0,1)$. Then, via this very argument, we conclude that contractive sequences are Cauchy. Having said this, this is a good time for us to review the definition of a contractive sequence.

Definition 7.2. Let (x_n) be a sequence. We say that (x_n) is contractive, provided there exists $C \in (0,1)$ such that

$$|x_{n+2} - x_{n+1}| \leq C\,|x_{n+1} - x_n|$$

for all $n \in \mathbb{N}$.

As stated above, being contractive is sufficient to establish convergence. However, it is not a necessary condition: there are plenty of convergent sequences that are not contractive.

Example 7.10. Let $x_1 := 2$ and define

$$x_{n+1} := 2 + \frac{1}{x_n}, \quad \forall n \geq \mathbb{N}.$$

Show that (x_n) is contractive and hence convergent. Determine $\lim x_n$.

Proof. Notice that, by induction, $x_n \geq 2$ for all $n \in \mathbb{N}$. Indeed, the base case is given and, if $x_n \geq 2 > 0$,

$$x_{n+1} = 2 + \underbrace{\frac{1}{x_n}}_{>0} > 2.$$

Let us next show that the sequence is contractive. Given $n \in \mathbb{N}$, we have

$$\begin{aligned} |x_{n+2} - x_{n+1}| &= \left| 2 + \frac{1}{x_{n+1}} - \left(2 + \frac{1}{x_{n-1}} \right) \right| \\ &= \left| \frac{1}{x_{n+1}} - \frac{1}{x_n} \right| \\ &= \frac{|x_{n+1} - x_n|}{x_n x_{n+1}} \\ &\leq \frac{|x_{n+1} - x_n|}{4}. \end{aligned}$$

It follows that (x_n) is contractive, and hence Cauchy by class results. Therefore, (x_n) is convergent with some limit $x \in \mathbb{R}$. In fact, since $x_n \geq 2$ for all $n \in \mathbb{N}$, we must have $x \geq 2$. Especially, $x \neq 0$ and so the quotient limit law (see Proposition 5.3), we have that

$$\lim \frac{1}{x_n} = \frac{1}{x}.$$

Since (x_{n+1}) is itself a subsequence of (x_n), it follows that

$$x = \lim x_{n+1} = \lim \left(2 + \frac{1}{x_n} \right) = 2 + \lim \left(\frac{1}{x_n} \right) = 2 + \frac{1}{x}.$$

Thus, the limit x satisfies the relation

$$x = 2 + \frac{1}{x}$$

whence $x^2 - 2x - 1 = 0$. By the quadratic formula, this implies that $x = 1 \pm \sqrt{2}$. Since $x \geq 2 > 0$, the only possibility is therefore $x = 1 + \sqrt{2}$. □

8 Eighth Tutorial

We first recall some notions about open and closed subsets of $\mathbb{R}$. A set $U \subseteq \mathbb{R}$ is called **open** if, for every $x \in O$, there exists $\varepsilon > 0$ such that

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) \subseteq U.$$

In symbolic terms, U is said to be open if and only if:

$$(\forall x \in U)\ (\exists \varepsilon > 0 \text{ such that } V_\varepsilon(x) \subseteq X)\,.$$

In other words, a set U is open if for every point $x \in U$ there exists an open interval $(x - \varepsilon, x + \varepsilon)$ contained in U.[18] Since an open interval $(x - \varepsilon, x + \varepsilon)$ contains all points "near x", the definition for an open set can be interpreted as follows:

> *A set $U \subseteq X$ is called open if, for every $x \in U$, the set U also contains all points near x.*[19]

A set $F \subseteq \mathbb{R}$ is said to be **closed** if $F^c = \mathbb{R} \setminus F$ is open in $\mathbb{R}$. Let us now summarize some properties of open sets that you will have proven in the lectures.

- If $\{U_\alpha\}_{\alpha \in I}$ is an indexed family of open sets, then $\bigcup_{\alpha \in I} U_\alpha$ is also open.
- If U_1 and U_2 are open subsets of $\mathbb{R}$, then so is $U_1 \cap U_2$. By induction, finite intersections of open sets are thus open.
- Both $\mathbb{R}$ and $\varnothing$ are open in $\mathbb{R}$.

The set $\mathcal{T}$ of all open subsets of $\mathbb{R}$ is therefore closed under unions, finite intersections, and contains both $\varnothing$ and $\mathbb{R}$. This makes $\mathcal{T}$ into what we call a *topology*. Since $\mathbb{R}$ has a topology (i.e. open subsets), it is called a *topological space*.

Example 8.1. Let us show that infinite intersections of open sets need not be open. For every $n \in \mathbb{N}$, let U_n be the open interval:

$$U_n := \left(-\frac{1}{n}, \frac{1}{n}\right).$$

Now consider the infinite family $\{U_n\}_{n=1}^{\infty}$, what can be said about $\bigcap_{n=1}^{\infty} U_n$? Well, after a moment's consideration, it is not difficult to see that

$$\bigcap_{n=1}^{\infty} U_n = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}. \tag{8.1}$$

To prove this, we first note the obvious inclusion $\{0\} \subseteq U_n$, for each $n \in \mathbb{N}$. Since this holds for all n, we have $\{0\} \subseteq \bigcap_{n=1}^{\infty} U_n$. Conversely, let $x \in \bigcap_{n=1}^{\infty} U_n$ be given; we will show that $x \neq 0$ is impossible. If $x \neq 0$, then either $x < 0$ or $x > 0$.

1. Case $x > 0$. By the Archimedean property, there exists $n \in \mathbb{N}$ so large that $\frac{1}{n} < x$. Thus, $x \notin U_n$.

[18] Note that the empty set $\varnothing$ is vacuously open by this definition as well. If $\varnothing$ were *not* open, then there must exist $x \in \varnothing$ such that $V_\varepsilon(x) \not\subseteq \varnothing$ for all $\varepsilon > 0$. Since this would imply $x \in \varnothing$, we have a contradiction.

[19] Please note that this is not rigorous, and is only intended to provide intuition.

2. Case $x < 0$. Then, $-x > 0$. As above, we can find $n \in \mathbb{N}$ such that $\frac{1}{n} > -x$. This implies that $x < -\frac{1}{n}$ whence $x \notin U_n$.

In either case, we get $x \notin U_n$ for some $n \in \mathbb{N}$. Since this contradicts the assumption that $x \in \bigcap_{n=1}^{\infty} U_n$, we conclude that $x \neq 0$ is impossible. Thus, $x \in \bigcap_{n=1}^{\infty} U_n$ implies $x = 0$, i.e. $\bigcap_{n=1}^{\infty} U_n \subseteq \{0\}$. To summarize, we have proven (12.1).

But $\{0\}$ is not open! If it were, then for $0 \in \{0\}$ we could find $\varepsilon > 0$ such that $(0 - \varepsilon, 0 + \varepsilon) \subseteq \{0\}$; which is clearly absurd. Hence $\bigcap_{n=1}^{\infty} U_n$ is an infinite intersection of open sets that is not open.

Example 8.2. You have proven that any intersection of closed sets is necessarily closed. You have also seen that finite unions of closed sets are closed. What about infinite unions of closed sets? First, note that any singleton $\{x\} \subset \mathbb{R}$ is closed. Indeed,

$$\mathbb{R} \setminus \{x\} = (-\infty, x) \cup (x, \infty)$$

is a union of open sets, and hence open. Consider the union $\bigcup_{x \in (0,1)} \{x\} = (0, 1)$ of infinitely many closed sets. We claim that $(0, 1)$ is not closed in $\mathbb{R}$. If it were, then $\mathbb{R} \setminus (0, 1) = (-\infty, 0] \cup [1, \infty)$ would be open in $\mathbb{R}$. But, for any $\varepsilon > 0$, the set

$$V_\varepsilon(0) = (-\varepsilon, \varepsilon) \not\subseteq (-\infty, 0] \cup [1, \infty).$$

This means that $\mathbb{R} \setminus (0, 1)$ is not open whence $(0, 1)$ is not closed.

By way of another example, we show that $\mathbb{N}$ is closed in $\mathbb{R}$. Indeed, $\mathbb{R} \setminus \mathbb{N}$ can be written as the union of open sets:

$$\mathbb{R} \setminus \mathbb{N} = (-\infty, 1) \cup \bigcup_{n=1}^{\infty} (n, n + 1).$$

Thus, $\mathbb{R} \setminus \mathbb{N}$ is open. A more interesting example is the following.

Example 8.3. We show that $\mathbb{Q}$ is neither open nor closed in $\mathbb{R}$. Suppose for a contradiction that $\mathbb{Q}$ is open in $\mathbb{R}$. Then, for any $q \in \mathbb{Q}$ there exists $\varepsilon > 0$ such that $V_\varepsilon(q) = (q - \varepsilon, q + \varepsilon) \subseteq \mathbb{Q}$. Recall from the previous tutorial that the irrationals are dense in $\mathbb{R}$. Thus, every open interval in $\mathbb{R}$ contains an irrational number. In particular, so does $V_\varepsilon(q)$. It follows that $V_\varepsilon(q)$ cannot be contained in $\mathbb{Q}$. We conclude that $\mathbb{Q}$ is *not* open.

To see that $\mathbb{Q}$ is not closed, we argue by contradiction. If $\mathbb{Q}$ were closed, then the set of irrationals $\mathbb{R} \setminus \mathbb{Q}$ would be open in $\mathbb{R}$. Then, for any irrational number ξ, we could find $\varepsilon > 0$ such that $V_\varepsilon(\xi) = (\xi - \varepsilon, \xi + \varepsilon) \subseteq \mathbb{R} \setminus \mathbb{Q}$. Since any open interval contains a rational number ($\mathbb{Q}$ is dense in $\mathbb{R}$), this is impossible. Hence, $\mathbb{Q}$ cannot be closed.

Remark 8.1. Since $\mathbb{Q}$ is countable, it is the countable union of singletons

$$\mathbb{Q} = \bigcup_{x \in \mathbb{Q}} \{x\}\,.$$

Since $\{x\}$ is a closed set for any $x \in \mathbb{R}$, this means that $\mathbb{Q}$ is a countable union of closed sets. Therefore, $\mathbb{Q}$ is an example of a countably infinite union of closed sets that is itself not closed.

8.1 Characterization of Open Sets

Consider a collection $\{I_\alpha\}_{\alpha \in I}$ of open intervals. Since each I_α is open, we know that $\bigcup_{\alpha \in I} I_\alpha$ is open in $\mathbb{R}$. Now, as it turns out, every open set is the union of open intervals. This itself is not too hard to prove. Indeed, given $U \subseteq \mathbb{R}$ open, for each $x \in U$, there exists $\varepsilon_x > 0$ such that $V_{\varepsilon_x}(x) = (x - \varepsilon, x + \varepsilon) \subseteq U$. Then, $\bigcup_{x \in U} V_{\varepsilon_x}(x) = U$ and we have expressed U as the union of open intervals. What is remarkable, is that we can write any open U as the *countable union* of disjoint open intervals. We prove this below.

Theorem 8.1 (Characterization of Open Sets)**.** *A subset $G \subseteq \mathbb{R}$ is open if and only if G is the countable union of disjoint open intervals.*

Proof. Since arbitrary (in particular countable) unions of open sets remain open, it is obvious that the countable union of open sets is open. Thus, we need only establish the " $\implies$ " direction. To this end, let $G \subseteq \mathbb{R}$ be open. If G is empty, it may be expressed vacuously as the empty union of open intervals. Thus, we may assume $G \neq \varnothing$. For each $x \in G$, using that G is open, there exists $\varepsilon > 0$ such that $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \subseteq G$. In particular,

$$(x - \varepsilon, x], [x, x + \varepsilon) \subseteq G$$

and so the sets

$$A_x := \{a \in \mathbb{R} : (a, x] \subseteq G\}\,, \quad B_x := \{b \in \mathbb{R} : [x, b) \subseteq G\}$$

are non-empty. If A_x is bounded from below, define $a_x := \inf A_x$ and otherwise set $a_x := -\infty$. Analogously, if B_x is bounded from above, put $b_x := \sup B_x$ and $b_x := \infty$ otherwise; finally, define the open interval $I_x := (a_x, b_x) \subseteq \mathbb{R}$. Notice that $a_x \notin G$. Indeed, if $a_x = -\infty$, this is trivial. If A_x is bounded and $a_x = \inf A_x \in G$, there would exist $\delta > 0$ such that $V_\delta(a_x) = (a_x - \delta, a_x + \delta) \subseteq G$. Now, since $a_x + \delta$ is not a lower bound of A_x, there exists $a \in A_x$ such that $a_x \leq a < a_x + \delta$; from $a \in A_x$ we get $(a, x] \subseteq G$ so that $(a_x - \delta, x] \subseteq (a_x - \delta, a_x + \delta) \cup (a, x] \subseteq G$ which contradicts the definition of a_x. One can show, in a similar vein, that $b_x \notin G$.

We claim first that $I_x \ni x$. Indeed, notice that $x - \varepsilon \in A_x$ and so $a_x \leq x - \varepsilon < x$. Similarly, $x + \varepsilon \in B_x$ so $b_x \geq x + \varepsilon > x$. Thus, $x \in I_x$ whence $G \subseteq \bigcup_{x \in G} I_x$. Next, we

show that $I_x \subseteq G$, for each $x \in G$. Indeed, given $y \in I_x = (a_x, b_x)$, we have $y > a_x$. Thus, there exists $a \in A_x$ with $a_x \leq a < y$. Similarly, $y < b_x$ implies there exists $b \in B_x$ such that $y < b \leq b_x$. Since $a \in A_x$, we have that $(a, x] \subseteq G$ and, using $b \in B_x$, there holds $[x, b) \subseteq G$. Hence, $y \in (a, b) = (a, x] \cup [x, b) \subseteq G$. As $y \in I_x$ was arbitrary, it follows that $I_x \subseteq G$ whence $\bigcup_{x \in G} I_x \subseteq G$. Combining this with our previous work, we infer that

$$\bigcup_{x \in G} I_x = G. \tag{8.2}$$

Next, given $x, y \in G$, we claim that either $I_x \cap I_y = \varnothing$ or $I_x = I_y$. To this end, suppose $I_x = (a_x, b_x)$ and $I_y = (a_y, b_y)$ are two of the open intervals in our construction, with $x, y \in G$, such that $I_x \cap I_y \neq \varnothing$. Then, there exists $z \in I_x \cap I_y$ and

$$a_x < z < b_y, \tag{8.3}$$

$$a_y < z < b_x. \tag{8.4}$$

If $a_x \neq a_y$, then either $a_x < a_y$ or $a_y < a_x$. In the former case, we find that $a_x < a_y < b_x$ through (8.4) whence $a_y \in I_x \subseteq G$ which is a contradiction. If $a_y < a_x$, then $a_y < a_x < b_y$ by virtue of (8.3) and so $a_x \in I_y \subseteq G$ which is again a contradiction. One can show that $b_x = b_y$ in a similar way whence $I_x = I_y$, as was asserted.

It remains to show that

$$\{I_x : x \in G\}$$

is countable. To see this, let $\{q_n\}_{n \in \mathbb{N}}$ be an enumeration of $\mathbb{Q}$. Given $x \in G$, let $n(x) \in \mathbb{N}$ denote the minimal index such that $q_{n(x)} \in I_x$ (by density of $\mathbb{Q}$ and the well ordering property, such an $n(x)$ exists). Consider then the injective mapping

$$f : \{I_x : x \in G\} \to \mathbb{N}, \quad I_x \mapsto n(x).$$

Notice that f is an injection because, assuming $f(I_x) = f(I_y)$, we have $n(x) = n(y)$ so that $q_{n(x)} = q_{n(y)}$. Since $q_{n(x)} \in I_x$ and $q_{n(y)} \in I_y$, it follows that $I_x \cap I_y \neq \varnothing$ and so $I_x = I_y$. Since $\{I_x : x \in G\}$ is in bijection with $\mathrm{Im}(f) \subseteq \mathbb{N}$, this completes the proof. □

8.2 The Cantor Set

We devote this tutorial to the construction of an infamous topological structure known as the *Cantor set*. It is this set which helped give rise to the subject known today as point-set topology. The Cantor set is very special in the sense that it defies intuition. For one, the Cantor set is a *closed uncountable set* that does not contain **any** open interval. Moreover, the Cantor set is both totally disconnected and nowhere dense (these terms will be defined rigorously in a few moments). Despite this, the Cantor set does not contain any *isolated points* (again, we will give precise meaning to this term shortly). In

short, the Cantor set serves as a prime example of how weak our ingrained topological intuition can be. It is very useful to keep the Cantor set in mind for counter examples to topological claims about $\mathbb{R}$.

8.2.1 Constructing the Cantor Set

We begin with the closed and bounded interval $[0, 1]$, which has length equal to 1. Let us put $A_0 := [0, 1]$. We can now subdivide A_0 into three **disjoint bounded** intervals, each of length $\frac{1}{3}$, according to the following rule:

$$A_0 := \left[0, \frac{1}{3}\right] \cup \left(\frac{1}{3}, \frac{2}{3}\right) \cup \left[\frac{2}{3}, 1\right].$$

Note that the three intervals given above are pairwise disjoint (i.e. none of these intervals have common points). The idea is now to *remove* what we will call the **middle open third interval**. In the case of A_0, this "middle open third interval" will simply be the open interval

$$\left(\frac{1}{3}, \frac{2}{3}\right).$$

Deleting this interval from A_0, we obtain a *new* subset of $[0, 1]$ given by

$$A_1 := \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right].$$

Note that A_1 is actually a subset of A_0. Moreover, since finite unions of closed sets are always closed, A_1 will be both closed and bounded. Now, A_1 consists of two intervals, each having length $\frac{1}{3}$. We will now remove the "middle open third interval" from each of these two sub-intervals. To do this, we decompose $[0, \frac{1}{3}]$ into thirds as follows:

$$\left[0, \frac{1}{3}\right] = \left[0, \frac{1}{9}\right] \cup \left(\frac{1}{9}, \frac{2}{9}\right) \cup \left[\frac{2}{9}, \frac{1}{3}\right].$$

Thus, when removing the "open middle third" interval from $\left[0, \frac{1}{3}\right]$, we would be left with

$$\left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right]. \tag{8.5}$$

Similarly, we break the second interval in A_1 into thirds:

$$\left[\frac{2}{3}, 1\right] = \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left(\frac{7}{9}, \frac{8}{9}\right) \cup \left[\frac{8}{9}, 1\right].$$

The "open middle third" interval corresponding to the interval above is therefore $\left(\frac{7}{9}, \frac{8}{9}\right)$. Deleting this from $\left[\frac{2}{3}, 1\right]$ leaves us with the following:

$$\left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]. \tag{8.6}$$

Thus, (12.1)-(8.6) consist of the sets leftover after deleting the "open middle third" intervals from the two subintervals of A_1, respectively. We then define

$$A_2 := \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right].$$

Again, A_2 is the disjoint union of finitely many closed intervals in $[0, 1]$. Hence, A_2 is closed and bounded. Note that every subinterval in the definition of A_2 has length equal to $\frac{1}{3^2}$. Furthermore, observe that there are 2^2 subintervals comprising a_2.

We now apply the last steps recursively. Given A_n, for some $n \in \mathbb{N}$, we remove the "open middle third" interval from every (disjoint) closed subinterval given in the definition of A_n. After removing an "open middle third" interval from a closed interval $[a, b]$, we are always left with two disjoint closed intervals. Therefore, upon removing the "open middle third" subintervals from each of the subintervals of A_n, we will be left with a finite family of disjoint closed intervals, each contained in A_n. We then define A_{n+1} to be the union of these closed intervals. The following properties will hold for this A_{n+1}.

- A_{n+1} is closed. Indeed, A_{n+1} is by definition the finite union of disjoint closed intervals.
- $A_{n+1} \subseteq A_n$. Certainly, every interval in the definition of A_{n+1} was obtained by removing a "middle open third" interval from some subinterval of A_n. In particular, every interval that makes up A_{n+1} is a subset of A_n. Since A_{n+1} is merely the union of these intervals, we get $A_{n+1} \subseteq A_n$.
- $0 \in A_n$ for every $n \in \mathbb{N}$. Indeed, this follows by observing that the endpoints of intervals are never removed when deleting the "open middle third" interval. Similarly, $1 \in A_n$.

Therefore, for every $n \geq 0$, we obtain a sequence of closed and bounded sets, denoted by $\{A_n\}_{n \in \mathbb{N}_0}$. Note that, by construction, $0 \in \bigcap_{n \in \mathbb{N}_0} A_n$. In particular, this intersection is non-empty.

Definition 8.1 (Cantor Set)**.** The Cantor set $\mathfrak{C}$ is the non-empty set

$$\mathfrak{C} := \bigcap_{n \in \mathbb{N}_0} A_n. \tag{8.7}$$

Note that $\mathfrak{C} \subseteq A_0 = [0,1]$. Moreover, since A_n is closed for each $n \in \mathbb{N}$, it is immediate from the intersection property of closed sets that $\mathfrak{C}$ is closed.

Remark 8.2. Explicitly, the Cantor set $\mathfrak{C}$ is given by

$$\mathfrak{C} = [0,1] \setminus \bigcup_{n=0}^{\infty} \bigcup_{k=0}^{3^n-1} \left(\frac{3k+1}{3^{n+1}}, \frac{3k+2}{3^{n+1}} \right).$$

Since the Cantor set is very difficult to picture, I have included a depiction from Wikipedia below.



Figure 1: Six iterations of the Cantor set

8.3 The Cantor Set is Uncountable

Given the title of this subsection, we can all guess what we will prove next.

Proposition 8.2. *The Cantor set $\mathfrak{C}$ defined in (8.7) is uncountable.*

Proof. Suppose, by way of contradiction, that $\mathfrak{C}$ is countable and enumerate its elements:

$$\mathfrak{C} := \{x_1, x_2, \ldots, x_n, \ldots\}.$$

By construction, $\mathfrak{C} = \bigcap_{n\in\mathbb{N}_0} A_n \subseteq A_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$. Since these two intervals are disjoint, the point x_1 only belongs to one of these two intervals. Let I_1 denote the interval **not** containing this point x_1. Dividing this subinterval I_1 by removing its "open middle third" interval, we obtain two disjoint closed subintervals of A_2 (each of length $\frac{1}{3^2}$). Again, since these two intervals are disjoint and $x_2 \in A_2$, we see that x_2 cannot belong to both of these subintervals. Let I_2 be any one of these two subintervals **not** containing x_2. Since I_2 was obtained by deleting points from I_1, we clearly have $I_2 \subseteq I_1 \subseteq [0,1]$.

We proceed recursively, obtaining a nested sequence

$$I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \cdots$$

of closed and bounded intervals, with $x_n \notin I_n$ for every $n \in \mathbb{N}$. By the *nested interval property* of $\mathbb{R}$, we know that $\bigcap_{n\in\mathbb{N}} I_n$ is non-empty. Let now $x \in \bigcap_{n\in\mathbb{N}} I_n$ so that $x \in I_n$ for all $n \in \mathbb{N}$. By construction, $I_n \subseteq A_n$ whence $x \in \bigcap_{n\in\mathbb{N}_0} A_n = \mathfrak{C}$. Thus, x belongs to the Cantor set $\mathfrak{C}$ and must be present in our enumeration. On the other hand, $x_n \notin I_n$ for every index n. Since $x \in I_n$, this forces $x \neq x_n$ for all $n \in \mathbb{N}$. We have therefore found an element of $\mathfrak{C}$ not present in our enumeration. Hence $\mathfrak{C}$ cannot be countable. □

8.4 The Topology of $\mathfrak{C}$

We now discuss the surprising topological aspects of the Cantor set. We have already seen that $\mathfrak{C}$ is closed and bounded (making it an example of what is known as a *compact* set). Now, we claim that $\mathfrak{C}$ does not contain any open interval (and hence does not contain any non-trivial interval).

Proposition 8.3. *The Cantor set $\mathfrak{C}$ does not contain any non-trivial interval.*

Proof. Here we have to look back at our construction of the Cantor set. At the nth stage of the construction, A_n is the *disjoint* union of 2^n closed intervals in $[0,1]$, each having length 3^{-n}. Thus, any interval contained in A_n must have length no larger than 3^{-n}. Hence, if I is a non-trivial interval contained in $\mathfrak{C}$, it must satisfy

$$|I| \leq \inf_{n\geq 1} 3^{-n}.$$

Here, $|I|$ simply denotes the length of I. By Bernoulli's inequality,

$$3^n > n \quad \text{for all } n \geq 1.$$

Thus,

$$0 \leq \inf_{n\geq 1} 3^{-n} \leq \inf_{n\geq 1} \frac{1}{n} = 0.$$

We then see that $|I| = 0$, which is absurd for any non-trivial interval. We conclude that $\mathfrak{C}$ does not contain any non-trivial interval. □

Corollary 8.4. *The Cantor set $\mathfrak{C}$ has empty interior. This makes $\mathfrak{C}$ and example of a nowhere dense set.*[20]

Proof. Let $\mathring{\mathfrak{C}}$ denote the interior of $\mathfrak{C}$. Then, $\mathring{\mathfrak{C}}$ is a subset of $\mathfrak{C}$ by definition. If $\mathring{\mathfrak{C}}$ were non-empty, then for any point $x \in \mathring{\mathfrak{C}}$ we could find $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq \mathring{\mathfrak{C}} \subseteq \mathfrak{C}$. Of course, this means that $\mathfrak{C}$ would contain an open interval. This contradiction shows that $\mathfrak{C}$ has empty interior. □

Remark 8.3. Note that $0 = \inf \mathfrak{C}$ and $1 = \sup \mathfrak{C}$.[21] Since $0, 1 \in \mathfrak{C}$, we see that $\mathfrak{C}$ contains both of its "endpoints". But this does not imply that $\partial\mathfrak{C} = \{0,1\}$! Indeed, as we shall see $\partial\mathfrak{C}$, contains much more than the two endpoints 0 and 1 of $\mathfrak{C}$.

We claim that $\partial\mathfrak{C} = \mathfrak{C}$. Since $\mathfrak{C}$ is closed, the inclusion $\partial\mathfrak{C} \subseteq \mathfrak{C}$ is immediate. Conversely, let $x \in \mathfrak{C}$ be given and fix $\varepsilon > 0$. Clearly, $V_\varepsilon(x) \cap \mathfrak{C} \neq \varnothing$ since $x \in \mathfrak{C}$. However, because $\mathfrak{C}$ does not contain any open interval, we have $V_\varepsilon(x) \not\subseteq \mathfrak{C}$. This implies that

$$V_\varepsilon(x) \cap \mathfrak{C} \neq \varnothing \quad \text{and} \quad V_\varepsilon(x) \cap \mathfrak{C}^c \neq \varnothing.$$

[20] A set $A \subseteq \mathbb{R}$ is said to be nowhere dense if the interior of $\overline{A}$ is empty.

[21] This is because $\mathfrak{C} \subseteq [0,1]$ and $0, 1 \in \mathfrak{C}$.

Since $\varepsilon > 0$ was arbitrary, it follows that $x \in \partial\mathfrak{C}$. Thus, $\mathfrak{C} \subseteq \partial\mathfrak{C}$ and $\partial\mathfrak{C} = \mathfrak{C}$. In particular, $\partial\mathfrak{C}$ is uncountable. Therefore, the Cantor set is an example of a closed set with boundary points that are **not** endpoints. In fact, every point in $\mathfrak{C}$ is a boundary point of $\mathfrak{C}$.

Definition 8.2. A subset X of $\mathbb{R}$ is said to be *totally disconnected* if, for any $x, y \in X$ with $x < y$, there exists $z \in \mathbb{R} \setminus X$ such that $x < z < y$.

One example of a totally disconnected set is $\mathbb{Q}$. Indeed, by the density of the irrationals, for any $x < y$ in $\mathbb{Q}$ one can always find an irrational number z with $z \in (x, y)$.

Proposition 8.5. *The Cantor set $\mathfrak{C}$ is totally disconnected.*

Proof. Let $x, y \in \mathfrak{C}$ be given and assume that $x < y$. We must show that there exists some $z \notin \mathfrak{C}$ with $x < z < y$. Let $\varepsilon := y - x > 0$ be the distance between x and y. Since $x, y \in \mathfrak{C}$, these two points must belong to every A_n. On the other hand, every subinterval making up A_n has length equal to $\frac{1}{3^n}$. Since

$$\inf_{n \geq 1} \frac{1}{3^n} = 0,$$

we can find N so large that $\frac{1}{3^N} < \varepsilon$. Since $x, y \in A_N$ and $|x - y| = \varepsilon > \frac{1}{3^N}$, x and y cannot belong to the same subinterval of A_N. Noticing that the subintervals of A_N are separated by open intervals, we can choose $z \notin A_N$ with $x < z < y$. Then, this z does not belong to $\mathfrak{C}$ and satisfies the condition of the proposition. □

Definition 8.3. Let X be a non-empty subset of $\mathbb{R}$. We say that a point $x \in X$ is an *isolated point* of X if there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \cap (X \setminus \{x\}) = \varnothing$.

In more familiar language, $x \in X$ is called an isolated point of X if there exists an ε-neighbourhood of x that does not contain points in X other than x.

Proposition 8.6. *The Cantor set $\mathfrak{C}$ has no isolated points.*

Proof. Fix $x \in \mathfrak{C}$ and let $\varepsilon > 0$ be given; we must show that $V_\varepsilon(x)$ intersects $\mathfrak{C}$ at a point other than x. Since $x \in \mathfrak{C}$, we must have $x \in A_n$ for all $n \geq 1$. Arguing as before, we can find $N \in \mathbb{N}$ so large that $3^{-N} < \varepsilon$. Consider this A_N, and recall that every subinterval of A_N has length equal to $3^{-N} < \varepsilon$. Let $I = [a, b]$ be the closed subinterval of A_N containing the point x. For any $y \in I$:

$$|x - y| \leq |b - a| .^{22}$$

Since I has length $3^{-N} < \varepsilon$, it follows that

$$|x - y| \leq 3^{-N} < \varepsilon$$

[22] First note that $x \leq b$ and $y \geq a$. Therefore, $x - y \leq b - a$. Similarly, one can show that $y - x \leq b - a$ whence $-(b - a) \leq x - y$.

whence $y \in V_\varepsilon(x)$. Or, rather, $I = [a,b] \subseteq V_\varepsilon(x)$. Now, I is a subinterval from the construction of A_N whence $I \subseteq A_n$ for all $n \leq N$. On the other hand, when removing the "open middle third" in the construction, the end points of the intervals are never removed. Therefore the end points a, b of I will belong to A_n for all $n \in \mathbb{N}$. This simply means that

$$a, b \in \mathfrak{C} = \bigcap_{n \in \mathbb{N}_0} A_n.$$

Since $a, b \in V_\varepsilon(x)$, it follows that $V_\varepsilon(x)$ intersects $\mathfrak{C}$ at a point other than x. □

9 Ninth Tutorial

This tutorial will be a bit of an anomaly amongst the others as we shall begin with a review of the midterm exam questions. Although this may not take too much time, we will have the time to cover fewer problems as a result. Furthermore, solutions to the midterm problems are not included here; please consult MyCourses for the professor's solutions. Below, we include only solutions to relevant practice exercises (some of which are important results in their own right).

9.1 Open and Closed Sets & Their Relationship with Sequences

We now explore two results which relate open/closed sets and the limits of convergent sequences. The first result, tells that the convergence of a sequence is completely determined by the open sets. The second asserts that a set is closed if and only if it contains the limits of all its *convergent* sequences.

Theorem 9.1. *Let (x_n) be a sequence and fix $x \in \mathbb{R}$. The following statements are equivalent:*

1. $\lim x_n = x$;
2. *for every open set U containing x, there exists $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$.*

Proof. Suppose that $\lim x_n = x$ and let U be an open set containing x; we must show that $x_n \in U$ for all sufficiently large n. Since U is open, there exists $\varepsilon > 0$ such that $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \subseteq U$. On the other hand, since $x_n \to x$, we can find $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

This is precisely the statement that $x_n \in V_\varepsilon(x) \subseteq U$, for all $n \geq N$. Thus, we have proven the first implication.

Conversely, assuming 2., we must show that $\lim x_n = x$. Given $\varepsilon > 0$, we note that $V_\varepsilon(x)$ is an open set containing x. Therefore, we can find $N \in \mathbb{N}$ such that $x_n \in V_\varepsilon(x)$ for all $n \geq N$. But,

$$x_n \in V_\varepsilon(x) \iff |x_n - x| < \varepsilon.$$

Thus, for all $n \geq N$, we must have $|x_n - x| < \varepsilon$. By definition, this means that $x_n \to x$. □

The above gives us an alternative (but equivalent) definition for the convergence of sequences. It is this second definition that we work with, when dealing with more general spaces.

Theorem 9.2. *Let $F \subseteq \mathbb{R}$. The following statements are equivalent:*

1. *F is closed;*
2. *if (x_n) is a sequence of points in F converging to some $x \in \mathbb{R}$, then $x \in F$.*

Proof. First suppose that F is closed, and let (x_n) be a sequence of points in F converging to some $x \in \mathbb{R}$. To establish 2., we must show that $x \in F$. Proceeding by contradiction, let us assume that $x \notin F$, i.e. $x \in F^c$. Since F is closed, F^c is open. In particular, it is an open set containing x. By the previous theorem, there is some $N \in \mathbb{N}$ such that $x_n \in F^c$ for all $n \geq N$. Obviously, this contradicts the assumption that (x_n) is a sequence of points *in* F. Therefore, we must have $x \in F$.

Conversely, suppose that criterion 2. holds true. By way of contradiction, suppose that F is not closed. This means that F^c is **not** open.[23] Thus, there exists a point $x \in F^c$ such that, for each $\varepsilon > 0$, one has $V_\varepsilon(x) \not\subseteq F^c$. Equivalently, $V_\varepsilon(x) \cap F \neq \varnothing$ for each $\varepsilon > 0$. So, for $n \geq 1$, we can therefore choose a point

$$x_n \in F \cap V_{1/n}(x) = F \cap \left(x - \frac{1}{n}, x + \frac{1}{n}\right).$$

This gives us a sequence (x_n) living in F. We now claim that $\lim x_n = x$. Indeed, by the above:

$$|x_n - x| < \frac{1}{n}, \quad \forall n \in \mathbb{N}.$$

This means that (x_n) is a sequence of points in F converging to x. However, by the hypothesis in 2., this means that $x \in F$. Since $x \notin F$, we have reached our contradiction. Hence, F must be closed. □

[23] Although it may be tempting, F not being closed does **not** mean that F is open! Not closed does not imply open!

9.2 Accumulation Points

Let (x_n) be a sequence; recall that a point $a \in \mathbb{R}$ is said to be an *accumulation point* of the sequence (x_n) provided there exists a subsequence (x_{n_k}) of (x_n) converging to a, i.e. $\lim x_{n_k} = a$. If (x_n) is bounded, then it follows from Bolzano-Weierstrass that (x_n) possesses an accumulation point. Furthermore, any convergent sequence can only have a single accumulation point, since all subsequences must also converge to the limit of (x_n). Finally, recall that, in the event that (x_n) is bounded, the set of accumulation points of (x_n) is compact (closed and bounded) with minimal and maximal elements, which we respectively write $\liminf x_n$ and $\limsup x_n$.

Proposition 9.3. *Let (x_n) be a bounded sequence in $\mathbb{R}$. Then, $\lim x_n$ exists if and only if the sequence (x_n) has exactly one accumulation point. Furthermore, in this case, $\lim x_n$ is this accumulation point.*

Proof. One direction is easy: if (x_n) converges to $x \in \mathbb{R}$, then so must any subsequence of (x_n). In particular, x can be the only accumulation point of (x_n).

Conversely, suppose that the sequence (x_n) has only one accumulation point, say, x. Then, every convergent subsequence (x_{n_k}) of (x_n) must converge to this point x. By way of contradiction, suppose that (x_n) does *not* converge to this point x. Then, there is some $\varepsilon_0 > 0$ such that, for every $N \in \mathbb{N}$, there exists $n \geq N$ with the property that $|x_n - x| \geq \varepsilon_0$.[24] In particular, we can find a subsequence (x_{n_k}) of (x_n) such that $\left|x_{n_k} - x\right| \geq \varepsilon_0$, for all $k \in \mathbb{N}$. Note that this subsequence is also a bounded sequence. By the Bolzano-Weierstrass theorem, we can find a subsequence $(x_{n_{k_j}})$ of our subsequence (x_{n_k}) that converges. Since $(x_{n_{k_j}})$ is then a convergent subsequence of (x_n), it must converge to x. But this is impossible for, as a subsequence of (x_{n_k}),

$$\left|x_{n_{k_j}} - x\right| \geq \varepsilon_0, \quad \text{for all } j \in \mathbb{N}.$$

This proves our claim. □

Example 9.1. Consider the sequence (x_n) of real numbers given by

$$x_n := \frac{(-1)^n n}{2n+1}.$$

Determine all accumulation points of (x_n) and calculate $\limsup x_n$ and $\liminf x_n$.

Solution. Consider the subsequences indexed by even and odd numbers, i.e. (x_{2n}) and

[24] To see this, just negate the definition of convergence to x.

(x_{2n-1}). Then, it is easy to see that

$$\lim x_{2n} = \lim\left(\frac{(-1)^{2n}(2n)}{2(2n)+1}\right) = \lim\frac{2n}{4n+1} = \frac{1}{2},$$
$$\lim x_{2n-1} = \lim\left(\frac{(-1)^{2n-1}(2n-1)}{2(2n-1)+1}\right) = -\lim\frac{2n-1}{4n-1} = -\frac{1}{2}.$$

Hence, $\pm\frac{1}{2}$ are accumulation points of (x_n). We now assert these are the *only* accumulation points. Indeed, let a be an accumulation point of (x_n) and take a subsequence (x_{n_k}) of (x_n) converging to a. Now, observe that every n_k is either odd or even, and as such (x_{n_k}) must share a common subsequence with (x_{2n}) or (x_{2n-1}). Since every subsequence of (x_{n_k}) must converge to a, and (x_{2n}), (x_{2n-1}) are themselves convergent, it follows that $a = \lim x_{n_k}$ must be equal to either $\lim x_{2n}$ or $\lim x_{2n-1}$. That is, $a = \pm\frac{1}{2}$.

We conclude that $\pm\frac{1}{2}$ are the sole accumulation points of (x_n). It follows then that

$$\limsup x_n = \frac{1}{2}, \quad \liminf x_n = -\frac{1}{2}.$$

□

Remark 9.1. The boundedness assumption in the previous proposition is necessary. Indeed, consider the sequence (x_n) in $\mathbb{R}$ given by

$$x_n = \begin{cases} 1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

Clearly, $\lim x_{2n} = \lim 1 = 1$ whence 1 is an accumulation point of (x_n). Let now (x_{n_k}) be a convergent subsequence of (x_n). In particular, (x_{n_k}) is bounded. Now, (x_{n_k}) either contains a subsequence of (x_{2n}), or a subsequence of (x_{2n+1}). However, since any subsequence of (x_{2n+1}) is unbounded, (x_{n_k}) must contain a subsequence of (x_{2n}). In particular, (x_{n_k}) contains a subsequence converging to 1. Because (x_{n_k}) is assumed to be convergent, all subsequences converge and have the same limit as (x_{n_k}). This tells us that $\lim x_{n_k} = 1$. In short, (x_n) has a single accumulations point, i.e. 1. However, (x_n) is unbounded and clearly cannot converge.

Proposition 9.4. *Let (x_n) be a bounded sequence in $\mathbb{R}$. Then, (x_n) converges if and only if*

$$\liminf x_n = \limsup x_n.$$

In this case,

$$\lim x_n = \liminf x_n = \limsup x_n.$$

Proof. Let (x_n) be a convergent sequence. By the previous proposition, (x_n) has but a single accumulation point, denoted a. Hence,

$$\liminf x_n = a = \limsup x_n.$$

Conversely, assume

$$\liminf x_n = \limsup x_n.$$

If $a \in \mathbb{R}$ is an accumulation point of (x_n), then

$$\limsup x_n = \liminf x_n \leq a \leq \limsup x_n = \liminf x_n$$

whence

$$\liminf x_n = \limsup x_n = a.$$

Thus, (x_n) has only one accumulation point and must converge to

$$\liminf x_n = \limsup x_n.$$

□

10 Tenth Tutorial

This tutorial is all about limits of functions. Before giving any problems and examples, let us first reiterate the $\varepsilon - \delta$ definition of the limit. To properly make sense of the behaviour of a function "near a point x", it is critical that the function be defined at *some* points near x. This is precisely what the concept of a cluster point formalizes:

Definition 10.1. Let $A \subseteq \mathbb{R}$. A point $a \in \mathbb{R}$ is called a *cluster point* of A if, for each $\delta > 0$, $V_\delta(a) \cap (A \setminus \{a\}) \neq \varnothing$. Or, equivalently (see Lemma 10.1 below), if

$$a \in \overline{A \setminus \{a\}}.$$

Plainly put, this guarantees that there are "enough points" in A near a to define a meaningful limit for a function f defined on A.

Lemma 10.1. *Let $S \subseteq \mathbb{R}$. Then, $x \in \overline{S}$ if and only if for all $\varepsilon > 0$ we have $V_\varepsilon(x) \cap S \neq \varnothing$.*

Proof. First, if $x \in \overline{S} = S \cup \partial S$, then either $x \in S$ or $x \in \partial S$. In the first case, clearly $x \in V_\varepsilon(x) \cap S$ for all $\varepsilon > 0$. In the latter case, for all $\varepsilon > 0$ we have

$$V_\varepsilon(x) \cap S \neq \varnothing \quad \text{and} \quad V_\varepsilon(x) \cap S^{\mathsf{c}} \neq \varnothing.$$

In both cases, we have $V_\varepsilon(x) \cap S \neq \varnothing$ for all $\varepsilon > 0$.

Conversely, suppose $x \notin \overline{S} = S \cup \partial S$. Then, $x \in S^c$ and there exists $\varepsilon > 0$ such that either

$$V_\varepsilon(x) \cap S = \varnothing \quad \text{or} \quad V_\varepsilon(x) \cap S^c = \varnothing.$$

Since $x \in S^c$, we must be in the case $V_\varepsilon(x) \cap S = \varnothing$, which concludes the proof. □

Corollary 10.2. *Let $S \subseteq \mathbb{R}$. Then $x \in \overline{S}$ if and only if there exists a sequence (x_n) in S such that $\lim x_n = x$.*

Proof. First, let $x \in \overline{S}$ and observe that, by our lemma, $V_{1/n}(x) \cap S \neq \varnothing$ for each $n \in \mathbb{N}$. Thus, given $n \in \mathbb{N}$, we may choose some x_n belonging to this intersection. Then, (x_n) is a sequence in S satisfying $x_n \in V_{1/n}(x)$ for all $n \in \mathbb{N}$, i.e.

$$|x_n - x| < \frac{1}{n}$$

whence $\lim x_n = x$ by the null sequence criterion.

Conversely, suppose there exists a sequence (x_n) living in S which converges to x. Since $x_n \in S \subseteq \overline{S}$, where $\overline{S}$ is closed in $\mathbb{R}$, it follows from Theorem 9.2 that $x \in \overline{S}$. □

10.1 Functional Limits

Definition 10.2. Let $A \subseteq \mathbb{R}$ and a be a cluster point of A. Let $f : A \to \mathbb{R}$ be a function and fix $L \in \mathbb{R}$. We say that $f(x)$ *converges* to L as $x \to a$, written

$$\lim_{x \to a} f(x) = L,$$

provided for each $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever } x \in A \text{ and } 0 < |x - a| < \delta.$$

Or, symbolically, if

$$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in A)(0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon).$$

Note that the value of $f(a)$ (if it even exists) is of no importance as the requirement that $0 < |x - a| < \delta$ excludes the possibility that $x = a$.

Remark 10.1. The $\varepsilon - \delta$ definition of the limit bears a significant resemblance to the $\varepsilon - N$ definition for sequences. Indeed, the ε serves as an error term to measure how close to the limit L the value $f(x)$ is. However, instead of looking at very large natural numbers, we care about values of x close to a. So, in the $\varepsilon - \delta$ definition, the δ-term measures how close to a the variable x is. Hence, the $\varepsilon - \delta$ definition formalizes the following intuitive statement:

If the distance between $f(x)$ and L can be made arbitrarily small by making x sufficiently close to a, then $\lim_{x\to c} f(x) = L$.

Remark 10.2. Notice that in our definition of the limit we do *not* require that a belong to the domain of the function f. We do however require that a be a cluster point of A, and this is non-negotiable. To see why, let $f : A \subseteq \mathbb{R} \to \mathbb{R}$ be a function and suppose $a \in \mathbb{R}$ is *not* a cluster point of A (this will happen, for example, if $a \notin \overline{A}$). Then, the $\varepsilon - \delta$ definition holds for **all** real numbers L (in other words, the limit is not well defined in this case). Certainly, since a is not a cluster point of A, there exists $\delta > 0$ so small that $V_\delta(a) \setminus \{a\}$ contains no points from A. Then, the implication

$$(\forall x \in A)\, (0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon)$$

is vacuously true for all $\varepsilon > 0$ and all $L \in \mathbb{R}$ since there are no points $x \in A$ satisfying $0 < |x - a| < \delta$. This brings us back to following interpretation: being a cluster point of the domain ensures there are enough points in the domain "near a" to properly analyze the function's limiting behaviour there.

Example 10.1. Prove that, for every $c > 0$, one has

$$\lim_{x\to c} \frac{1}{\sqrt{x}} = \frac{1}{c}.$$

Of course, we are are viewing $\frac{1}{\sqrt{x}}$ as a function with domain equal to $(0, \infty)$.

Proof. Given $\varepsilon > 0$ and $x > 0$ with $x \neq c$, we have the estimate

$$\begin{aligned}\left|\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{c}}\right| = \frac{\left|\sqrt{x} - \sqrt{c}\right|}{\sqrt{x}\sqrt{c}} &= \frac{|x - c|}{\sqrt{x}\sqrt{c}\left(\sqrt{x} + \sqrt{c}\right)} \\ &\leq \frac{|x - c|}{x\sqrt{c}}.\end{aligned}$$

Let now

$$\delta := \min\left\{\frac{c^{3/2}}{2}\varepsilon, \frac{c}{2}\right\} > 0.$$

If $0 < |x - c| < \delta$, then $x \neq c$ and

$$-\delta < x - c < \delta.$$

But then, $\delta \leq \frac{c}{2}$ would imply

$$x > c - \delta \geq \frac{c}{2}$$

so that

$$\left|\frac{1}{\sqrt{x}}-\frac{1}{\sqrt{c}}\right| \leq \frac{|x-c|}{x\sqrt{c}} < \frac{\delta}{\sqrt{c}}\frac{2}{c} = \frac{2\delta}{c^{3/2}} \leq \varepsilon.$$

This shows that

$$\frac{1}{\sqrt{x}} \to \frac{1}{\sqrt{c}} \quad \text{as } x \to c.$$

□

Example 10.2. We show that

$$\lim_{x\to a}\frac{1}{x^2} = \frac{1}{a^2}$$

for all $a \neq 0$. Obviously, $f(x) = \frac{1}{x^2}$ is considered as a function with domain $\mathbb{R} \setminus \{0\}$. Let now $\varepsilon > 0$ be given. If $\delta > 0$ and $x \neq 0$ satisfies $0 < |x-a| < \delta$, then we have the estimate

$$\left|\frac{1}{x^2}-\frac{1}{a^2}\right| = \frac{\left|x^2-a^2\right|}{x^2a^2} = \frac{|x-a|\,|x+a|}{x^2a^2} < \delta\frac{|x|+|a|}{x^2a^2}.$$

If in addition $\delta \leq |a|$, then

$$|x| \leq |x-a|+|a| < \delta+|a| \leq 2\,|a|\,.$$

Hence,

$$\left|\frac{1}{x^2}-\frac{1}{a^2}\right| < \delta\frac{|x|+|a|}{x^2a^2} \leq \frac{3\delta\,|a|}{x^2a^2} = \frac{3\delta}{x^2\,|a|}.$$

If we assume further that $\delta \leq \frac{|a|}{2}$, then

$$||x|-|a|| \leq |x-a| < \delta$$

implies $|a|-\delta < |x| < |a|+\delta$ whence $|x| > \frac{|a|}{2}$. Rather,

$$\frac{1}{x^2} < \frac{4}{a^2}$$

so that

$$\left|\frac{1}{x^2}-\frac{1}{a^2}\right| < \frac{3\delta}{x^2\,|a|} \leq \frac{12\delta}{|a|^3}.$$

Finally, if

$$\delta \leq \frac{\varepsilon\,|a|^3}{12},$$

then

$$\left|\frac{1}{x^2} - \frac{1}{a^2}\right| < \frac{12\delta}{|a|^3} \leq \varepsilon.$$

Therefore, we define

$$\delta := \min\left\{\frac{|a|}{2}, \frac{\varepsilon\,|a|^3}{12}\right\}.$$

By our earlier calculations, for every $x \neq 0$ with $0 < |x - a| < \delta$, one has

$$\left|\frac{1}{x^2} - \frac{1}{a^2}\right| < \varepsilon.$$

This completes the proof.

10.2 The Sequential Criterion for the Limit

The sequential criterion is a result which describes functional limits in terms of sequential limits. Although it will not always be possible to easily apply the sequential criterion, it does offer a method by which we may leverage the significant list of limit laws we have developed for sequences.

Theorem 10.3 (Sequential Criterion)**.** *Let $f : A \subseteq \mathbb{R} \to \mathbb{R}$ be a function and let $a \in A$ be a cluster point. Fix $L \in \mathbb{R}$. The following statements are equivalent:*

(i) $\lim_{x \to a} f(x) = L$.

(ii) For every sequence (x_n) in A, with $x_n \neq a$ for all $n \in \mathbb{N}$ and $\lim x_n = a$, there holds $\lim f(x_n) = L$.

Thus, we may establish criterion (ii) instead of verifying that the $\varepsilon - \delta$ definition holds when trying to prove that $\lim_{x \to a} f(x) = L$. Let us illustrate this by repeating the last two examples with the sequential criterion,

Example 10.3. Define $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ by $f(x) := \frac{1}{x^2}$. Let $a \neq 0$ and use the sequential criterion to show that

$$\lim_{x \to a} f(x) = \frac{1}{a^2}.$$

Proof. Let (x_n) be a sequence in $\mathbb{R} \setminus \{0\}$, such that $x_n \neq a$ for all $n \in \mathbb{N}$, and $x_n \to a$. By the product limit law for convergent sequences, we have that $x_n^2 \to a^2 \neq 0$. But then, we may apply the quotient limit law for sequences (since $x_n^2 \to a^2 \neq 0$) to deduce that

$$f(x_n) = \frac{1}{x_n^2} \to \frac{1}{a^2}, \quad \text{as } n \to \infty.$$

Since (x_n) was an arbitrary sequence in $\mathbb{R} \setminus \{0\}$, never equal to a, and converging to a, it follows from the sequential criterion that $\lim_{x \to a} f(x) = \frac{1}{a^2}$. □

Example 10.4. Use the sequential criterion to show that

$$\lim_{x\to a}\frac{1}{\sqrt{x}}=\frac{1}{\sqrt{a}},\quad \forall a>0.$$

Proof. Let (x_n) be an arbitrary sequence in $(0,\infty)$, with $x_n\neq a$ for all $n\in\mathbb{N}$, such that $\lim x_n = a$. From previous exercises with sequences, we know that $\sqrt{x_n}\to\sqrt{a}$, where $\sqrt{a}\neq 0$. Finally, by applying the quotient limit law for sequences, we find that

$$\lim f(x_n)=\lim\frac{1}{\sqrt{x_n}}=\frac{1}{\sqrt{a}}$$

□

10.3 Some More Examples

Example 10.5. Given $a\neq -1$, we claim that

$$\lim_{x\to a}\frac{x}{1+x}=\frac{a}{1+a}.$$

Again, here we are viewing $\frac{x}{1+x}$ as a function $\mathbb{R}\setminus\{-1\}\to\mathbb{R}$.

Proof. Let $\varepsilon>0$ be given and fix $a\neq -1$. Henceforth, we shall always assume that $x\neq -1$. If for some $\delta>0$ one has $0<|x-a|<\delta$, then

$$\left|\frac{1}{1+x}-\frac{1}{1+a}\right|=\frac{|x-a|}{|1+x|\,|1+a|}<\frac{\delta}{|1+x|\,|1+a|}.$$

Next, we have to "rid ourselves" of the $|1+x|$ term. To this end, suppose additionally that

$$\delta\leq\frac{|1+a|}{2}.$$

Then,

$$|1+x|\geq|1+a|-|x-a|>|1+a|-\delta\geq\frac{|1+a|}{2}>0.$$

Therefore,

$$\left|\frac{1}{1+x}-\frac{1}{1+a}\right|<\frac{\delta}{|1+x|\,|1+a|}\leq\frac{2\delta}{(1+a)^2}.$$

Thus, if

$$\delta\leq\varepsilon\frac{(1+a)^2}{2}$$

then

$$\left|\frac{1}{1+x}-\frac{1}{1+a}\right| < \frac{2\delta}{(1+a)^2} \leq \varepsilon.$$

Finally, we define

$$\delta := \min\left\{\frac{|1+a|}{2}, \varepsilon\frac{(1+a)^2}{2}\right\}.$$

By the calculations we carried out above, we conclude that

$$\left|\frac{1}{1+x}-\frac{1}{1+a}\right| < \varepsilon$$

for all $x \neq -1$ with $0 < |x-a| < \delta$. □

Remark 10.3. Using the sequential criterion for the limit, we can easily conclude that the limit

$$\lim_{x\to -1}\frac{x}{1+x}$$

does not exist. Indeed, consider the sequence (x_n) in $\mathbb{R}$ defined by

$$x_n := -1-\frac{1}{n}.$$

Clearly, $x_n \neq -1$ for all $n \in \mathbb{N}$ and $\lim x_n = -1$. Furthermore, for each $n \in \mathbb{N}$ we calculate:

$$f(x_n) = \frac{x_n}{1+x_n} = \frac{-1-\frac{1}{n}}{-\frac{1}{n}} = n\left(1+\frac{1}{n}\right) \geq n.$$

Hence, we get that $\lim f(x_n)$ does not exist in $\mathbb{R}$. This shows that $\lim\limits_{x\to -1}\dfrac{x}{1+x}$ does not exist either.

11 Eleventh Tutorial

We have now come to the topic of continuity. For the moment, let us fix a set $A \subseteq \mathbb{R}$ and a point $a \in A$. We say that a function $f : A \to \mathbb{R}$ is *continuous* at the point a if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x)-f(a)| < \varepsilon$$

whenever $x \in A$ satisfies $|x-a| < \delta$. Or, in symbolic terms:

$$(\forall \varepsilon > 0)\ (\exists \delta > 0)(\forall x \in A)(|x-a| < \delta \implies |f(x)-f(a)| < \varepsilon).$$

Note that in contrast to the $\varepsilon - \delta$ definition of the limit, we do not require that $|x - a|$ be positive. Equivalently, we allow for $x = a$. It is a routine exercise to show that the continuity of f at a implies the continuity of $|f|$ at a. Indeed, the proof is immediate from the reverse triangle inequality:

$$||f(x)| - |f(a)|| \leq |f(x) - f(a)|\,.$$

This is highly reminiscent of a result we had for sequences, and as such it is natural to check whether the converse holds.

Example 11.1. Consider the function $f : [0, 1] \to \mathbb{R}$ given by the rule

$$f(x) := \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ -1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Clearly, $|f(x)| = |\pm 1| = 1$ for all $x \in [0, 1]$. Hence, $|f|$ is continuous. However, f will turn out to be discontinuous at every point $x \in [0, 1]$. To see this, we fix a point $x \in [0, 1]$ and distinguish the only two possible cases.

1. Suppose that $x \in \mathbb{Q}$ so that $f(x) = 1$. By their density, we can select a sequence of irrational numbers (x_n) from $[0, 1]$ such that $\lim x_n = x$. Since every $x_n \notin \mathbb{Q}$, we see that
$$\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} (-1) = -1 \neq 1 = f(x).$$
2. Suppose that $x \notin \mathbb{Q}$, whence $f(x) = -1$. As above, choose a sequence (x_n) in $\mathbb{Q} \cap [0, 1]$ that converges to x as $n \to \infty$. Since every x_n is rational, we see that
$$\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} 1 = 1 \neq -1 = f(x).$$

In either case, we see (by the sequential criterion for continuity) that the function f is nowhere continuous on $[0, 1]$.

Example 11.2. Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) := \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Discuss the continuity of f at all points in $\mathbb{R}$.

Solution. We first show that f is continuous at 0. Indeed, observe first that $|f(x)| \leq |x|$ for all $x \in \mathbb{R}$. Consequently, given $\varepsilon > 0$, we may select $\delta := \varepsilon$ so that, for all $x \in \mathbb{R}$ with $|x - 0| = |x| < \delta$ there holds

$$|f(x) - f(0)| = |f(x)| \leq |x| < \delta = \varepsilon.$$

Hence, f is continuous at 0. Let now $a \neq 0$, we claim that f is discontinuous at a. Here, there are two cases to consider.

(i) If $a \in \mathbb{Q}$ then, since $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$, we may choose a sequence of irrationals (ξ_n) converging to a, as $n \to \infty$. But, $f(a) = a \neq 0$ and yet

$$\lim f(\xi_n) = \lim 0 = 0.$$

Thus, f fails the sequential criterion for continuity at $a \in \mathbb{Q} \setminus \{0\}$.

(ii) If $a \in (\mathbb{R} \setminus \mathbb{Q}) \setminus \{0\}$ then, using that $\mathbb{Q}$ is dense in $\mathbb{R}$, we may select a sequence of rational numbers (r_n) converging to a, as $n \to \infty$. But, $f(a) = 0$ whilst

$$\lim f(r_n) = \lim r_n = a \neq 0.$$

Thus, f fails the sequential criterion for continuity at a.

□

Let us now see how the continuity of a functions helps to locally preserve the "sign" of a function. In other words, we shall see that a continuous function does not change sign too rapidly near a fixed point.

Proposition 11.1. *Let $A \subseteq \mathbb{R}$ and assume that $f : A \to \mathbb{R}$ is continuous at a point $a \in A$. If $f(a) > 0$, there exists a δ-neighbourhood $V_\delta(a)$ such that*

$$f(x) > \frac{f(a)}{2} > 0, \quad \forall x \in A \cap V_\delta(a).$$

In particular, $f(x) > 0$ for all $x \in A \cap V_\delta(a)$.

Remark 11.1. Loosely speaking, if $f : A \to \mathbb{R}$ is continuous at a and $f(a) > 0$, the result above says that $f(x)$ is positive at all points in A "near" a.

Proof. Let us define

$$\varepsilon := \frac{f(a)}{2} > 0.$$

Since f is continuous at a, we can find $\delta > 0$ such that

$$|f(x) - f(a)| < \varepsilon = \frac{f(a)}{2}$$

for all $x \in A$ with $|x - a| < \delta$. That is, the above holds for all $x \in V_\delta(x) \cap A$. However, for any such x, we trivially have

$$f(a) - f(x) \leq |f(x) - f(a)| < \frac{f(a)}{2}$$

whence it follows that

$$f(a) - \frac{f(a)}{2} < f(x).$$

Or, rather, we find that

$$f(x) > \frac{f(a)}{2} > 0$$

for all $x \in V_\delta(x) \cap A$. □

Corollary 11.2. *Let $A \subseteq \mathbb{R}$ and assume that $f : A \to \mathbb{R}$ is continuous at a point $a \in A$. If $f(a) < 0$, there exists a δ-neighbourhood $V_\delta(a)$ such that $f(x) < 0$ for all $x \in A \cap V_\delta(a)$.*

Proof. Apply the previous result to the function $g := -f$. □

The next reformulation is particularly simple and useful.

Proposition 11.3. *Let $f : A \subseteq \mathbb{R} \to \mathbb{R}$ be a function and assume that f is continuous at a point $a \in A$. If $f(a) \neq 0$, then one can find $\delta > 0$ such that*

$$|f(x)| > \frac{|f(a)|}{2} > 0$$

for all $x \in V_\delta(a) \cap A$. Especially, $f(x) \neq 0$ for all $x \in V_\delta(a) \cap A$.

Proof. Apply Proposition 11.1 to $h := |f|$. □

With this, we can rapidly establish the next claim. It often appears as an exercise and as such we find it instructive to include it here.

Corollary 11.4. *Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and consider the zero-set*

$$S_0 := \{x \in \mathbb{R} : f(x) = 0\} = f^{-1}(\{0\}).$$

Then, S_0 is closed in $\mathbb{R}$.

First Proof. We shall show that

$$S_0^c := \{x \in \mathbb{R} : f(x) \neq 0\}$$

is open. To this end, let $x \in S_0^c$ be given. Since $f(x) \neq 0$ and f is continuous at x, there exists $\delta > 0$ such that $f(x) \neq 0$ on all of $V_\delta(x)$ (see Proposition 11.3). Then, $V_\delta(x) \subseteq S_0^c$ whence S_0^c is seen to be open. □

Second Proof. We shall use Theorem 9.2 to show S_0 is closed. To this end, let (x_n) be a sequence in S_0 converging to some $x \in \mathbb{R}$. We will be done provided we can show $x \in S_0$. Now, by the sequential criterion for continuity, and the fact that f is continuous at x, we must have $f(x) = \lim f(x_n) = \lim 0 = 0$ whence $f(x) = 0$ and $x \in S_0$. □

11.1 The Location of Roots & Intermediate Value Theorems

Although we have seen this result within the lectures, we provide an alternative proof of the location of roots theorem that does not rely on the nested interval theorem. Instead, we argue directly using only the sequential criterion for continuity and the supremum property of $\mathbb{R}$. Of course, once we have established the location of roots theorem, the intermediate value theorem is deduced in the exact same was as in the lectures.

Lemma 11.5. *Let $A \subseteq \mathbb{R}$ be non-empty and bounded from above. There exists a sequence (a_n) in A such that $\lim a_n = \sup A$.*

Proof. Given $n \in \mathbb{N}$, the quantity $\sup A - \frac{1}{n}$ is not an upper bound of A. Thus, there exists $a_n \in A$ such that

$$\sup A - \frac{1}{n} < a_n \leq \sup A < \sup A + \frac{1}{n}.$$

The sequence (a_n) resulting from this selection must then satisfy

$$|a_n - \sup A| < \frac{1}{n}, \quad \forall n \in \mathbb{N}$$

whence (a_n) converges to $\sup A$ by the null sequence criterion. □

Theorem 11.6 (Location of Roots Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. Assume that $f(a) < 0$ and $f(b) > 0$. Define $W := \{x \in [a, b] : f(x) < 0\}$ and put $c := \sup W$. Then, $f(c) = 0$.*

Proof. Observe that $a \in W$ and thus $a \leq c$. In addition, $W \subseteq [a, b]$ implies that b is an upper bound of W. Since c is the least upper bound of W, we must have $c \leq b$ and therefore $c \in [a, b]$. Now, by Lemma 11.5, there exists a sequence (x_n) in W converging to c, as $n \to \infty$. By the sequential criterion for continuity, and owing to the fact that $x_n \in W$ for all $n \in \mathbb{N}$:

$$f(c) = \lim f(x_n) \leq 0 \quad \text{because } f(x_n) < 0 \text{ for all } n \in \mathbb{N}.$$

Having now shown that $f(c) \leq 0$, it automatically follows that $c < b$.

It remains only to rule out the possibility of $f(c) < 0$. Arguing by contradiction, if $f(c) < 0$, there exists by Corollary 11.2 some $\delta > 0$ such that $f(x) < 0$ for all $x \in [a, b] \cap V_\delta(c)$. Without loss of generality, we may assume that $\delta \leq b - c$. Then, if $x \in (c, c + \delta)$ we have $x > c \geq a$ and $x < c + \delta \leq b$ and that $f(x) < 0$. But this would imply that $x \in W$ which contradicts $x > c$. Consequently, we must have $f(c) = 0$ and the proof is complete. □

Corollary 11.7. *Let $f : [a, b] \to \mathbb{R}$ be continuous. For any $y \in \mathbb{R}$ between $f(a)$ and $f(b)$, there exists $x \in [a, b]$ such that $f(x) = y$.*

11.2 Thomae's Function

In this subsection we discuss a special function whose continuity points are precisely the irrationals within some interval. This function, named Thomae's function, is an important example of a "pathological function", i.e. a function which defies some aspect of "common sense". The takeaway here is, although continuity is meant to reflect an intuitive notion, there are still exotic functions satisfying this definition.

Given any number $x \in (0,1) \cap \mathbb{Q}$, we may *uniquely* express $x = \frac{p}{q}$, with $p, q \in \mathbb{N}$, and $\gcd(p,q) = 1$. We shall call this the *standard representation* of $x \in \mathbb{Q} \cap (a,b)$. With this in mind, let us define

$$\mathsf{T} : (0,1) \to \mathbb{R}, \quad \mathsf{T}(x) := \begin{cases} 0 & \text{if } x \in (\mathbb{R} \setminus \mathbb{Q}) \cap (0,1), \\ \frac{1}{q} & \text{if } x \in \mathbb{Q} \cap (0,1) \text{ and } x = \frac{p}{q} \text{ in standard form.} \end{cases}$$

A key observation regarding this function is given below in the form of a lemma.

Lemma 11.8. *Given any $\varepsilon > 0$, there exist at most finitely many $x \in (0,1)$ with the property that $\mathsf{T}(x) \geq \varepsilon$. Furthermore, every such x must be rational.*

Proof. If $\mathsf{T}(x) \geq \varepsilon > 0$, then we must have $x \in \mathbb{Q} \cap (0,1)$. Writing $x = p/q$ in standard form, it follows that $\mathsf{T}(x) = 1/q \geq \varepsilon$ whence $q \leq \frac{1}{\varepsilon}$. Since $q \in \mathbb{N}$, this leaves only finitely many possibilities for q. Since $x = p/q \in (0,1)$, we must have $p < q$ so only finitely many p are possible as well. Hence, there are only finitely many $x = p/q \in \mathbb{Q} \cap (0,1)$ that can satisfy $\mathsf{T}(x) \geq \varepsilon$ and the claim is proven. □

This gives rise to the following classification of the continuity points of T.

Theorem 11.9. *Thomae's function $\mathsf{T} : (0,1) \to \mathbb{R}$ is continuous on $(\mathbb{R} \setminus \mathbb{Q}) \cap (0,1)$ and discontinuous at all points in $\mathbb{Q} \cap (0,1)$.*

Proof. We first demonstrate the continuity of T on $(\mathbb{R} \setminus \mathbb{Q}) \cap (0,1)$. Let $\varepsilon > 0$ be given and fix $\xi \in (\mathbb{R} \setminus \mathbb{Q}) \cap (0,1)$; note also that $\mathsf{T}(\xi) = 0$. By Lemma 11.8, the set

$$Z := \{x \in (0,1) : \mathsf{T}(x) \geq \varepsilon\}$$

is a finite subset of $\mathbb{Q}$. Since $\xi \notin \mathbb{Q}$, it follows that $|\xi - z| > 0$ for all $z \in Z$. Because Z is finite, we may define

$$\delta := \min\{|z - \xi| : z \in Z\} > 0.$$

Now, if $x \in (0,1)$ is such that $|x - \xi| < \delta$, we must have that $x \notin Z$ (because then $\delta \leq |x - \xi|$). By definition, this means that $\mathsf{T}(x) < \varepsilon$. But,

$$|\mathsf{T}(x) - \mathsf{T}(\xi)| = |\mathsf{T}(x)| = \mathsf{T}(x) < \varepsilon.$$

Hence, T is continuous at any $\xi \in (\mathbb{R} \setminus \mathbb{Q}) \cap (0,1)$.

Finally, let us show T is discontinuous at all $r \in \mathbb{Q} \cap (0,1)$. Indeed, given $r \in \mathbb{Q} \cap (0,1)$, we have by definition that $\mathsf{T}(r) \neq 0$. However, since $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$, we may also choose a sequence (ξ_n) of irrationals such that $\lim x_n = r \in (0,1)$. Since $(0,1)$ is open, there exists $N \in \mathbb{N}$ such that $\xi_n \in (0,1)$ for all $n \geq N$. Then, $(\xi_n)_{n=N}^{\infty}$ is a sequence in $(0,1) \cap (\mathbb{R} \setminus \mathbb{Q})$ converging to r, as $n \to \infty$. However,

$$\lim \mathsf{T}(\xi_n) = \lim 0 = 0 \neq \mathsf{T}(r).$$

Thus, T fails the sequential criterion for continuity at every $r \in \mathbb{Q} \cap (0,1)$. □

11.3 Applications of the Extreme and Intermediate Value Theorems

We continue to solve some classic exercises related to continuity on intervals and compacta. Before solving these, let us recall a very important result from the lectures.

Theorem 11.10 (Extreme Value Theorem)**.** *Let $K \subseteq \mathbb{R}$ be compact and $f : K \to \mathbb{R}$ continuous. Then, f is bounded and $f(K)$ is compact. Furthermore, f achieves a global maximum and a global minimum on K.*

The compactness of the domain here is absolutely vital. For example, the function $f(x) = x$ is continuous on $\mathbb{R}$ but $f(\mathbb{R}) = \mathbb{R}$ is unbounded (and not compact) and, of course, $f(x) = x$ achieves neither a maximum nor a minimum on $\mathbb{R}$.

Proposition 11.11. *Let $A \subseteq \mathbb{R}$ be a non-empty compact and let $f : A \to \mathbb{R}$ be a continuous function. Assume that f has the following property: for all $x \in A$, there exists $y \in A$ such that $|f(y)| \leq \frac{|f(x)|}{2}$. Then, there exists a point $c \in A$ such that $f(c) = 0$.*

Proof. Since compositions preserve continuity, $|f|$ is continuous on A as well. Then, since A is compact, the extreme value theorem asserts that $|f|$ achieves a minimum at some $c \in A$. That is, we have $|f(x)| \geq |f(c)|$ for all $x \in A$. However, by assumption, for this $c \in A$ there must also exist $c' \in A$ with the property that

$$|f(c')| \leq \frac{|f(c)|}{2}.$$

But then,

$$|f(c)| \leq |f(c')| \leq \frac{|f(c)|}{2}$$

implies that $2\,|f(c)| \leq |f(c)|$ whence $|f(c)| \leq 0$ so that $|f(c)| = 0$. That is, $f(c) = 0$. □

Proposition 11.12. *Let $D \subseteq \mathbb{R}$ be a set with dense complement (i.e. D^{c} is dense in $\mathbb{R}$). A function $f : \mathbb{R} \to D$ is continuous if and only if it is constant.*

Proof. Suppose for a contradiction that $f : \mathbb{R} \to D$ is continuous but non-constant. Then, there exist distinct $d_1, d_2 \in D$ along with $x_1, x_2 \in \mathbb{R}$ such that

$$d_1 = f(x_1) < f(x_2) = d_2.$$

That is, if f is non-constant, it achieves at least two distinct values. Since D^{c} is dense in $\mathbb{R}$, there exists $e \in D^{\mathsf{c}}$ such that $d_1 < e < d_2$. By the intermediate value theorem, there must therefore exist x between x_1 and x_2 such that $f(x) = e \notin D$, which contradicts the fact that f only takes values in D. It follows that f must be constant.

The converse is trivial since any constant function is automatically continuous. □

Example 11.3. Let $A \subseteq \mathbb{R}$ be compact and $f : A \to \mathbb{R}$ continuous. If $f(x) > 0$ for all $x \in A$, show there exists $\alpha > 0$ such that $f(x) \geq \alpha$ for all $x \in A$.

Solution. By the extreme value theorem, f achieves a global minimum at some $c \in A$. That is, $f(x) \geq f(c) > 0$ for all $x \in A$ (here, $f(c) > 0$ because $c \in A$). Defining $\alpha := f(c) > 0$ yields at once the assertion. □

12 Twelfth Tutorial

Our main focus in this tutorial will be *uniform continuity*. As such, let us begin with a recollection of this concept. Given a subset $A \subseteq \mathbb{R}$ and a function $f : A \to \mathbb{R}$, we say that f is *uniformly continuous* on A if for each $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $x, y \in A$ satisfy $|x - y| < \delta$. In symbolic terms:

$$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x, y \in A)(|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon).$$

At a first glance this may seem to be the same as saying that f is continuous on A; but this is not the case. By way of clarification, let us compare the definitions of continuity and uniform continuity.

- We say that f is continuous at a point $y \in A$ if

  $$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in A)(|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon).$$

  Thus, we say that f is continuous on A if

  $$(\forall y \in A)(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in A)(|x - a| < \delta \implies |f(x) - f(y)| < \varepsilon).$$

  In this expression, the variable y appears before the "$\exists\delta$", which means that δ may depend on the point y. However, this is not the case in the definition of uniform continuity. For f to be uniformly continuous on A, the $\delta > 0$ cannot depend on x or y.

- Continuity is a *local property* and uniform continuity is a *global property*. We know what it means for the function f to be continuous at an individual point $a \in A$, but we have only defined uniform continuity for the entire set A. Namely, the continuity of f at a single point a depends only on the behaviour of f in a neighbourhood of a. On the other hand, uniform continuity depends on how f behaves on the entire set A.

Let us also recall the following fundamental result from class.

Theorem 12.1 (Uniform Continuity Theorem)**.** *Let $K \subset \mathbb{R}$ be compact and let $f : K \to \mathbb{R}$ be continuous. Then, f is uniformly continuous on A.*

If K is not compact, this implication may fail. However, there are still many nice continuous functions that turn out to be uniformly continuous on unbounded intervals. One example is the function

$$f : (0, \infty) \to \mathbb{R}, \quad x \mapsto \sqrt{x}.$$

Example 12.1. As seen in class, the function f described above is uniformly continuous on $[0, \infty)$. Indeed, let us establish the following useful inequality:

$$\left|\sqrt{x} - \sqrt{y}\right| \leq \sqrt{|x - y|}, \quad \forall x, y \geq 0. \tag{12.1}$$

By symmetry, it is enough to show that the above holds whenever $0 \leq y \leq x$. For such x, y, the inequality in (12.1) reduces to proving that

$$\begin{aligned}
\sqrt{x} - \sqrt{y} \leq \sqrt{x - y} &\iff \left(\sqrt{x} - \sqrt{y}\right)^2 \leq (x - y) \\
&\iff x - 2\sqrt{xy} + y \leq x - y \\
&\iff -2\sqrt{xy} + y \leq -y \\
&\iff 2y \leq 2\sqrt{xy} \\
&\iff y \leq \sqrt{xy}.
\end{aligned}$$

This last inequality holds true because $y \leq x$ implies

$$y = \sqrt{y^2} = \sqrt{y \cdot y} \leq \sqrt{x \cdot y}.$$

Thus, we have proven (12.1). It now becomes trivial to show that $\sqrt{x}$ is uniformly continuous on $[0, \infty)$. To see this, let $\varepsilon > 0$ be given and take $\delta := \varepsilon^2 > 0$. If $x, y \in [0, \infty)$ are such that $|x - y| < \delta$, then (12.1) tells us that

$$|f(x) - f(y)| = \left|\sqrt{x} - \sqrt{y}\right| \leq \sqrt{|x - y|} < \sqrt{\delta} = \varepsilon.$$

12.1 Sequences and Uniform Continuity

Let us begin by recalling the sequential characterization of uniform continuity. Although not particularly useful in establishing uniform continuity, its negation offers a sometimes straightforward way of demonstrating *non-uniform* continuity.

Theorem 12.2 (Sequential Criterion for Uniform Continuity)**.** *Let $f : A \subseteq \mathbb{R} \to \mathbb{R}$ be a function. The following statements are equivalent.*

1. *f is uniformly continuous on A.*
2. *For any sequences $(x_n), (y_n)$ in A such that $\lim(x_n - y_n) = 0$, there holds that*

$$\lim(f(x_n) - f(y_n)) = 0.$$

By negating the above, we obtain the following criterion for non-uniform continuity.

Corollary 12.3 (Sequential Criterion for Non-Uniform Continuity)**.** *Let $f : A \subseteq \mathbb{R} \to \mathbb{R}$ be a function. The following statements are equivalent.*

1. *f is not uniformly continuous on A.*
2. *There exists $\varepsilon_0 > 0$ and two sequences $(x_n), (y_n)$ in A such that $\lim(x_n - y_n) = 0$, but*

$$|f(x_n) - f(y_n)| \geq \varepsilon_0$$

for all $n \in \mathbb{N}$.

You have seen in the lectures that uniformly continuous functions take Cauchy sequences to Cauchy sequences. In fact, the proof makes for a good exercise that helps to internalize the definitions. Is the converse necessarily true? To help answer this question, we consider a continuous function on an interval that is not uniformly continuous.

Lemma 12.4. *Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) := x^2$. The function f is not uniformly continuous on $\mathbb{R}$, but is uniformly continuous on any compact interval $[-a, a]$.*

Proof. The only non-trivial claim is that f is not uniformly continuous on all of $\mathbb{R}$. To see this, we argue by contradiction. If f is uniformly continuous on $\mathbb{R}$, then for $\varepsilon := 1$ we can find $\delta > 0$ such that

$$\left|x^2 - y^2\right| < 1$$

whenever $|x - y| < \delta$. Given $x \in \mathbb{R}$ we consider the point

$$y := x + \frac{\delta}{2}.$$

It is clear that $|x - y| < \delta$, and we must therefore have

$$\left| x^2 - \left(x + \frac{\delta}{2} \right)^2 \right| < 1.$$

However, the expression above reduces to the following:

$$\left| x\delta + \frac{\delta^2}{4} \right| < 1.$$

Since the function $x \mapsto x\delta + \frac{\delta^2}{4}$ is unbounded on $\mathbb{R}$, this is a contradiction. □

Second Proof. We show that $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$ by means of the sequential criterion for non-uniform continuity. Indeed, let (x_n) and (y_n) be the sequences

$$x_n := n + \frac{1}{n}, \quad y_n := n.$$

Then,

$$x_n - y_n = \frac{1}{n} \xrightarrow{n \to \infty} 0\,.$$

However,

$$|f(x_n) - f(y_n)| = \left| x_n^2 - y_n^2 \right| = \left| n^2 + 2 + \frac{1}{n^2} - n^2 \right| = 2 + \frac{1}{n^2} > 2$$

for all $n \in \mathbb{N}$. □

Having exhibited a continuous function $\mathbb{R} \to \mathbb{R}$ that is not uniformly continuous, we can answer our original question.

Example 12.2. We claim that the function $f : \mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^2$ takes Cauchy sequences in $\mathbb{R}$ to Cauchy sequences, despite not being uniformly continuous on $\mathbb{R}$. To see this, let (x_n) be a Cauchy sequence. Since Cauchy sequences are bounded, the sequence (x_n) is contained within some compact interval $[-a, a]$. That is, (x_n) is a Cauchy sequence in $[-a, a]$. By continuity on all of $\mathbb{R}$, the function f is uniformly continuous on $[-a, a]$. Since uniformly continuous functions take Cauchy sequences to Cauchy sequences, we see that $(f(x_n))$ is Cauchy in $\mathbb{R}$.

Thus, we have found a continuous map $\mathbb{R} \to \mathbb{R}$ that takes Cauchy sequences to Cauchy sequences, even though it is *not* uniformly continuous on $\mathbb{R}$.

Example 12.3. Prove that $f(x) := \sin\left(x^2\right)$ is *not* uniformly continuous on $\mathbb{R}$.

Proof. We shall employ the sequential criterion for non-uniform continuity. To this end, let us define

$$x_n := \sqrt{2\pi n + \frac{\pi}{2}}, \quad y_n := \sqrt{2\pi n}.$$

Then, the sequences (x_n) and (y_n) satisfy, for each $n \in \mathbb{N}$,

$$\begin{aligned} 0 < x_n - y_n = \sqrt{2\pi n + \frac{\pi}{2}} - \sqrt{2\pi n} &= \frac{\sqrt{2\pi n + \frac{\pi}{2}} - \sqrt{2\pi n}}{\sqrt{2\pi n + \frac{\pi}{2}} + \sqrt{2\pi n}} \left[\sqrt{2\pi n + \frac{\pi}{2}} + \sqrt{2\pi n} \right] \\ &= \frac{\pi/2}{\sqrt{2\pi n + \frac{\pi}{2}} + \sqrt{2\pi n}} \\ &< \frac{\pi}{2\sqrt{2\pi n}} \\ &= \frac{\sqrt{\pi}}{2^{3/2}\sqrt{n}} \xrightarrow{n\to\infty} 0. \end{aligned}$$

By the squeeze theorem, $\lim(x_n - y_n) = 0$. However,

$$|f(x_n) - f(y_n)| = \left| \sin\left(2\pi n + \frac{\pi}{2}\right) - \sin(2\pi n) \right| = 1 > 0, \quad \forall n \in \mathbb{N}.$$

□

Moving away from examples, let us now prove some general statements.

Proposition 12.5. *Let $A \subseteq \mathbb{R}$ and $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(1) $f + g$ is uniformly continuous as a function $A \to \mathbb{R}$.

(2) If f and g are bounded on A, then the product fg is uniformly continuous on A.

Proof. We begin with the first claim. Let $\varepsilon > 0$ be given and let $\delta_1 > 0$ be such that

$$|f(x) - f(y)| < \frac{\varepsilon}{2}$$

whenever $x, y \in A$ satisfy $|x - y| < \delta_1$. Similarly, because g is uniformly continuous on A, we can find $\delta_2 > 0$ with the property that $|x - y| < \delta_2$ (with $x, y \in A$) implies

$$|g(x) - g(y)| < \frac{\varepsilon}{2}.$$

Taking $\delta := \min\{\delta_1, \delta_2\} > 0$, we see that for any $x, y \in A$ with $|x - y| < \delta$ we have:

$$\begin{aligned} |(f+g)(x) - (f+g)(y)| &= |f(x) + g(x) - f(y) - g(y)| \\ &\leq |f(x) - f(y)| + |f(y) - g(y)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

To prove the second claim, let us assume in addition that f, g are bounded on A, i.e. there exists $M > 0$ such that

$$|f(x)| \leq M \quad \text{and} \quad |g(x)| \leq M$$

for all $x \in A$. Let now $\varepsilon > 0$ be given. Using that f is uniformly continuous on A, there exists $\delta_1 > 0$ such that

$$|f(x) - f(y)| < \frac{\varepsilon}{2M}$$

whenever $x, y \in A$ satisfy $|x - y| < \delta_1$. Similarly, there exist $\delta_2 > 0$ such that

$$|g(x) - g(y)| < \frac{\varepsilon}{2M}$$

for all $x, y \in A$ such that $|x - y| < \delta_2$. Define $\delta := \min\{\delta_1, \delta_2\}$ and let $x, y \in A$ be such that $|x - y| < \delta$. Then, because $|x - y| < \delta_1$ and $|x - y| < \delta_2$, there holds that

$$\begin{aligned}
|(fg)(x) - (fg)(y)| &= |f(x)g(x) - f(y)g(x) + f(y)g(x) - f(y)g(y)| \\
&\leq |f(x)g(x) - f(y)g(x)| + |f(y)g(x) - f(y)g(y)| \\
&= |g(x)|\,|f(x) - f(y)| + |f(y)|\,|g(x) - g(y)| \\
&\leq M\,|f(x) - f(y)| + M\,|g(x) - g(y)| \\
&< M\frac{\varepsilon}{2M} + M\frac{\varepsilon}{2M} \\
&= \varepsilon.
\end{aligned}$$

□

Remark 12.1. In the second claim of the last proposition, the boundedness assumption may not be relaxed. Indeed, as we saw in Lemma 12.4, the function $f(x) = x^2$ is *not* uniformly continuous on A, despite being the product of two uniformly continuous functions. Certainly, $f(x) = x^2 = g(x)^2$, where $g(x) = x$ is uniformly continuous on $\mathbb{R}$ (however, it is unbounded there).[25]

12.2 Uniformly Continuous Functions Are Bounded

Given a non-compact *bounded* subset A of $\mathbb{R}$, a continuous function $f : A \to \mathbb{R}$ need not be bounded. Indeed, this is the case for the continuous map

$$f : (0, 1] \to \mathbb{R}, \quad x \mapsto \frac{1}{x}.$$

[25] The function $g(x)$ is easily seen to be uniformly continuous on $\mathbb{R}$. To see this, let $\varepsilon > 0$ and let $\delta := \varepsilon$. For any $x, y \in \mathbb{R}$ with $|x - y| < \delta$, we have $|g(x) - g(y)| = |x - y| < \delta = \varepsilon$. In fact, g is Lipschitz continuous on $\mathbb{R}$, which we shall discuss shortly.

Thus, continuous functions on bounded domains need not themselves be bounded, in general. However, it is true that uniformly continuous functions on bounded domains are bounded. We recall the proof of this fact below for the sake of completeness.

Theorem 12.6. *Let $A \subset \mathbb{R}$ be bounded and suppose that $f : A \to \mathbb{R}$ is uniformly continuous on A. Then, f is bounded on A.*

Proof. We argue by contradiction. If f were unbounded on A, we could find a sequence (x_n) in A such that $|f(x_n)| > n$ for each $n \in \mathbb{N}$. Since A is bounded, the sequence (x_n) is bounded. Thus, we can find a subsequence (x_{n_k}) that converges in $\mathbb{R}$. Equivalently, (x_{n_k}) is a Cauchy sequence in A such that

$$\left|f(x_{n_k})\right| > n_k \geq k$$

for all $k \in \mathbb{N}$. However, since f is uniformly continuous on A, we see that $(f(x_{n_k}))$ is Cauchy in $\mathbb{R}$. In particular, the sequence $(f(x_{n_k}))_k$ is bounded. This contradicts the fact that

$$\left|f(x_{n_k})\right| > n_k \geq k.$$

Hence, we conclude that f is bounded on A. □

12.3 Lipschitz Continuity

Lipschitz continuity is an additional continuity condition one can impose on a function. For the sake of completeness, let us recall what it means for a function $f : A \to \mathbb{R}$ to be Lipschitz continuous.

Definition 12.1. Let $A \subseteq \mathbb{R}$. A function $f : A \to \mathbb{R}$ is said to be *Lipschitz continuous* on A if there exists $L > 0$ such that

$$|f(x) - f(y)| \leq L\,|x - y|$$

for all $x, y \in A$. The set of all Lipschitz continuous functions $A \to \mathbb{R}$ is often denoted by $\mathrm{Lip}(A; \mathbb{R})$ or $\mathrm{Lip}(A)$.

It is automatic that Lipschitz continuous functions are uniformly continuous on their domains. Indeed, given $f \in \mathrm{Lip}(A)$ and $\varepsilon > 0$, we define $\delta := \varepsilon / L > 0$ and observe that, for all $x, y \in A$ with $|x - y| < \delta$, there holds

$$|f(x) - f(y)| \leq L\,|x - y| < L\delta = \varepsilon.$$

One can extend the concept of a Lipschitz function to that of a Hölder continuous function. We make this precise below.

Definition 12.2. Let $A \subseteq \mathbb{R}$, $f : A \to \mathbb{R}$ a function, and fix $0 < \alpha \leq 1$. We say that f is Hölder continuous on A with exponent $\alpha > 0$ if there exists $K > 0$ such that

$$|f(x) - f(y)| \leq K\,|x - y|^{\alpha}$$

for all $x, y \in A$.

Notice that 1-Hölder continuity is precisely the Lipschitz condition for f on A. Lipschitz functions (and more generally Hölder continuous functions) are useful because they are automatically *uniformly* continuous on their domains.[26] Furthermore, the Lipschitz condition (resp. Hölder condition) is often easy to establish by direct algebraic manipulations and some straightforward inequalities, making them especially practical in applications.

Example 12.4. Let $f : A \subseteq \mathbb{R} \to \mathbb{R}$ be Hölder continuous on A with exponent α. Prove that f is uniformly continuous on A.

Proof. Since f is Hölder continuous on A, there exists $K > 0$ and $\alpha \in (0, 1]$ such that

$$|f(x) - f(y)| \leq K\,|x - y|^{\alpha}, \quad \forall x, y \in A.$$

Define $\delta := (\varepsilon/K)^{1/\alpha}$ and notice that, for all $x, y \in A$ with $|x - y| < \delta$, there holds

$$|f(x) - f(y)| \leq K\,|x - y|^{\alpha} < K\delta^{\alpha} = \varepsilon.$$

Hence, f is uniformly continuous on A. □

Let us now give some examples of Lipschitz/Hölder continuous functions..

1. By the reverse triangle inequality, the function $f(x) := |x|$ is Lipschitz on $\mathbb{R}$. Indeed, note that
$$||x| - |y|| \leq |x - y|$$
for all $x, y \in \mathbb{R}$.

[26] In the case of Lipschitz continuity, much more can be said. As it turns out, Lipschitz continuity is strong enough to enforce differentiability at "most points" in the domain. For example, it is possible to prove that any Lipschitz continuous function $f : \mathbb{R} \to \mathbb{R}$ is differentiable at all x belonging to some set Z whose complement is "small" in the volumetric sense (i.e. measure). Moreover, $|f'(x)|$ is bounded by the Lipschitz constant of f. It is because of this phenomenon that Lipschitz functions (and also Hölder continuous functions) play an important role in the theory of weak derivatives and Sobolev spaces, as well as enjoying many applications in the analysis of differential equations.

2. In class (see also (12.1)) we showed that

$$\left|\sqrt{x}-\sqrt{y}\right| \leq \sqrt{|x-y|} = |x-y|^{1/2}, \quad \forall x, y \in [0,\infty).$$

Thus, we were really proving that $f(x) := \sqrt{x}$ is Hölder continuous on $[0,\infty)$ with exponent $\alpha = \frac{1}{2}$.

3. The function $f(x) = \sqrt{x}$ is *not* Lipschitz on $[0,1]$. To see this, we argue by contradiction. If it were Lipschitz on $[0,1]$, we could find $L > 0$ such that

$$\left|\sqrt{x}-\sqrt{y}\right| \leq L\,|x-y|, \quad \forall x, y \in [0,1].$$

Thus,

$$\sup_{\substack{x,y\in[0,1]\\ x\neq y}} \frac{\left|\sqrt{x}-\sqrt{y}\right|}{|x-y|} \leq L < \infty.$$

In particular,

$$\sup_{\substack{x\in[0,1]\\ x\neq 0}} \frac{\sqrt{x}}{x} = \sup_{\substack{x\in[0,1]\\ x\neq 0}} \frac{1}{\sqrt{x}} < \infty.$$

Since this is absurd, we have a contradiction.

Example 12.5. Prove that the function $f(x) := \dfrac{1}{x^2+1}$ is Lipschitz, and hence uniformly continuous, on $\mathbb{R}$.

Proof. Let $x, y \in \mathbb{R}$ be given and compute

$$\left|\frac{1}{1+x^2}-\frac{1}{1+y^2}\right| = \frac{\left|x^2-y^2\right|}{(1+x^2)(1+y^2)} = \frac{|x+y|}{(1+x^2)(1+y^2)}\,|x-y|.$$

Now, for any $x, y \in \mathbb{R}$ we have

$$\begin{aligned}\frac{|x+y|}{(1+x^2)(1+y^2)} &\leq \frac{|x|+|y|}{(1+x^2)(1+y^2)} \\ &= \frac{|x|}{(1+x^2)(1+y^2)} + \frac{|y|}{(1+x^2)(1+y^2)} \\ &< \frac{|x|}{1+x^2} + \frac{|y|}{1+y^2}.\end{aligned}$$

Now, observe that if $|x| \leq 1$ then

$$\frac{|x|}{x^2+1} \leq \frac{|x|}{1} = |x| \leq 1,$$

while if $|x| > 1$,

$$\frac{|x|}{x^2+1} < \frac{|x|}{x^2} = \frac{1}{|x|} < 1.$$

It therefore follows that

$$\frac{|x|}{x^2+1} \leq 1$$

and, similarly,

$$\frac{|y|}{y^2+1} \leq 1.$$

Returning to our previous work, we find that for all $x, y \in \mathbb{R}$

$$\left|\frac{1}{1+x^2} - \frac{1}{1+y^2}\right| < \left(\frac{|x|}{1+x^2} + \frac{|y|}{1+y^2}\right)|x-y| \leq 2\,|x-y|\,.$$

Thus, f is Lipschitz on $\mathbb{R}$. □

Example 12.6. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) := \sqrt{x^2+1}$. Show that f is Lipschitz, and hence uniformly continuous, on $\mathbb{R}$.

Proof. As before, let $x, y \in \mathbb{R}$ be arbitrary and write

$$\begin{aligned}
|f(x) - f(y)| = \left|\sqrt{x^2+1} - \sqrt{y^2+1}\right| &= \frac{\left|\sqrt{x^2+1} - \sqrt{y^2+1}\right|\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|}{\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|} \\
&= \frac{\left|x^2 - y^2\right|}{\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|} \\
&= \frac{|x+y|}{\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|}\,|x-y| \\
&\leq \frac{|x|+|y|}{\sqrt{x^2+1} + \sqrt{y^2+1}}\,|x-y| \\
&< \left(\frac{|x|}{\sqrt{x^2+1}} + \frac{|y|}{\sqrt{y^2+1}}\right)|x-y|\,.
\end{aligned}$$

Observe that for all $z \in \mathbb{R}$, we have $|z| = \sqrt{z^2} \leq \sqrt{z^2+1}$ whence

$$\frac{|z|}{\sqrt{z^2+1}} \leq 1, \quad \forall z \in \mathbb{R}.$$

Using this in the estimates we derived above, we infer that for all $x, y \in \mathbb{R}$ there holds

$$|f(x) - f(y)| < \left(\frac{|x|}{\sqrt{x^2+1}} + \frac{|y|}{\sqrt{y^2+1}} \right) |x - y| \leq (1+1)\,|x-y| = 2\,|x-y|\,.$$

□

Example 12.7. Show that $f(x) = \sin x$ is Lipschitz continuous on $\mathbb{R}$. Deduce from this that $g(x) = \cos x$ is also Lipschitz on $\mathbb{R}$. You may use, without proof, trigonometric identities and that $|\sin z| \leq |z|$ for all $z \in \mathbb{R}$.

Proof. For any $x, y \in \mathbb{R}$, the sum to angle formula states that

$$\sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right).$$

Therefore, since $|\cos z| \leq 1$ for all $z \in \mathbb{R}$, we find that

$$\begin{aligned} |\sin x - \sin y| = 2\left|\cos\left(\frac{x+y}{2}\right)\right|\left|\sin\left(\frac{x-y}{2}\right)\right| &\leq 2\left|\sin\left(\frac{x-y}{2}\right)\right| \\ &\leq 2\left|\frac{x-y}{2}\right| \\ &= |x-y|\,. \end{aligned}$$

where, in the before last step, we used that $|\sin z| \leq |z|$ for all $z \in \mathbb{R}$.

To see that $g(x) = \cos x$ is Lipschitz continuous as a consequence, first observe that $f(x + \frac{\pi}{2}) = \sin\left(x + \frac{\pi}{2}\right) = \cos(x) = g(x)$, for all $x \in \mathbb{R}$. Therefore,

$$\begin{aligned} |\cos x - \cos y| = \left|\sin\left(x + \frac{\pi}{2}\right) - \sin\left(y + \frac{\pi}{2}\right)\right| &\leq \left|\left(x + \frac{\pi}{2}\right) - \left(y + \frac{\pi}{2}\right)\right| \\ &= |x-y|\,. \end{aligned}$$

□

12.4 Uniform Continuity on Unbounded Intervals

Theorem 12.7. *Let $f : [0, \infty) \to \mathbb{R}$ be continuous and assume there exists $a > 0$ such that f is uniformly continuous on $[a, \infty)$. Then, f is uniformly continuous on $[0, \infty)$.*

Proof. Consider the compact interval $[0, 2a] \subset \mathbb{R}$, on which f is continuous. Thus, f is uniformly continuous on $[0, 2a]$. So, given $\varepsilon > 0$, there exists $\delta_1 > 0$ such that one has

$$|f(x) - f(y)| < \varepsilon, \tag{12.2}$$

whenever $x, y \in [0, 2a]$ are such that $|x - y| < \delta_1$. Similarly, because f is uniformly continuous on $[a, \infty)$, there exists corresponding $\delta_2 > 0$ such that

$$|f(x) - f(y)| < \varepsilon, \tag{12.3}$$

for all $x, y \in [a, \infty)$ satisfying $|x - y| < \delta_2$. Define $\delta := \min(\delta_1, \delta_2, a)$ and let $x, y \in [0, \infty)$ be such that $|x - y| < \delta$; we claim that $|f(x) - f(y)| < \varepsilon$. Without loss of generality, we may assume that $0 \leq y \leq x$. Then, we distinguish the following two possible cases:

(i) Case $x \leq 2a$. Then, $0 \leq y \leq x \leq 2a$ implies that $x, y \in [0, 2a]$. Since we have $|x - y| < \delta \leq \delta_1$, it follows from (12.2) that $|f(x) - f(y)| < \varepsilon$.

(ii) Case $x > 2a$. Then, using that $|x - y| < \delta \leq a$, we must have

$$-a < x - y < a$$

whence $y > x - a > 2a - a = a$. Thus, $x, y \in [a, \infty)$ and $|x - y| < \delta \leq \delta_2$ implies by (12.3) that $|f(x) - f(y)| < \varepsilon$.

We obtain that $|f(x) - f(y)| < \varepsilon$ whenever $x, y \in [0, \infty)$ satisfies $|x - y| < \delta$. □

Example 12.8. Prove that $f(x) = \sqrt{x}$ is Lipschitz continuous on $[1, \infty)$. Deduce from this, using the theorem above, that f is uniformly continuous on $[0, \infty)$.

Solution. We already know from a class example that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$, but this offers an alternative method of demonstrating this.

First, to see that f is Lipschitz on $[1, \infty)$, let $x, y \geq 1$ and write

$$\begin{aligned} |f(x) - f(y)| = \left|\sqrt{x} - \sqrt{y}\right| &= \left|\sqrt{x} - \sqrt{y}\right| \frac{\left|\sqrt{x} + \sqrt{y}\right|}{\left|\sqrt{x} + \sqrt{y}\right|} \\ &= \frac{|x - y|}{\left|\sqrt{x} + \sqrt{y}\right|} \\ &= \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \\ &\leq \frac{|x - y|}{\sqrt{1} + \sqrt{1}} \\ &= \frac{|x - y|}{2}. \end{aligned}$$

Thus, f is Lipschitz on $[1, \infty)$, and hence uniformly continuous there. Furthermore, $f(x) = \sqrt{x}$ is continuous on $[0, \infty)$. Thus, it follows from the last theorem (with $a = 1$) that f is uniformly continuous on all of $\mathbb{R}$. □

13 Review Tutorial

For the purposes of review, we include below a multitude of solved exercises covering much of the material from the course. Given that the focus of the final exam will be weighted in favour of the post-midterm material, we focus mainly on problems involving topology, properly divergent sequences, applications of continuity, and uniform continuity. We should point out that some of the problems here are taken from the ungraded assignment problems, and tutorial exercises we did not have the time to cover.

In addition, despite our best efforts, no set of review problems can perfectly exhaust all concepts covered in class. Therefore, I strongly recommend you review all class definitions, major theorems, and solve the ungraded assignment problems.

Example 13.1. Let (x_n) be a convergent sequence of real numbers with limit x. Let $f : \mathbb{N} \to \mathbb{N}$ be a bijection and define a new sequence (y_n) by the rule $y_n := x_{f(n)}$. Show that (y_n) is convergent and determine its limit.

Proof. We claim that (y_n) is convergent and $\lim y_n = x$. To this end, let $\varepsilon > 0$ be given. Using that $x_n \to x$, as $n \to \infty$, we see that there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N. \tag{13.1}$$

Now, since f is a bijection, the pre-image of any natural number under f is a finite set (indeed, because f is bijective, $f^{-1}(\{n\})$ consists of exactly one point). Consequently,

$$f^{-1}\left(\{1, \ldots, N-1\}\right) = \bigcup_{n=1}^{N-1} f^{-1}(\{n\})$$

is a finite subset of $\mathbb{N}$, since it is the finite union of finite sets. Thus, there exists[27] $M \in \mathbb{N}$ with the property that

$$n < M, \quad \forall n \in f^{-1}\left(\{1, \ldots, N-1\}\right) \subseteq \mathbb{N}. \tag{13.2}$$

Let now $n \geq M$ and observe that, by virtue of (13.2), we have $n \notin f^{-1}\left(\{1, \ldots, N-1\}\right)$. In other words, given $n \geq M$, there holds $f(n) \notin \{1, \ldots, N-1\}$, i.e. $f(n) \geq N$. By an application of (13.1), it follows that

$$|y_n - x| = \left|x_{f(n)} - x\right| < \varepsilon,$$

for all $n \geq M$. □

[27] If $f^{-1}\left(\{1, \ldots, N-1\}\right)$ is empty, then any $M \in \mathbb{N}$ will do. If $f^{-1}\left(\{1, \ldots, N-1\}\right)$ is non-empty, then it is finite and must admit a maximal element m; defining $M := m+1 \in \mathbb{N}$ then yields the required quantity.

Exercise 13.1. Examining the last example, it is reasonable to ask which assumptions on f are truly necessary for the claim to hold. Namely:

(a) If $f : \mathbb{N} \to \mathbb{N}$ is only assumed to be injective, then does the conclusion of Example 13.1 continue to hold?

(b) What about if we *only* require that $f : \mathbb{N} \to \mathbb{N}$ be such that $f^{-1}(\{n\})$ is finite for all $n \in \mathbb{N}$?

Hint: Examine the part of the proof of Example 13.1 where we use the bijectivity of f.

Example 13.2. Let $\mathcal{B}$ denote the set of all binary sequences, i.e. all sequences (a_n) such that $a_n \in \{0, 1\}$ for all $n \in \mathbb{N}$. Prove that $\mathcal{B}$ is uncountable.

Proof. We argue by contradiction. Suppose that $\mathcal{B}$ is countable, then we can enumerate the elements of $\mathcal{B}$:

$$\begin{aligned}
(a_n^{(1)}) &= a_1^{(1)}, a_2^{(1)}, a_3^{(1)} \dots \\
(a_n^{(2)}) &= a_1^{(2)}, a_2^{(2)}, a_3^{(2)} \dots \\
&\vdots \\
(a_n^{(m)}) &= a_1^{(m)}, a_2^{(m)}, a_3^{(m)} \dots \\
&\vdots
\end{aligned}$$

Here, each $(a_n^{(m)})$ represents a sequence in $\mathcal{B}$. To derive a contradiction, we will construct an element in $\mathcal{B}$ (i.e. a binary sequence) that does not belong to the list above. For a given $n \in \mathbb{N}$, let b_n be given by the following formula:

$$b_n := \begin{cases} 1 & \text{if } a_n^{(n)} = 0, \\ 0 & \text{if } a_n^{(n)} = 1. \end{cases}$$

This defines a sequence (b_n) in $\mathcal{B}$. We claim that (b_n) is not equal to any $(a_n^{(m)})$ appearing in the list above. If $(b_n) = (a_n^{(m)})$ for some $m \geq 1$, then $b_m = a_m^{(m)}$ for this m. But this would directly contradict the construction of (b_n). Hence, $(b_n) \notin \mathcal{B}$ which contradicts our assumption that $\mathcal{B}$ was countable. □

Example 13.3. Let $S \subseteq \mathbb{R}$ be non-empty and bounded from above. Let $s := \sup S$ and show that there exists an increasing sequence (x_n) with $x_n \in S$ for all $n \in \mathbb{N}$ and $\lim x_n = s$.

Proof. First, if $s \in S$, then we may define $x_n := s$ for all $n \in \mathbb{N}$ which is obviously an increasing (non-strict!) sequence converging to s, as $n \to \infty$. Thus, we need only handle the case $s \notin S$. Then, since $s - 1$ is not an upper bound of S, there exists $x_1 \in S$ such that

$$s - 1 < x_1 < s.$$

Notice that $x_1 < s$ because $x_1 \leq s$ and $x_1 \in S$ whilst $s \notin S$. Now, having chosen x_n, we define x_{n+1} as follows. Clearly, $\max\left(s - \frac{1}{n+1}, x_n\right) < s$ and thus $\max\left(s - \frac{1}{n+1}, x_n\right)$ is not an upper bound of S. Consequently, there exists $x_{n+1} \in S$ such that

$$\max\left(s - \frac{1}{n+1}, x_n\right) < x_{n+1} < s,$$

where again we have used that $x_{n+1} \in S$ and $s \notin S$ to gain this last strict inequality. Proceeding inductively, we obtain a sequence (x_n) in S satisfying

$$s - \frac{1}{n} < x_n < s < s + \frac{1}{n}, \quad \forall n \in \mathbb{N}.$$

Thus, $|x_n - s| < \frac{1}{n}$ for all $n \in \mathbb{N}$ whence $x_n \to s$ by the null sequence criterion. Furthermore, by construction, $x_{n+1} > x_n$ for all $n \in \mathbb{N}$. □

Example 13.4. Prove from the definition that $\mathring{E} = \operatorname{int}(E)$ is open, for arbitrary $E \subseteq \mathbb{R}$.

Solution. We show $\mathring{E}$ is open. To this end, let $x \in \mathring{E} = E \setminus \partial E$ be given and notice that $x \notin \partial E$ implies the existence of $\varepsilon > 0$ having the property that either $V_\varepsilon(x) \cap E = \varnothing$, or $V_\varepsilon(x) \cap E^{\mathsf{c}} = \varnothing$. Since $V_\varepsilon(x) \cap E$ is non-empty (this intersection contains x), we must be in the case $V_\varepsilon(x) \cap E^{\mathsf{c}} = \varnothing$, i.e. $V_\varepsilon(x) \subseteq E$. Moreover, $V_\varepsilon(x) \subseteq \mathring{E}$ since every $y \in V_\varepsilon(x)$ is *not* a boundary point of E. To see this, notice that since $V_\varepsilon(x) \ni y$ is open, there exists $\delta > 0$ such that $V_\delta(y) \subseteq V_\varepsilon(x) \subseteq E$ whence $V_\delta(y) \cap E^{\mathsf{c}} = \varnothing$ and $y \notin \partial E$. We conclude that $V_\varepsilon(x) \subseteq \mathring{E}$ which concludes our claim. □

Example 13.5. Show that if $E \subseteq \mathbb{R}$ is open, then $E \cap \partial E = \varnothing$. In contrast, if $E \subseteq \mathbb{R}$ is closed, then, $\partial E \subseteq E$.

Proof. First suppose E is open and assume, by way of contradiction, that $E \cap \partial E$ contains a point x. Since E is open and $x \in E$, there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq E$. On the other hand, $x \in \partial E$ means, by definition, that $V_\varepsilon(x) \cap E \neq \varnothing$ *and* $V_\varepsilon(x) \cap E^{\mathsf{c}} \neq \varnothing$ for *all* $\varepsilon > 0$. This is impossible since, for the $\varepsilon > 0$ we found previously, we have $V_\varepsilon(x) \subseteq E$ which implies $V_\varepsilon(x) \cap E^{\mathsf{c}} = \varnothing$.

Next, suppose E is closed and let $x \in \partial E$ be arbitrary. If $x \in E^{\mathsf{c}}$, where E^{c} is open (E is closed) then there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq E^{\mathsf{c}}$, i.e. $V_\varepsilon(x) \cap E = \varnothing$. However, since $x \in \partial E$, we must have that $V_\varepsilon(x) \cap E \neq \varnothing$. Consequently, we have $x \in E$ and we infer that $\partial E \subseteq E$. □

Example 13.6. Let $E \subseteq \mathbb{R}$ be bounded, i.e. there exists $M \in \mathbb{N}$ such that $|x| \leq M$ for all $x \in E$. Show that $\overline{E}$ is compact.

Proof. Since $\overline{E}$ is closed, it is enough to show that $\overline{E}$ is bounded. Indeed, since E is bounded, there exists $M > 0$ such that $|x| \leq M$ for all $x \in E$. In other words, we have the inclusion $E \subseteq [-M, M]$. Now, recall that

$$\overline{E} = \bigcap_{\substack{F \supseteq E \\ F \text{ closed}}} F.$$

Using that $[-M, M]$ is a closed set containing E, it follows that $\overline{E} \subseteq [-M, M]$ whence $\overline{E}$ is bounded. It follows that $\overline{E}$ is compact. □

Example 13.7. Let $S \subseteq \mathbb{R}$. Then, $x \in \overline{S}$ if and only if for all $\varepsilon > 0$ we have $V_\varepsilon(x) \cap S \neq \varnothing$.

Proof. First, if $x \in \overline{S} = S \cup \partial S$, then either $x \in S$ or $x \in \partial S$. In the first case, clearly $x \in V_\varepsilon(x) \cap S$ for all $\varepsilon > 0$. In the latter case, for all $\varepsilon > 0$ we have

$$V_\varepsilon(x) \cap S \neq \varnothing \quad \text{and} \quad V_\varepsilon(x) \cap S^{\mathsf{c}} \neq \varnothing.$$

In both cases, we have $V_\varepsilon(x) \cap S \neq \varnothing$ for all $\varepsilon > 0$.

Conversely, suppose $x \notin \overline{S} = S \cup \partial S$. Then, $x \in S^{\mathsf{c}}$ and there exists $\varepsilon > 0$ such that either

$$V_\varepsilon(x) \cap S = \varnothing \quad \text{or} \quad V_\varepsilon(x) \cap S^{\mathsf{c}} = \varnothing.$$

Since $x \in S^{\mathsf{c}}$, we must be in the case $V_\varepsilon(x) \cap S = \varnothing$, which concludes the proof. □

Example 13.8. Let $S \subseteq \mathbb{R}$. Then $x \in \overline{S}$ if and only if there exists a sequence (x_n) in S such that $\lim x_n = x$.

Proof. First, let $x \in \overline{S}$ and observe that, by the last example, $V_{1/n}(x) \cap S \neq \varnothing$ for each $n \in \mathbb{N}$. Thus, given $n \in \mathbb{N}$, we may choose some x_n belonging to this intersection. Then, (x_n) is a sequence in S satisfying $x_n \in V_{1/n}(x)$ for all $n \in \mathbb{N}$, i.e.

$$|x_n - x| < \frac{1}{n}$$

whence $\lim x_n = x$ by the null sequence criterion.

Conversely, suppose there exists a sequence (x_n) living in S which converges to x. Since $x_n \in S \subseteq \overline{S}$, where $\overline{S}$ is closed in $\mathbb{R}$, it follows from Theorem 9.2 (closed sets contain the limits of their convergent sequences) that $x \in \overline{S}$. □

Example 13.9. If $S \subseteq \mathbb{R}$ is open or closed, then ∂S has empty interior.

Proof. Let us first observe the case where S is open. Arguing by contradiction, suppose there exists a point $x \in \text{int}(\partial S) \subseteq \partial S$. Then, since $\text{int}(\partial S)$ is open, there exists $\delta > 0$ such that $V_\delta(x) \subseteq \text{int}(\partial S) \subseteq \partial S$. In particular, $x \in \partial S$ so that $V_\delta(x) \cap S \neq \varnothing$ and there exists $y \in V_\delta(x) \cap S$. Now, $y \in V_\delta(x)$ implies that $y \in \partial S$. On the other hand, since S is open, there exists $\varepsilon > 0$ such that $V_\varepsilon(y) \subseteq S$, which contradicts $y \in \partial S$. We conclude that ∂S has non-empty interior if S is open.

It remains to handle the case where S is closed. Here, we appeal to the symmetry of the boundary with respect to complements. That is, we shall utilize the fact that

$$\begin{aligned}\partial S &= \{x \in \mathbb{R} : \forall \varepsilon > 0,\ V_\varepsilon(x) \cap S \neq \varnothing \text{ and } V_\varepsilon(x) \cap S^{\mathsf{c}} \neq \varnothing\} \\ &= \{x \in \mathbb{R} : \forall \varepsilon > 0,\ V_\varepsilon(x) \cap S^{\mathsf{c}} \neq \varnothing \text{ and } V_\varepsilon(x) \cap (S^{\mathsf{c}})^{\mathsf{c}} \neq \varnothing\} \\ &= \partial S^{\mathsf{c}}.\end{aligned}$$

If S is closed, then S^{c} is open so that $\text{int}(\partial S^{\mathsf{c}})$ is empty by the first part. This implies that $\text{int}(\partial S) = \text{int}\,(\partial S^{\mathsf{c}})$ is empty and we are done. □

Example 13.10. Let $S \subseteq \mathbb{R}$. Then, $\partial\,(\partial S) = \partial S$ if and only if ∂S has empty interior.

Proof. Assume $\partial(\partial S) = \partial S$. By your assignment 7 problem 3, ∂S is closed. Therefore, by the last example, $\partial(\partial S)$ has empty interior whence ∂S has empty interior as well.

Conversely, suppose that ∂S has empty interior, i.e.

$$\varnothing = \text{int}(\partial S) = \partial S \setminus \partial\,(\partial S)\,.$$

Since ∂S is closed, we must have $\partial(\partial S) \subseteq \partial S$ and therefore it follows from the above that $\partial(\partial S) = \partial S$. □

Remark 13.1. There are sets (neither open nor closed) whose boundary has non-empty interior. One example is $\mathbb{Q}$. Indeed, since $\mathbb{Q}$ and $\mathbb{Q}^{\mathsf{c}}$ are both dense in $\mathbb{R}$, it follows that, for every $x \in \mathbb{R}$,

$$V_\varepsilon(x) \cap \mathbb{Q} \neq \varnothing \text{ and } V_\varepsilon(x) \cap \mathbb{Q}^{\mathsf{c}} \neq \varnothing, \quad \forall \varepsilon > 0.$$

In other words, every $x \in \mathbb{R}$ is a boundary point of $\mathbb{Q}$, i.e. $\partial\mathbb{Q} = \mathbb{R}$ which is open and has interior $\text{int}(\mathbb{R}) = \mathbb{R}$.

Example 13.11. Let $p > 0$ be a real number and define $x_1 := \sqrt{p}$. Following this, set $x_{n+1} := \sqrt{p + x_n}$. Show that (x_n) is convergent and determine its limit. (Hint: show that $x_n \leq 1 + 2\sqrt{p}$ for all $n \in \mathbb{N}$).

Proof. The hint suggests we establish an upper bound for (x_n), and thus it seems a good idea to attempt an application of the monotone convergence theorem. In fact, since the hint suggests an upper bound, it is reasonable to suspect the sequence will be increasing.

To this end, let us verify that (x_n) is (strictly) increasing, i.e. $x_{n+1} > x_n$ for all $n \in \mathbb{N}$. We establish this fact by induction. For our base case ($n = 1$), we simply observe that

$$x_2 = \sqrt{p + x_1} = \sqrt{p + \sqrt{p}} > \sqrt{p} = x_1$$

because $\sqrt{x}$ is strictly increasing on $[0, \infty)$. For our inductive step, assume that $x_{n+1} > x_n$ for some $n \in \mathbb{N}$. Then,

$$x_{n+2} = \sqrt{p + x_{n+1}} > \sqrt{p + x_n} = x_{n+1}.$$

Thus, (x_n) is (strictly) increasing. To prove the upper bound provided in the hint, we also argue by induction. Again, the base case $n = 1$ is given:

$$x_1 = \sqrt{p} < 1 + 2\sqrt{p}.$$

Assuming now that $x_n < 1 + 2\sqrt{p}$ for some $n \in \mathbb{N}$, it follows that

$$\begin{aligned} x_{n+1} = \sqrt{p + x_n} < \sqrt{p + 1 + 2\sqrt{p}} &= \sqrt{p + 2\sqrt{p} + 1} \\ &= \sqrt{(\sqrt{p} + 1)^2} \\ &= \sqrt{p} + 1 \\ &< 1 + 2\sqrt{p}. \end{aligned}$$

Thus, by induction, (x_n) is bounded above. Consequently, the monotone convergence theorem asserts that the sequence (x_n) is convergent with limit $x \geq x_1 > 0$. As a subsequence of (x_n), we must have that $\lim x_{n+1} = x$ as well. Furthermore, if $x_n \to x$ then $x_n + p \to x + p$ and $\sqrt{x_n + p} \to \sqrt{x + p}$. Therefore,

$$x = \lim x_{n+1} = \lim \sqrt{p + x_n} = \sqrt{p + x}.$$

Thus, $x^2 = p + x$ which implies that

$$x = \frac{1 \pm \sqrt{1 + 4p}}{2}.$$

Since $x \geq 0$, we conclude that

$$x = \frac{1 + \sqrt{1 + 4p}}{2}.$$

□

Example 13.12. Let (x_n) and (y_n) be bounded sequences.

(a) Prove that $\limsup (x_n + y_n) \leq \limsup x_n + \limsup y_n$.

(b) Give an example of two sequences (x_n) and (y_n) such that this inequality is strict.

(c) Show that $\liminf (x_n + y_n) \leq \liminf x_n + \limsup y_n$.

Proof.

(a) Let $n \in \mathbb{N}$ be arbitrary. Then, for all $k \geq n$,

$$x_k \leq \sup \left\{x_j : j \geq n\right\} \quad \text{and} \quad y_k \leq \sup \left\{x_j : j \geq n\right\}.$$

Consequently,

$$x_k + y_k \leq \sup \left\{x_j : j \geq n\right\} + \sup \left\{y_j : j \geq n\right\}.$$

Since $k \geq n$ is arbitrary, it follows from the above that

$$\sup \left\{x_k + y_k : k \geq n\right\} \leq \sup \left\{x_j : j \geq n\right\} + \sup \left\{y_j : j \geq n\right\}.$$

Taking limits on both sides, we deduce from the limit laws that

$$\begin{aligned}
\limsup (x_n + y_n) &= \lim_{n\to\infty} \left(\sup_{k\geq n} \{x_k + y_k\}\right) \leq \lim_{n\to\infty} \left(\sup_{j\geq n} \left\{x_j\right\} + \sup_{j\geq n} \left\{y_j\right\}\right) \\
&= \lim_{n\to\infty} \sup_{j\geq n} x_j + \lim_{n\to\infty} \sup_{j\geq n} y_j \\
&= \limsup x_n + \limsup y_n.
\end{aligned}$$

(b) Define $x_n := (-1)^n$ and $y_n := (-1)^{n+1}$. Then, it is easy to see that $\limsup x_n = 1$ and $\limsup y_n = 1$. However,

$$x_n + y_n = (-1)^n + (-1)^{n+1} = (-1)^n - (-1)^n = 0$$

so $\limsup(x_n + y_n) = \lim(x_n + y_n) = \lim 0 = 0 < \limsup x_n + \limsup y_n = 2$.

(c) We argue directly via the accumulation point definition (distinct from the MyCourses assignment solutions). Since $\liminf y_n$ is the minimum accumulation point of (x_n), there exists a subsequence (x_{n_k}) of (x_n) converging to $\liminf x_n$, i.e. such that $\lim x_{n_k} = \liminf x_n$. Then, (y_{n_k}) is a subsequence of (y_n) and is therefore bounded; an application of Bolzano-Weierstrass yields a further subsequence $(y_{n_{k_j}})$ of (y_{n_k}) converging to some y. Notice that y is an accumulation point of (y_n) (since $(y_{n_{k_j}})$ is a subsequence of (y_n) too) and so $y \leq \limsup y_n$. Now, $(x_{n_{k_j}})$ is a subsequence of the *convergent* subsequence (x_{n_k}), and therefore

$$\lim x_{n_{k_j}} = \lim x_{n_k} = \liminf x_n.$$

Finally, it follows that $(x_{n_{k_j}}+y_{n_{k_j}})$ is a convergent subsequence of (x_n+y_n) whence

$$\begin{aligned}\liminf(x_n+y_n) \leq \lim\left(x_{n_{k_j}}+y_{n_{k_j}}\right) &= \lim x_{n_{k_j}} + \lim y_{n_{k_j}} \\ &= \liminf x_n + y \\ &\leq \liminf x_n + \limsup y_n.\end{aligned}$$

□

Example 13.13. Let (x_n) be a bounded sequence.

(a) Prove that (x_n) converges if and only if (x_n) has a single accumulation point.

(b) Deduce from this that (x_n) converges if and only if

$$\limsup x_n = \liminf x_n.$$

Proof.

(a) Suppose first (x_n) is convergent. Then, every subsequence of (x_n) must converge to $\lim x_n$. Since every accumulation point is the limit of some subsequence of (x_n), it follows that $\lim x_n$ is the only accumulation point of (x_n).

Conversely, suppose that (x_n) has a single accumulation point, which we denote by x. Arguing by contradiction, suppose that (x_n) does *not* converge to x. Then, there exists $\varepsilon_0 > 0$ such that, for all $N \in \mathbb{N}$, there exists $n \geq N$ with the property that $|x_n - x| \geq \varepsilon_0 > 0$. In other words, there are infinitely many terms x_n of the sequence such that $|x_n - x| \geq \varepsilon_0$. Thus, there exists a subsequence (x_{n_k}) of (x_n) having the property that

$$\left|x_{n_k} - x\right| \geq \varepsilon_0 > 0, \quad \forall k \in \mathbb{N}.$$

Since (x_{n_k}) is itself bounded, it has a convergent subsequence $(x_{n_{k_j}})$ converging to some real number. Because $(x_{n_{k_j}})$ is also a subsequence of (x_n), which has only x as an accumulation point, we must have $\lim x_{n_{k_j}} = x$. However, being a subsequence of (x_{n_k}),

$$\left|x_{n_{k_j}} - x\right| \geq \varepsilon_0 > 0, \quad \forall j \in \mathbb{N}.$$

However, this contradicts $\lim x_{n_{k_j}} = x$. It follows that (x_n) converges to x, as $n \to \infty$.

(b) By (a), (x_n) converges if and only if (x_n) has a single accumulation point. However, (x_n) has a single accumulation point if and only if

$$\liminf x_n = \limsup x_n,$$

since $\liminf x_n \leq \limsup x_n$ are themselves accumulation points of (x_n).

□

Example 13.14. Let (x_n) be a sequence of positive real numbers. Show that $\lim x_n = 0$ if and only if $\lim \frac{1}{x_n} = \infty$.

Proof. Suppose first $x_n \to 0$, as $n \to \infty$. Let $M \in \mathbb{R}$ be arbitrary. If $M \leq 0$, then $\frac{1}{x_n} > 0 \geq M$ for all $n \geq 1$. If $M > 0$, using that $\lim x_n = 0$, we find $N \in \mathbb{N}$ such that

$$x_n = |x_n - 0| < \frac{1}{M}, \quad \forall n \geq N.$$

Then,

$$\frac{1}{x_n} > M, \quad \forall n \geq N.$$

Thus, we have proven that for each $M \in \mathbb{R}$ there exists $N \in \mathbb{N}$ such that $\frac{1}{x_n} > M$ for all $n \geq N$. That is, $\lim \frac{1}{x_n} = \infty$.

Conversely, assume $\lim \frac{1}{x_n} = \infty$ and let $\varepsilon > 0$ be given. There exists $N \in \mathbb{N}$ such that

$$\frac{1}{x_n} > \frac{1}{\varepsilon}, \quad \forall n \geq N.$$

That is, $|x_n - 0| = x_n < \varepsilon$ for all $n \geq N$. It follows that $\lim x_n = 0$. □

Example 13.15. Let (x_n) and (y_n) be positive sequences such that

$$\lim \frac{x_n}{y_n} = 0.$$

(a) If $\lim x_n = \infty$, then $\lim y_n = \infty$.

(b) If (y_n) is bounded, then $\lim x_n = 0$.

Proof.

(a) Let $M \in \mathbb{R}$ be arbitrary, and choose $N_1 \in \mathbb{N}$ such that $x_n > M$ for all $n \geq N_1$. Using that $\lim x_n/y_n = 0$, there exists $N_2 \in \mathbb{N}$ with the property that

$$\frac{x_n}{y_n} = \left| \frac{x_n}{y_n} - 0 \right| < 1, \quad \forall n \geq N_2.$$

Especially, $x_n < y_n$ for all $n \geq N_2$. Setting $N := \max(N_1, N_2) \in \mathbb{N}$, it follows that, for all $n \geq N$,

$$M < x_n < y_n.$$

Since $M \in \mathbb{R}$ was arbitrary, it follows that $\lim y_n = \infty$.

(b) Since (y_n) is bounded, there exists $C > 0$ such that $y_n \leq C$, for all $n \in \mathbb{N}$. Thus, given any $n \in \mathbb{N}$, we find that

$$0 < x_n = \frac{x_n}{y_n} y_n \leq C \frac{x_n}{y_n}.$$

Since $\lim \frac{x_n}{y_n} = 0$, an appeal to the squeeze theorem implies that $\lim x_n = 0$.

□

Example 13.16. We show that

$$\lim_{x \to a} \frac{1}{x^2} = \frac{1}{a^2}$$

for all $a \neq 0$. Obviously, $f(x) = \frac{1}{x^2}$ is considered as a function with domain $\mathbb{R} \setminus \{0\}$. Let now $\varepsilon > 0$ be given. If $\delta > 0$ and $x \neq 0$ satisfies $0 < |x - a| < \delta$, then we have the estimate

$$\left| \frac{1}{x^2} - \frac{1}{a^2} \right| = \frac{\left| x^2 - a^2 \right|}{x^2 a^2} = \frac{|x - a| \, |x + a|}{x^2 a^2} < \delta \frac{|x| + |a|}{x^2 a^2}.$$

If in addition $\delta \leq |a|$, then

$$|x| \leq |x - a| + |a| < \delta + |a| \leq 2 \, |a| \, .$$

Hence,

$$\left| \frac{1}{x^2} - \frac{1}{a^2} \right| < \delta \frac{|x| + |a|}{x^2 a^2} \leq \frac{3\delta \, |a|}{x^2 a^2} = \frac{3\delta}{x^2 \, |a|}.$$

If we assume further that $\delta \leq \frac{|a|}{2}$, then

$$||x| - |a|| \leq |x - a| < \delta$$

implies $|a| - \delta < |x| < |a| + \delta$ whence $|x| > \frac{|a|}{2}$. Rather,

$$\frac{1}{x^2} < \frac{4}{a^2}$$

so that

$$\left| \frac{1}{x^2} - \frac{1}{a^2} \right| < \frac{3\delta}{x^2 \, |a|} \leq \frac{12\delta}{|a|^3}.$$

Finally, if

$$\delta \leq \frac{\varepsilon |a|^3}{12},$$

then

$$\left|\frac{1}{x^2} - \frac{1}{a^2}\right| < \frac{12\delta}{|a|^3} \leq \varepsilon.$$

Therefore, we define

$$\delta := \min\left\{\frac{|a|}{2}, \frac{\varepsilon |a|^3}{12}\right\}.$$

By our earlier calculations, for every $x \neq 0$ with $0 < |x - a| < \delta$, one has

$$\left|\frac{1}{x^2} - \frac{1}{a^2}\right| < \varepsilon.$$

This completes the proof.

Example 13.17. Given $a \neq -1$, prove using the $\varepsilon - \delta$ definition that

$$\lim_{x \to a} \frac{x}{1+x} = \frac{a}{1+a}.$$

Here, we are viewing $\frac{x}{1+x}$ as a function $\mathbb{R} \setminus \{-1\} \to \mathbb{R}$.

Proof. Let $\varepsilon > 0$ be given and fix $a \neq -1$. Henceforth, we shall always assume that $x \neq -1$. If for some $\delta > 0$ one has $0 < |x - a| < \delta$, then

$$\left|\frac{x}{1+x} - \frac{a}{1+a}\right| = \frac{|x-a|}{|1+x|\,|1+a|} < \frac{\delta}{|1+x|\,|1+a|}.$$

Next, we have to "rid ourselves" of the $|1+x|$ term. To this end, suppose additionally that

$$\delta \leq \frac{|1+a|}{2}.$$

Then, by the reverse triangle inequality,

$$|1+x| \geq |1+a| - |x-a| > |1+a| - \delta \geq \frac{|1+a|}{2} > 0.$$

Therefore,

$$\left|\frac{x}{1+x} - \frac{a}{1+a}\right| < \frac{\delta}{|1+x|\,|1+a|} \leq \frac{2\delta}{(1+a)^2}.$$

Thus, if additionally

$$\delta \leq \varepsilon \frac{(1+a)^2}{2}$$

then

$$\left|\frac{x}{1+x}-\frac{a}{1+a}\right|<\frac{2\delta}{(1+a)^2}\leq\varepsilon.$$

Finally, we define

$$\delta:=\min\left\{\frac{|1+a|}{2},\varepsilon\frac{(1+a)^2}{2}\right\}.$$

By the calculations we carried out above, we conclude that

$$\left|\frac{x}{1+x}-\frac{a}{1+a}\right|<\varepsilon$$

for all $x\neq -1$ with $0<|x-a|<\delta$. □

Example 13.18. Use the sequential criterion for the limit to prove that

$$\lim_{x\to -1}\frac{x}{1+x}$$

does not exist.

Proof. Indeed, consider the sequence (x_n) in $\mathbb{R}$ defined by

$$x_n:=-1-\frac{1}{n}.$$

Clearly, $x_n\neq -1$ for all $n\in\mathbb{N}$ and $\lim x_n=-1$. Furthermore, for each $n\in\mathbb{N}$ we calculate:

$$f(x_n)=\frac{x_n}{1+x_n}=\frac{-1-\frac{1}{n}}{-\frac{1}{n}}=n\left(1+\frac{1}{n}\right)\geq n.$$

Hence, we get that $\lim f(x_n)$ does not exist in $\mathbb{R}$. This shows that $\lim\limits_{x\to -1}\dfrac{x}{1+x}$ does not exist either. □

Example 13.19. Let $f:\mathbb{R}\to\mathbb{R}$ be an additive function, i.e.

$$f(x+y)=f(x)+f(y),\quad \forall x,y\in\mathbb{R}.$$

Prove that if f is continuous at some $x_0\in\mathbb{R}$, then f is uniformly continuous on $\mathbb{R}$.

Proof. First, observe that $f(-x)=f(x)$ for all $x\in\mathbb{R}$. Indeed,

$$f(0)=f(0+0)=f(0)+f(0)=2f(0)\implies f(0)=0.$$

Then, $0=f(x-x)=f(x)+f(-x)$ forces $f(-x)=-f(x)$.

Let now $\varepsilon > 0$ be given and choose $\delta > 0$ such that $|z - x_0| < \delta$ implies

$$|f(z) - f(x_0)| < \varepsilon. \tag{13.3}$$

Let now $x, y \in \mathbb{R}$ satisfy $|x - y| < \delta$. Then,

$$\begin{aligned} f(x) = f(x + x_0 - y - x_0 + y) &= f(x + x_0 - y - x_0) + f(y) \\ &= f(x + x_0 - y) - f(x_0) + f(y) \end{aligned}$$

implies

$$|f(x) - f(y)| = |f(x + x_0 - y) - f(x_0)|.$$

However, $|(x + x_0 - y) - x_0| = |x - y| < \delta$ whence

$$|f(x) - f(y)| = |f(x + x_0 - y) - f(x_0)| < \varepsilon$$

by applying (13.3) with $z = x + x_0 - y$. □

Example 13.20. Let $K \subseteq \mathbb{R}$ be compact and let $f : K \to K$ be a continuous function such that

$$|f(x) - f(y)| < |x - y|, \quad \forall x, y \in K \text{ with } x \neq y.$$

Prove that there exists a unique point $c \in K$ such that $f(c) = c$.

Proof. First, observe that any $c \in K$ satisfying the assertion must be unique. Indeed, let $c, d \in K$ satisfy $f(c) = c$ and $f(d) = d$. If $c \neq d$, then

$$|c - d| = |f(c) - f(d)| < |c - d|$$

which is absurd. Thus, $c = d$.

Now, we show that c exists. Consider the function

$$g(x) := |f(x) - x| \geq 0$$

on K. By composition, g is continuous and therefore achieves a global minimum at some point $c \in K$. If $f(c) \neq c$, then

$$g(f(c)) = |f(f(c)) - f(c)| < |f(c) - c|,$$

contradicting the fact that c is a minimum of g on K. □

Example 13.21. Let $A \subseteq \mathbb{R}$ be a bounded set and let $f : A \to \mathbb{R}$ be uniformly continuous on A. Show that f is bounded on A, i.e. there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in A$.

Proof. We argue by contradiction. If f is not bounded, then, for all $M > 0$, there exists some $x \in A$ with $|f(x)| > M$. In symbolic terms,

$$(\forall M > 0)(\exists x \in A \text{ such that } |f(x)| > M).$$

In particular, for each $n \in \mathbb{N}$, there exists $x_n \in A$ with the property that $|f(x_n)| > n$. Now, since A is bounded and (x_n) is a sequence in A, there exists a convergent subsequence (x_{n_k}) of (x_n). Equivalently, (x_n) has a Cauchy subsequence (x_{n_k}). Then, using that uniformly continuous functions preserve Cauchy sequences, we infer that $(f(x_{n_k}))$ is a bounded sequence. However, since (x_{n_k}) is a subsequence of our original sequence (x_n), we have

$$\left|f(x_{n_k})\right| > n_k \geq k, \quad \forall k \in \mathbb{N}$$

which means that $(f(x_{n_k}))$ is unbounded – a contradiction. □

Remark 13.2. If f is merely assumed to be continuous, this conclusion may not necessarily hold. Indeed, let $A = (0, 1)$ and $f : (0, 1) \to \mathbb{R}$ given by $f(x) := \frac{1}{x}$. Then, f is continuous on all of $(0, 1)$, but f is not bounded on $(0, 1)$.

Example 13.22. Prove that $f(x) = x^2$ is uniformly continuous on every compact subset of $\mathbb{R}$, but is not uniformly continuous on $\mathbb{R}$.

Proof. Recall from class that any continuous function, defined on a compact set, must be uniformly continuous there. Hence, since $f(x) = x^2$ is continuous on $\mathbb{R}$, it must be uniformly continuous on each compact subset of $\mathbb{R}$. Thus, the only non-trivial claim is that f is not uniformly continuous on all of $\mathbb{R}$. To see this, we argue by contradiction. If f is uniformly continuous on $\mathbb{R}$, then for $\varepsilon := 1$ we can find $\delta > 0$ such that

$$\left|x^2 - y^2\right| < 1$$

whenever $|x - y| < \delta$. Given $x \in \mathbb{R}$ we consider the point

$$y := x + \frac{\delta}{2}.$$

It is clear that $|x - y| < \delta$, and we must therefore have

$$\left|x^2 - \left(x + \frac{\delta}{2}\right)^2\right| < 1.$$

However, the expression above reduces to the following:

$$\left|x\delta + \frac{\delta^2}{4}\right| < 1.$$

Since the function $x \mapsto x\delta + \frac{\delta^2}{4}$ is unbounded on $\mathbb{R}$, this is a contradiction. □

Second Proof. We show that $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$ by means of the sequential criterion for non-uniform continuity. Indeed, let (x_n) and (y_n) be the sequences

$$x_n := n + \frac{1}{n}, \quad y_n := n.$$

Then,

$$x_n - y_n = \frac{1}{n} \xrightarrow{n\to\infty} 0\,.$$

However,

$$|f(x_n) - f(y_n)| = \left|x_n^2 - y_n^2\right| = \left|n^2 + 2 + \frac{1}{n^2} - n^2\right| = 2 + \frac{1}{n^2} > 2$$

for all $n \in \mathbb{N}$. □

Example 13.23. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) := \sqrt{x^2+1}$. Show that f is Lipschitz, and hence uniformly continuous, on $\mathbb{R}$.

Proof. As before, let $x, y \in \mathbb{R}$ be arbitrary and write

$$\begin{aligned}
|f(x) - f(y)| = \left|\sqrt{x^2+1} - \sqrt{y^2+1}\right| &= \frac{\left|\sqrt{x^2+1} - \sqrt{y^2+1}\right|\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|}{\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|} \\
&= \frac{\left|x^2 - y^2\right|}{\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|} \\
&= \frac{|x+y|}{\left|\sqrt{x^2+1} + \sqrt{y^2+1}\right|}\,|x-y| \\
&\le \frac{|x|+|y|}{\sqrt{x^2+1} + \sqrt{y^2+1}}\,|x-y| \\
&< \left(\frac{|x|}{\sqrt{x^2+1}} + \frac{|y|}{\sqrt{y^2+1}}\right)|x-y|\,.
\end{aligned}$$

Observe that for all $z \in \mathbb{R}$, we have $|z| = \sqrt{z^2} \le \sqrt{z^2+1}$ whence

$$\frac{|z|}{\sqrt{z^2+1}} \le 1, \quad \forall z \in \mathbb{R}.$$

Using this in the estimates we derived above, we infer that for all $x, y \in \mathbb{R}$ there holds

$$|f(x) - f(y)| < \left(\frac{|x|}{\sqrt{x^2+1}} + \frac{|y|}{\sqrt{y^2+1}}\right)|x-y| \le (1+1)\,|x-y| = 2\,|x-y|\,.$$

□

Example 13.24. Prove that the function $f(x) := \dfrac{1}{x^2+1}$ is Lipschitz, and hence uniformly continuous, on $\mathbb{R}$.

Proof. Let $x, y \in \mathbb{R}$ be given and compute

$$\left|\frac{1}{1+x^2} - \frac{1}{1+y^2}\right| = \frac{\left|x^2 - y^2\right|}{(1+x^2)(1+y^2)} = \frac{|x+y|}{(1+x^2)(1+y^2)}\,|x-y|\,.$$

Now, for any $x, y \in \mathbb{R}$ we have

$$\begin{aligned}\frac{|x+y|}{(1+x^2)(1+y^2)} &\leq \frac{|x|+|y|}{(1+x^2)(1+y^2)} \\ &= \frac{|x|}{(1+x^2)(1+y^2)} + \frac{|y|}{(1+x^2)(1+y^2)} \\ &< \frac{|x|}{1+x^2} + \frac{|y|}{1+y^2}.\end{aligned}$$

Observe that if $|x| \leq 1$ then

$$\frac{|x|}{x^2+1} \leq \frac{|x|}{1} = |x| \leq 1,$$

while if $|x| > 1$,

$$\frac{|x|}{x^2+1} < \frac{|x|}{x^2} = \frac{1}{|x|} < 1.$$

It therefore follows that

$$\frac{|x|}{x^2+1} \leq 1$$

and, similarly,

$$\frac{|y|}{y^2+1} \leq 1.$$

Returning to our previous work, we find that for all $x, y \in \mathbb{R}$

$$\left|\frac{1}{1+x^2} - \frac{1}{1+y^2}\right| < \left(\frac{|x|}{1+x^2} + \frac{|y|}{1+y^2}\right)|x-y| \leq 2\,|x-y|\,.$$

Thus, f is Lipschitz on $\mathbb{R}$. □